POLYTECHNIQUE MONTRÉAL
affiliée à l'Université de Montréal

Numerical Modeling of Three-Dimensional Multilayer In-Flight Ice Accretion Using a Level-Set Method

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Thèse présentée en vue de l'obtention du diplôme de *Philosophiæ Doctor*
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a été dûment acceptée par le jury d'examen constitué de :

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Éric LAURENDEAU, membre et directeur de recherche
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XX XX, membre externe

DEDICATION

À Mama, Baba et Leila
je vous aime . . .

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A PhD is an incredible journey, filled with ups and downs, challenges and achievements, and a true rollercoaster of emotions. Although it is, in many ways, an individual endeavor, reaching the end is far from a solitary accomplishment. It is the result of a collective effort and the invaluable support of supervisors, family, friends, and colleagues who accompany us along the way.

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RÉSUMÉ

Les simulations tridimensionnelles et multicouches d'accrétion de givre présentent d'importants défis numériques puisqu'elles nécessitent le couplage successif de modèles aérodynamiques, d'impact de gouttelettes, de thermodynamique et d'évolution géométrique, tout en mettant à jour de façon répétée une géométrie givrée de plus en plus complexe. En particulier, les méthodes basées sur le déplacement explicite de la surface et le remaillage répété de maillages conformes à la géométrie peuvent devenir sensibles à la qualité du maillage lorsque des structures tridimensionnelles complexes se développent. L'objectif de cette thèse est d'améliorer la robustesse des simulations tridimensionnelles et multicouches d'accrétion de givre par le développement et l'évaluation de méthodologies numériques complémentaires au sein de CHAMPS, un logiciel multiphysique développé à l'interne à Polytechnique Montréal.

Dans un premier temps, une méthodologie lagrangienne de suivi de particules est développée pour la prédiction de l'impact des gouttelettes. L'approche combine une intégration implicite des trajectoires, une détection accélérée des intersections, une injection adaptative des gouttelettes ainsi que des techniques d'interpolation à la surface afin d'améliorer l'efficacité numérique et la prédiction de la quantité d'eau collectée à la surface. Les effets associés aux grosses gouttelettes surfondues (SLD), notamment la déformation des gouttelettes, l'éclaboussement et le suivi des gouttelettes secondaires, sont intégrés aux formulations eulérienne et lagrangienne. Les prédictions obtenues montrent un bon accord avec les données de référence et démontrent une bonne concordance entre les approches eulérienne et lagrangienne. La comparaison met également en évidence leur complémentarité : la formulation eulérienne offre un coût de calcul plus prévisible et une parallélisation efficace pour les grands problèmes tridimensionnels, tandis que la formulation lagrangienne donne directement accès aux trajectoires individuelles et se prête particulièrement bien à l'intégration de phénomènes physiques à l'échelle des gouttelettes.

Dans un deuxième temps, une méthodologie d'évolution géométrique par level set est développée pour l'accrétion de givre tridimensionnelle et multicouche. L'interface givre–air est représentée implicitement par l'isocontour nul d'un champ de distance signée défini sur un maillage volumique auxiliaire. La méthodologie advecte l'interface selon une vitesse déterminée à partir des quantités de givrage, puis extrait l'interface afin d'obtenir une surface conforme à la géométrie résultante. Les cas de référence démontrent que l'approche permet de reproduire les formes de givre attendues tout en offrant une meilleure robustesse géométrique que le déplacement explicite des nœuds de surface. Cet avantage devient partic-

ulièrement important lorsque la complexité géométrique augmente, puisque la représentation implicite évite l'enchevêtrement du maillage surfacique et réduit le recours à des corrections géométriques heuristiques au cours des couches successives d'accrétion.

Finalement, le cadre numérique de givrage est étendu à une formulation par frontière immergée, dans laquelle les solutions aérodynamique et d'impact de gouttelettes sont découplées d'un maillage volumique conforme à la géométrie. Les cas de vérification et d'accrétion montrent un accord encourageant entre les prédictions obtenues avec les approches à maillage conforme et à frontière immergée pour l'impact des gouttelettes et la croissance du givre. Les résultats mettent également en évidence l'importance de la résolution locale du maillage ainsi que du traitement aérodynamique et thermique près de la paroi. Le raffinement adaptatif du maillage apparaît notamment comme un élément essentiel pour étendre efficacement la méthodologie par frontière immergée à des configurations tridimensionnelles complexes.

Dans son ensemble, cette thèse établit un ensemble d'outils numériques complémentaires visant à améliorer la robustesse et la flexibilité des simulations tridimensionnelles d'accrétion de givre. Les résultats montrent que l'évolution géométrique par level set constitue une base robuste pour les simulations multicouches, tandis que le suivi lagrangien des gouttelettes et les méthodes par frontière immergée offrent des capacités supplémentaires pour la modélisation détaillée des gouttelettes et le traitement de géométries complexes.

ABSTRACT

Three-dimensional multilayer ice accretion simulations present significant numerical challenges because they require the successive coupling of aerodynamic, droplet-impingement, thermodynamic, and geometry-evolution models while repeatedly updating an increasingly complex iced geometry. In particular, explicit surface-displacement techniques and repeated body-fitted remeshing can become sensitive to mesh quality as complex 3D ice features develop. The objective of this thesis is to improve the robustness of 3D multilayer ice accretion simulations through the development and assessment of complementary numerical methodologies within CHAMPS, an in-house multiphysics software being developed at Polytechnique Montréal.

First, a Lagrangian particle-tracking methodology is developed for droplet-impingement prediction. The approach combines implicit trajectory integration, accelerated intersection detection, adaptive droplet seeding, and surface interpolation techniques to improve computational efficiency and the prediction of the amount of water collected at the surface. Supercooled Large Droplet (SLD) effects, including droplet deformation, splashing, and secondary-droplet tracking, are incorporated into both Eulerian and Lagrangian formulations. The resulting predictions show good agreement with reference data and demonstrate consistency between the Eulerian and Lagrangian approaches. The comparison also highlights their complementary characteristics: the Eulerian formulation provides more predictable computational cost and efficient parallelization for large 3D problems, whereas the Lagrangian formulation provides direct access to individual droplet trajectories and is particularly attractive for incorporating detailed droplet-scale physics.

Second, a continuous level-set methodology is developed for 3D multilayer ice accretion. The ice–air interface is represented implicitly as the zero contour of a signed-distance field defined on an auxiliary volume mesh. The methodology advects the interface using a velocity determined from the icing quantities, and subsequently extracts a conformal surface representing the resulting geometry. Benchmark calculations demonstrate that the approach can reproduce reference ice shapes while providing increased geometric robustness compared with explicit surface-node displacement. This advantage becomes particularly important as geometric complexity increases, since the implicit representation avoids surface-mesh entanglement and reduces the need for heuristic geometric corrections during successive accretion layers.

Finally, the icing framework is extended to an immersed-boundary formulation in which

the aerodynamic and droplet-impingement solutions are decoupled from a body-fitted volume mesh conforming to the evolving geometry. Verification and ice-accretion calculations demonstrate encouraging agreement between body-fitted and immersed-boundary predictions for droplet impingement and ice growth. The results also highlight the importance of local mesh resolution and near-wall aerodynamic and thermal treatments. In particular, adaptive mesh refinement is identified as an important component for efficiently extending the immersed-boundary methodology to complex 3D configurations.

Overall, this thesis establishes a set of complementary numerical tools for improving the robustness and flexibility of 3D ice accretion simulations. The results indicate that level-set geometry evolution provides a robust foundation for multilayer simulations, while Lagrangian droplet tracking and immersed-boundary methods provide additional capabilities for detailed droplet modeling and the treatment of complex evolving geometries, respectively.

TABLE OF CONTENTS

DEDICATION iv

ACKNOWLEDGEMENTS v

RÉSUMÉ vi

ABSTRACT viii

LIST OF TABLES xiv

LIST OF FIGURES xv

LIST OF SYMBOLS AND ACRONYMS xxi

LIST OF APPENDICES xxv

CHAPTER 1 INTRODUCTION 1
- 1.1 Context 1
- 1.2 Ice Morphologies 2
- 1.3 Regulations & Certification 3
- 1.4 Ice Accretion Numerical Modeling 5
- 1.5 Problem Statement 6
  - 1.5.1 Motivation for the Present Work 8
  - 1.5.2 Objectives of the Present Study 9
- 1.6 Thesis Outline 11

CHAPTER 2 LITERATURE REVIEW 13
- 2.1 Overview of Numerical Ice Accretion Modeling 13
- 2.2 Aerodynamics 15
- 2.3 Dispersed Phase Modeling 17
  - 2.3.1 Eulerian Droplet Formulation 19
  - 2.3.2 Lagrangian Droplet Formulation 20
  - 2.3.3 Supercooled Large Droplet Modeling 21
- 2.4 Thermodynamic Modeling 24
- 2.5 Geometry Evolution 27
  - 2.5.1 Interface Tracking 27
  - 2.5.2 Interface Capturing 28

2.6 Geometry Representation 30
2.6.1 Geometrical Methods 32
2.6.2 Continuous-Forcing Methods 33
2.6.3 Discrete-Forcing Methods 35
2.7 Synthesis and Positioning of the Present Work 37

CHAPTER 3 A Ray-Tracing-Enhanced Lagrangian Particle Tracking Framework for Ice Accretion Applications 40
3.1 Introduction 40
3.2 Lagrangian Droplet Tracking for Ice Accretion 41
3.3 Improvements in Lagrangian Particle Tracking 45
3.3.1 Ray–Quadrilateral Intersections in CFD Applications 45
3.3.2 Adaptive Seeding 48
3.3.3 Collection Efficiency Computations and SLD Effects 49
3.3.4 Parallelism 53
3.4 Results 54
3.4.1 Scalability 55
3.4.2 Case IPW1 - 112: NACA64A008 Horizontal Tail 56
3.4.3 Case IPW1 - 122: Three Element Airfoil 58
3.4.4 Case HLPW5 - 2.1: CRM Wing-Body 62
3.5 Conclusion 64

CHAPTER 4 Article : Three-Dimensional Multilayer Ice Accretion Using a Level-Set Method 65
4.1 Introduction 65
4.2 CHAMPS Icing Framework 68
4.3 Level-Set Ice Accretion Formulation 69
4.3.1 Level-Set Governing Equations and Initialization 69
4.3.2 Level-Set Advection: Velocity Formulation, Surface-Quantity Propagation, and Time Integration 70
4.3.3 Ice Mass Computation and Local Interface Reconstruction 73
4.4 Geometry Extraction and Smoothing 74
4.5 Results 77
4.5.1 Three-Horns Ice Accretion Case 78
4.5.2 Case IPW1 241 79
4.5.3 Case IPW1 242 80
4.5.4 Case IPW1 361 85

4.5.5 Case IPW1 362 86
4.6 Conclusion 88

CHAPTER 5 An Immersed-Boundary Approach for Multilayer Ice Accretion Simulations 90
5.1 Introduction 90
5.2 Immersed-Boundary Representation 91
5.3 Immersed-Boundary Flow Treatment 93
5.4 Immersed-Boundary Droplet Treatment 96
5.4.1 Eulerian Formulation: Governing Equations and Penalization Strategy 97
5.4.2 Lagrangian Immersed-Surface Tracking 98
5.5 Immersed-Boundary Icing Coupling 99
5.6 Results 100
5.6.1 Cylinder Impingement Verification 100
5.6.2 SLD Impingement 101
5.6.3 Case 01 101
5.6.4 Case 04 103
5.6.5 Case RG15 104
5.7 Conclusion 107

CHAPTER 6 GENERAL DISCUSSIONS 109
6.0.1 Eulerian and Lagrangian Droplet Impingement 109
6.0.2 Algebraic and Level-Set Ice Accretion 112
6.0.3 Body-Fitted and Immersed-Boundary Approaches 114

CHAPTER 7 CONCLUSION 116
7.1 Summary of Works 116
7.2 Limitations 118
7.3 Future Research 119

REFERENCES 120

APPENDICES 135
A.1 Mass Balance 135
A.2 Energy Balance 135
B.1 Mesh Refinement Study 138
B.2 SLD Model Details 140
C.1 Lagrangian Droplet Integration 141

C.2 Level-Set Transport 141
C.2.1 Spatial Convergence 142
C.2.2 Temporal Convergence 142
C.3 Eulerian Immersed-Boundary Droplet Formulation 143

LIST OF TABLES

Table 2.1	Comparison of representative numerical aircraft icing solvers.	15
Table 2.2	SLD splashing models.	24
Table 3.1	Performance comparison of ray–quadrilateral intersection methods.	48
Table 3.2	Summary of the droplet-impingement benchmark cases.	54
Table 4.1	Icing Velocity Formulations	71
Table 4.2	Summary of the IPW1 benchmark cases.	78
Table 4.3	Comparison of the maximum ice height relative to the clean geometry.	80
Table 4.4	Comparison of the upper and lower horn angles for Case 242, together with the relative error with respect to the experimental reference.	85
Table 5.1	Summary of the benchmark cases.	100
Table 6.1	Reference conditions for the IPW3 droplet-impingement benchmark cases.	110
Table 6.2	NACA0012 grid hierarchy.	110
Table 6.3	Representative computational performance of the Eulerian and Lagrangian droplet formulations for the ONERA M6 configuration.	111
Table B.1	Summary of the NACA 0012 droplet-impingement verification case.	138
Table B.2	Difference Between Eulerian vs. Lagrangian results in terms of Collection Efficiency.	138
Table B.3	SLD Splashing Models Details	140

LIST OF FIGURES

Figure 1.1 Examples of ice accretion on aircraft components. 2

Figure 1.2 Ice accretion on the leading edge of the Common Research Model (CRM) hybrid wing configuration tested in the NASA Icing Research Tunnel. Photos taken from [1]. 3

Figure 1.3 Comparison of single-step and multilayer ice accretion predictions on a NACA 23012 airfoil at iso-physical parameters. 6

Figure 1.4 Mesh entanglement on a 30° swept NACA 0012 wing during multilayer ice accretion. 7

Figure 1.5 Mesh representation around a cylinder using (a) a body-fitted mesh and (b) an immersed-boundary mesh. 9

Figure 2.1 Schematic of the multilayer ice accretion framework with representative solutions at each stage. 14

Figure 2.2 Schematic representation of the Lagrangian droplet-seeding procedure. Droplets are released from an upstream plane and tracked until they impinge on the surface. 18

Figure 2.3 Examples of droplet deformation and splashing phenomena relevant to SLD conditions. 22

Figure 2.4 Control volume diagrams for mass and energy balance equations. Adapted from [2]. 25

Figure 2.5 Schematic representation of surface-node displacement along the local normal according to the predicted ice thickness. 27

Figure 2.6 Schematic representation of the level-set function. The ice–air interface Γ is implicitly defined by the zero level set ($\phi = 0$), separating the solid domain Ω_S ($\phi < 0$) from the surrounding air domain Ω_F ($\phi > 0$). . . 29

Figure 2.7 Illustration of the boundary cut-cell method [3]. 32

Figure 2.8 Schematic representation of the solid and fluid domains using the mask function χ. 34

Figure 2.9 Image-point and ghost-cell construction near an immersed boundary [4]. 36

Figure 3.1 Example of Eulerian solver results : Countour of water volume fraction α with droplet trajectories streamlines. 42

Figure 3.2 Droplet tracking techniques. 43

Figure 3.3	Cut through the computational mesh of the CRM wing–body configuration from HLPW5, illustrating the hexahedral refinement around the aircraft geometry.	46
Figure 3.4	Steps for ray–bilinear patch intersection.	47
Figure 3.5	Hybrid Eulerian–Lagrangian backtracking for initialization of the Lagrangian seeding region [5].	49
Figure 3.6	Example of adaptive refinement of the Lagrangian seeding grid around the region associated with impacting droplet trajectories.	50
Figure 3.7	Collection-efficiency computations. Primary and auxiliary droplets are represented in blue and red, respectively.	51
Figure 3.8	Schematic representation of secondary-droplet parcel tracking and re-impingement following splashing.	52
Figure 3.9	Parallel scalability of the Lagrangian solver: speedup and parallel efficiency relative to the two-node baseline.	55
Figure 3.10	Collection-efficiency results for Case IPW1-112: impact-zone treatment, droplet-size distribution, Eulerian–Lagrangian comparison, and SLD-model assessment.	57
Figure 3.11	Description of Case IPW1 - 122.	59
Figure 3.12	Aggregated impingement results of all IPW1 participants for Case IPW1-122.	59
Figure 3.13	Collection-efficiency results for Case IPW1-122. The upper row compares Eulerian and Lagrangian impingement predictions; the lower row assesses the SLD models and secondary-droplet re-impingement.	61
Figure 3.14	Locations of the collection-efficiency cuts on the CRM wing and fuselage.	62
Figure 3.15	Eulerian and Lagrangian collection-efficiency results for Case HLPW5-2.1 at the fuselage and five spanwise wing cuts.	63
Figure 4.1	Schematic of the multilayer ice accretion framework.	66
Figure 4.2	Representation of the level-set function ϕ. The ice region is defined by $\phi < 0$ (blue), while the surrounding air corresponds to $\phi > 0$. The interface is implicitly represented by the zero level set $\phi = 0$.	69
Figure 4.3	Contours of the level-set function ϕ obtained from the Eikonal equation, representing the wall-distance field.	70
Figure 4.4	Contours of the propagated scalar field φ, here representing the ice thickness extended from the clean surface into the volume. The field is transported along the ϕ-normal direction.	72

Figure 4.5 Illustration of local interface reconstruction within simplicial elements based on the sign of the level-set function ϕ. Red nodes denote positive values ($\phi > 0$), whereas blue nodes denote negative values ($\phi < 0$). The interface $\phi = 0$ is reconstructed by linear interpolation along the intersected edges. In 2D, the negative subregion is either the full triangle, a quadrilateral, or a triangle. In 3D, the negative subregion is either the full tetrahedron, a truncated tetrahedron, a clipped polyhedron, or a smaller tetrahedral subregion. These clipped regions are used to evaluate the local ice area or volume contribution without requiring a full surface extraction. 74

Figure 4.6 Adapted unstructured mesh around the iced surface of the geometry, showing local refinement in order to accurately capture the interface in the level-set framework. 75

Figure 4.7 Comparison of the effect of [6] smoothing on the extracted ice surface. Smoothing improves interface regularity in the first layer and prevents the accumulation of geometric distortions in the final ice shape. . . . 77

Figure 4.8 Comparison of the ice fronts obtained using Lagrangian node displacement and the proposed level-set approach for the manufactured three-horn configuration. 79

Figure 4.9 Comparison of the predicted aerodynamic and icing-related surface quantities for IPW1 Case 241 with the range of results submitted by IPW1 participants. 80

Figure 4.10 Predicted ice shapes for IPW1 Case 241. The clean and experimental profiles are shown with the IPW1 participants' results. CHAMPS level-set predictions using 1, 10, and 20 accretion layers are included to assess the effect of the multilayer approach. 81

Figure 4.11 Comparison of the predicted aerodynamic and icing-related surface quantities for IPW1 Case 242 with the range of results submitted by IPW1 participants. 82

Figure 4.12 Comparison of the predicted ice shapes for Case 242 obtained with different prescribed roughness values. Increasing k_s/c modifies the heat-transfer distribution and improves the predicted horn orientation. . . 83

Figure 4.13 Comparison of glaze ice predictions for Case 242: temporal convergence with respect to the number of accretion layers, variation of the heat transfer coefficient (HTC) with roughness magnitude k_s, and influence of k_s on the predicted ice shape. 83

Figure 4.14 Definition of the horn angles on the experimental ice shape. 84
Figure 4.15 Comparison of the predicted aerodynamic and icing-related surface quantities for IPW1 Case 361 against the results submitted by IPW1 participants. 85
Figure 4.16 Predicted ice shapes for Case 361. The effect of the number of accretion layers on the final ice shape is shown for 1, 2, and 5 layers, together with the experimental ice shape and the IPW1 participants' results. A 3D view of the predicted accreted geometry is also presented. 86
Figure 4.17 Predicted 3D ice shape for IPW1 Case 362 after 10 accretion layers. The clipped views illustrate the spanwise scallop structures formed along the leading edge and the corresponding adapted surface mesh after level-set extraction. 87
Figure 4.18 Predicted ice shapes for IPW1 Case 362. The experimental ice shape, clean airfoil profile, and IPW1 participant results are shown for reference. The level-set predictions obtained with 1 and 10 accretion layers are compared, together with the ice shape extracted from a single spanwise slice and the maximum combined cross-section (MCCS) extracted from the 3D ice shape. 88
Figure 5.1 Signed level-set field ϕ around a circular cylinder. The zero level set defines the cylinder surface, while positive values correspond to the fluid region and negative values to the solid region. 92
Figure 5.2 Comparison of AABB-tree and quadtree search structures for a circular immersed surface. The AABB tree recursively partitions the surface elements and encloses each group within a bounding box, whereas the quadtree recursively partitions the surrounding spatial domain. 93
Figure 5.3 Schematic representation of the ghost-cell treatment near an immersed boundary. The standard approach (a) defines ghost-cell values using image points along the local wall-normal direction, while the modified approach (b) introduces additional ghost cells to impose the RANS wall treatment at a prescribed distance from the immersed interface. 94
Figure 5.4 Mask functions used in the penalized Eulerian droplet formulation: (a) solid–fluid mask χ, (b) droplet-direction mask χ_d, and (c) combined penalization indicator $\chi\chi_d$, which restricts the penalization to the overlap between the immersed-solid and incoming-droplet regions. 98

Figure 5.5 Droplet tracking in an immersed-boundary framework. The trajectory is advanced through the background mesh until it intersects the reconstructed immersed surface. 99
Figure 5.6 Collection efficiency distributions on the cylinder obtained with Eulerian and Lagrangian formulations on body-fitted and immersed-boundary meshes. 102
Figure 5.7 Collection efficiency distributions for the Papadakis SLD impingement case obtained with the Eulerian and Lagrangian formulations, with and without SLD effects. 103
Figure 5.8 Predicted ice shapes for Case 01 using body-fitted and immersed-boundary formulations. The clean and experimental profiles are shown for reference, together with predictions obtained using 1 and 5 accretion layers. 104
Figure 5.9 Predicted ice shapes and convective heat transfer coefficient distributions for Case 04 using body-fitted and immersed-boundary formulations. The clean and experimental profiles are shown for reference in the ice-shape comparisons, together with predictions obtained using 1 and 5 accretion layers. 105
Figure 5.10 Comparison of the collection-efficiency distributions predicted by the Eulerian and Lagrangian immersed-boundary formulations for the RG15 configuration. 106
Figure 5.11 Predicted 3D ice accretion on the RG15 wing: (a) evolution of the ice geometry through successive accretion layers and (b) clean wing and predicted ice geometry. 107
Figure 6.1 Spatial convergence of the integrated collected-water mass obtained with the Eulerian and Lagrangian droplet formulations for the NACA0012 IPW3 configuration using the MVD of 20 μm. 110
Figure 6.2 Comparison of algebraic and level-set ice shapes under identical icing conditions after five accretion layers. 113
Figure 6.3 3D level-set ice shapes obtained for different numerical and physical configurations. 114
Figure 6.4 Example of the prescribed refined region surrounding the NACA0012 geometry and the anticipated ice-accretion region. 115
Figure B.1 Mesh refinement effect on the Eulerian vs. Lagrangian results in terms of the collection efficiency. 139

Figure C.1 Temporal convergence of the Lagrangian droplet trajectory integration using BDF2 with a BDF1 startup step. 142
Figure C.2 Spatial and temporal convergence of the level-set transport solver for the uniform translation of a circular level-set field: (a) spatial convergence with mesh refinement and (b) temporal convergence of the second-order Rosenbrock–Wanner scheme with time-step refinement. . 143
Figure C.3 Spatial convergence of the integrated collection efficiency for the Eulerian immersed-boundary cylinder impingement problem. 144

LIST OF SYMBOLS AND ACRONYMS

Acronyms

AABB	Axis-aligned bounding box
AIAA	American Institute of Aeronautics and Astronautics
AMR	Adaptive mesh refinement
AOA	Angle of attack
BDF	Backward differentiation formula
BF	Body-fitted
CFD	Computational fluid dynamics
CFR	Code of Federal Regulations
CHAMPS	Chapel Multiphysics Software
CRM	Common Research Model
FAA	Federal Aviation Administration
FAR	Federal Airworthiness Regulations
FR	Freezing rain
GARP	Geometric Approach to Ray–Patch Intersections
GC	Ghost cell
GMRES	Generalized minimal residual method
HLPW5	Fifth High-Lift Prediction Workshop
HPC	High-performance computing
HTC	Heat-transfer coefficient
IBM	Immersed-boundary method
IM	Image point

IPS	Ice-protection system
IPW	Ice Prediction Workshop
IRT	Icing Research Tunnel
LES	Large-eddy simulation
LWC	Liquid water content
MCCS	Maximum combined cross section
MOF	Moment of fluid
MPI	Message Passing Interface
MVD	Median volume diameter
NUMA	Non-uniform memory access
ODE	Ordinary differential equation
PDE	Partial differential equation
RANS	Reynolds-averaged Navier–Stokes
ROW	Rosenbrock–Wanner
SA	Spalart–Allmaras
SLD	Supercooled large droplet
SSD	Standard supercooled droplet
SWIM	Shallow-Water Icing Model
VOF	Volume of fluid

Symbols

A	Area [m^2]
C_D	Droplet drag coefficient [-]
C_{ref}	Reference chord length [m]
D	Droplet diameter [m]
F	Droplet drag-response coefficient [s^{-1}]
f_m	Mass-loss coefficient [-]
Fr	Froude number [-]
$\boldsymbol{g}$	Gravitational acceleration [$\mathrm{m\,s}^{-2}$]
h	Water-film thickness [m]
h_{ice}	Local ice thickness [m]
H_{mod}	Wall-modeling height [m]
HTC	Convective heat-transfer coefficient [$\mathrm{W\,m}^{-2}\,\mathrm{K}^{-1}$]
k	Thermal conductivity [$\mathrm{W\,m}^{-1}\,\mathrm{K}^{-1}$]
K	Droplet inertia parameter [-]
M	Mach number [-]
$\dot{m}$	Mass flow rate [$\mathrm{kg\,s}^{-1}$]
$\dot{m}_{\mathrm{ice}}$	Local ice mass-accretion rate [$\mathrm{kg\,m}^{-2}\,\mathrm{s}^{-1}$]
$\boldsymbol{n}$	Outward unit-normal vector [-]
$\boldsymbol{n}_\phi$	Level-set interface unit-normal vector [-]
p	Pressure [Pa]
Pr	Prandtl number [-]
$\dot{Q}$	Heat flux [$\mathrm{W\,m}^{-2}$]
r	Recovery factor [-]
R	Specific gas constant [$\mathrm{J\,kg}^{-1}\,\mathrm{K}^{-1}$]
Re	Reynolds number [-]
Re_d	Droplet Reynolds number [-]
t	Time [s]
T	Temperature [K]
T_0	Stagnation temperature [K]
T_∞	Freestream temperature [K]
$\boldsymbol{u}$	Velocity vector [$\mathrm{m\,s}^{-1}$]
$\boldsymbol{u}_a$	Air velocity vector [$\mathrm{m\,s}^{-1}$]
$\boldsymbol{u}_d$	Droplet velocity vector [$\mathrm{m\,s}^{-1}$]
U_∞	Freestream velocity [$\mathrm{m\,s}^{-1}$]

$\boldsymbol{V}$	Interface-growth velocity [$\mathrm{m\,s^{-1}}$]
We	Weber number [-]
$\boldsymbol{x}_d$	Droplet position vector [m]
$\boldsymbol{x}_{IB}$	Position of the immersed-boundary point [m]
$\boldsymbol{x}_V$	Position of the neighboring volume point [m]
y^+	Nondimensional wall distance [-]
α	Nondimensional water concentration [-]
β	Collection efficiency [-]
γ	Droplet-bin weight [-]
Δt	Time step [s]
$\Delta t_{\mathrm{layer}}$	Accretion time associated with one ice layer [s]
ϵ	Sticking rate [-]
η	Immersed-boundary penalization parameter [-]
κ	Interface curvature [$\mathrm{m^{-1}}$]
μ	Dynamic viscosity [$\mathrm{Pa\,s}$]
μ_a	Air dynamic viscosity [$\mathrm{Pa\,s}$]
ρ	Density [$\mathrm{kg\,m^{-3}}$]
ρ_a	Air density [$\mathrm{kg\,m^{-3}}$]
ρ_d	Droplet density [$\mathrm{kg\,m^{-3}}$]
ρ_w	Water density [$\mathrm{kg\,m^{-3}}$]
σ	Surface tension [$\mathrm{N\,m^{-1}}$]
ϕ	Level-set signed-distance function [m]
$\boldsymbol{\varphi}_{IB}$	Flow variable at the immersed-boundary point
$\boldsymbol{\varphi}_V$	Flow variable at the neighboring volume point
χ	Solid-mask function [-]

LIST OF APPENDICES

Appendix A THERMODYNAMIC BALANCE EQUATIONS 135
Appendix B DROPLET-IMPINGEMENT VERIFICATION AND SLD MODELS 138
Appendix C NUMERICAL VERIFICATION 141

CHAPTER 1 INTRODUCTION

This chapter introduces the context of aircraft icing, its impact on flight safety, and its role in certification. It reviews the main ice morphologies, regulatory requirements, and certification tools, before presenting the challenges of numerical ice accretion modeling. The chapter concludes by defining the thesis objectives and outlining the organization of the thesis.

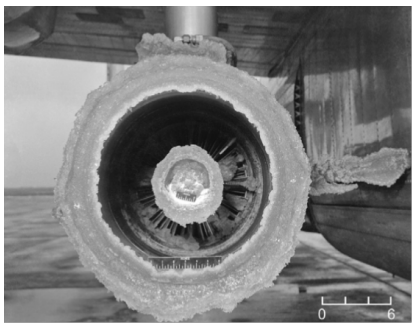
1.1 Context

In the context of climate change and increasingly stringent environmental regulations, the aeronautical industry is moving toward the development of more efficient and lower-emission aircraft. This transition introduces new aerodynamic, structural, and operational challenges, while maintaining flight safety as the primary requirement [7]. One by-product is the need for reliable certification procedures capable of assessing aircraft performance and safety over a wide range of operating conditions.

Among the phenomena considered during aircraft certification, in-flight icing presents significant challenges to aircraft safety and performance [8]. Ice accretion occurs when an aircraft flies through clouds containing supercooled water droplets[1]. When these droplets impinge on exposed surfaces and the local thermal conditions allow freezing, ice can accumulate on the aircraft geometry. The resulting ice shapes may significantly modify the aerodynamic configuration of the aircraft, leading to increased drag, reduced lift, altered stall behavior, and degraded controllability [8–11]. While aircraft icing is commonly associated with large exposed components such as wings, empennages, aircraft nose, and engines, it may also affect smaller but critical components, including probes and sensors, as illustrated in Figure 1.1. Ice accretion on these components can alter the aircraft response and produce erroneous measurements comprising flight safety.

Aircraft icing was not initially regarded as a primary concern during the early years of powered flight. Because aircraft were not yet equipped for routine instrument flight, pilots generally avoided clouds, and encounters with icing conditions were mostly inadvertent. This situation changed in the mid-1920s, when the U.S. Air Mail Service began operating scheduled day-and-night routes between New York and Chicago. These operations exposed pilots to icing conditions on a regular basis, and several serious incidents and accidents in 1926 and 1927 highlighted the need for practical solutions to the icing problem [8]. Early research

[1] Supercooled water droplets are liquid water droplets that remain unfrozen despite being at temperatures below the freezing point.



(a) Aircraft engine [12]

(b) Pitot probe [13]

Figure 1.1 Examples of ice accretion on aircraft components.

efforts were therefore mainly directed toward ice-protection systems (IPS)[2], including heated aircraft components, waterproof coatings, greased propellers, electrically heated pitot-static tubes, and pneumatic de-icing boots [8]. Although some of these solutions proved useful in specific applications, they remained largely empirical and did not provide a complete understanding of the physical mechanisms governing ice accretion.

Over the following decades, repeated claims that aircraft icing had been effectively solved proved premature, as operational experience continued to reveal unresolved challenges and the need for dedicated experimental investigation under controlled conditions. This need motivated the development of controlled icing research capabilities, culminating in the construction of the Icing Research Tunnel (IRT) in Cleveland in the 1940s [8]. The IRT enabled systematic investigations of supercooled cloud conditions, droplet impingement, aerodynamic performance degradation, and the performance of anti-icing and de-icing systems. These experimental studies played a central role in improving the physical understanding of aircraft icing and later provided essential data for the development, validation, and certification of numerical icing prediction methods [14]. These early efforts subsequently motivated the development of similar experimental icing facilities, which are discussed in Section 1.3.

1.2 Ice Morphologies

Depending on the atmospheric and aerodynamic conditions, different ice morphologies may develop on aircraft surfaces. At low temperatures, rime ice forms when supercooled droplets freeze almost immediately upon impact. This rapid solidification produces a relatively smooth, streamlined, and opaque ice shape. The opaque appearance of rime ice is asso-

[2]Ice protection systems are divided into two categories: anti-icing systems are designed to prevent ice formation on protected surfaces, whereas de-icing systems allow ice to form temporarily before removing it.

ciated with air trapped within the ice during the rapid freezing process.

As the temperature increases toward the freezing point, glaze ice may form. In this regime, the impinging droplets do not freeze completely upon impact. Instead, part of the liquid water remains on the surface, forming a thin water film that can run back before freezing downstream. Glaze ice is therefore usually denser and more transparent than rime ice and often leads to more complex ice shapes, including horn-like structures. On swept configurations, it can also produce 3D features such as ice feathers and larger-scale structures commonly referred to as scallops or lobster tails because of their characteristic resemblance to these forms (Fig. 1.2).

A third regime, mixed ice, occurs as a transition between rime and glaze conditions. In this case, glaze-like ice may form near the stagnation region, where the impingement rate is high and the water does not freeze completely on impact. Farther downstream, the remaining droplets or runback water may freeze more rapidly, producing rime-like deposits. Mixed ice therefore combines features of both rime and glaze ice.

Figure 1.2 Ice accretion on the leading edge of the Common Research Model (CRM) hybrid wing configuration tested in the NASA Icing Research Tunnel. Photos taken from [1].

1.3 Regulations & Certification

The diversity of ice morphologies and the severity of their aerodynamic effects make in-flight icing a critical consideration in aircraft certification. Regulatory authorities require aircraft to demonstrate safe operation within prescribed icing environments, which are defined through

envelopes of atmospheric and flight conditions. A major international reference for the certification of transport-category airplanes is Title 14 of the Code of Federal Regulations (CFR), Part 25, adopted by the Federal Aviation Administration (FAA). Within this framework, Appendix C [15] has historically defined the reference atmospheric icing envelopes used for the design and certification of IPS. These envelopes, in use since 1964, describe continuous maximum and intermittent maximum icing conditions in terms of parameters such as liquid water content, temperature, altitude, cloud horizontal extent, and representative droplet diameter [16]. They were primarily established for conventional cloud icing conditions, where the droplet median volume diameter (MVD) is typically limited to the range of standard supercooled droplets (SSD), up to approximately 50 μm.

However, several in-service events demonstrated that aircraft could also encounter icing environments outside the original Appendix C envelopes. A major example is the crash of American Eagle Flight 4184, an ATR-72, near Roselawn, Indiana, on October 31, 1994 [17]. The accident investigation identified icing as a central factor in the loss of control and highlighted the safety implications of supercooled large droplet (SLD) conditions, including freezing drizzle and freezing rain. These environments may contain droplets significantly larger than those considered in Appendix C, with droplet diameters reaching several hundred micrometres, typically up to about 500 μm in freezing-drizzle conditions, and up to approximately 1000 μm in freezing-rain conditions.

This recognition led to additional regulatory developments, culminating in the introduction of Appendix O for SLD icing conditions [18]. Appendix O extends the certification envelope to include freezing drizzle and freezing rain, thereby addressing icing scenarios in which large droplets may splash, rebound, or break up before freezing. These mechanisms may lead to ice formation downstream of the protected region. Consequently, aircraft seeking certification for flight in icing conditions must demonstrate safe operation under these conditions.

The certification of IPS may rely on several complementary approaches, including flight testing, icing wind-tunnel testing, and numerical simulation [19]. Flight tests provide the direct assessment of aircraft behavior in realistic operating conditions and may be conducted either in natural icing environments or using artificial ice shapes. However, such tests are expensive, difficult to reproduce, and may involve safety risks, particularly when degraded aerodynamic performance or controllability must be evaluated.

Icing wind tunnels therefore became an essential complementary experimental tool for studying ice accretion and evaluating IPS performance under controlled conditions. Facilities such as the NASA IRT, the National Research Council (NRC) Low-Speed Icing Wind Tunnel in Canada [20], the Rail Tec Arsenal (RTA) facility in Austria [21], and the Office National

d'Études et de Recherches Aérospatiales (ONERA) icing facilities [22] in France have contributed extensively to the development of experimental databases for icing research and certification support. Compared with flight testing, wind-tunnel experiments offer better control of atmospheric parameters, and detailed access to local measurements. Nevertheless, they remain costly and face important scaling limitations, especially as aircraft components increase in size. Reproducing full-scale icing conditions often requires preserving several competing similarity parameters, including the relationships between droplet size, leading-edge radius, liquid water content, velocity, temperature, and accretion time. In practice, the model geometry or operating conditions must often be adjusted to fit within the facility constraints, and these adjustments can themselves require dedicated extensive analyses [23].

In addition, ice accretion experiments are subject to variability from one test to another, owing to uncertainties in cloud calibration, droplet distributions, thermal conditions, surface roughness, and measurement procedures. These limitations have motivated the development of numerical icing tools as a complementary approach to flight and wind-tunnel testing.

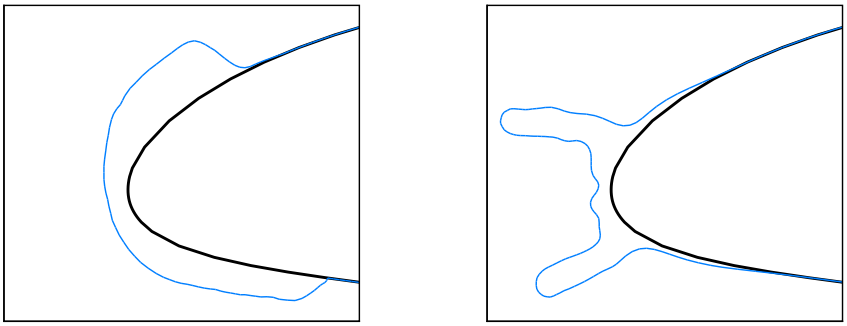
1.4 Ice Accretion Numerical Modeling

Ice accretion results from the interaction of several physical processes, including the aerodynamic flow around the aircraft, the transport and impingement of supercooled water droplets, the local thermodynamic balance at the surface, and the progressive modification of the geometry due to ice growth.

Although ice accretion is an unsteady phenomenon, most computational icing solvers rely on a quasi-steady approximation. This assumption is justified by the separation between the short characteristic time scales of the airflow and droplet motion and the much longer time scale associated with ice growth. In practice, the total exposure time is divided into one or several discrete accretion steps. Within each step, all the physical processes are simulated under steady-state conditions [24].

The simplest implementation consists of performing the entire accretion process in a single step. This approach is attractive because it limits the computational cost and avoids repeated geometry and mesh updates. However, it assumes that the flow, droplet trajectories, and surface thermodynamic solutions remain representative throughout the full exposure time, which can reduce the accuracy of the predicted ice shape. A more representative strategy consists of using a multilayer approach, in which the geometry is updated several times during the accretion process and the physical fields are recomputed on the evolving surface. The influence of these successive updates on the predicted ice shape is illustrated in

Fig. 1.3. Consequently, the computational cost increases with the number of accretion steps, as the aerodynamic, droplet-impingement, and thermodynamic solutions must be repeatedly recomputed on the evolving geometry. For 3D configurations, the size of the computational meshes and the repeated solution of these coupled physical models can result in substantial computational requirements.



(a) Single-step ice shape. (b) Multilayer ice shape.

Figure 1.3 Comparison of single-step and multilayer ice accretion predictions on a NACA 23012 airfoil at iso-physical parameters.

By accounting for the progressive modification of the geometry, the multilayer approach provides a stronger coupling between ice growth and the surrounding physical fields. This improvement comes at the cost of additional computations and repeated geometry and mesh updates, which introduce further computational robustness and accuracy challenges. In particular, successive geometry deformation or remeshing may progressively deteriorate the mesh quality, leading to highly skewed, tangled, or invalid elements. This degradation in mesh quality can adversely affect the numerical solvers by increasing numerical dissipation and stiffness, thereby reducing solution accuracy and convergence performance. In more severe cases, highly distorted or invalid elements may lead to solver instability and ultimately numerical failure. Such deterioration of the mesh elements is illustrated in Fig. 1.4 on a 30° swept NACA 0012 wing.

1.5 Problem Statement

Two-dimensional icing simulations have played a central role in the development and validation of ice accretion models. They remain computationally efficient and are particularly useful for airfoil configurations, parametric studies, and comparisons with well-controlled experimental databases. In some applications, they are also extended to approximate 3D effects

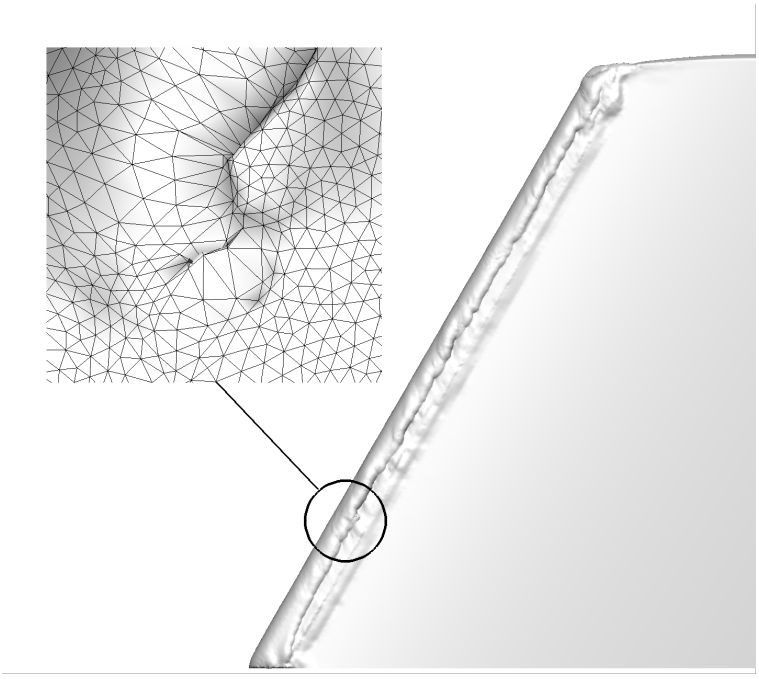


Figure 1.4 Mesh entanglement on a 30° swept NACA 0012 wing during multilayer ice accretion.

through 2.5D approaches [25]. However, such methods remain limited when the ice accretion process is governed by genuinely 3D flow features, spanwise transport, complex geometry effects, or non-uniform impingement patterns.

Fully 3D simulations provide a more general framework for predicting ice accretion on realistic aircraft components and complex configurations. Nevertheless, they are significantly more demanding in terms of computational resources and numerical robustness. The large computational meshes required to resolve 3D geometries, combined with the repeated solution of the aerodynamic, droplet, and thermodynamic models in a multilayer framework, make high-performance computing (HPC) an important component of practical 3D ice accretion simulations. Efficient parallel algorithms and scalable numerical methods are therefore required to maintain tractable computational times as the problem size and number of accretion steps increase. These difficulties become particularly important in multilayer simulations, where the geometry must be updated repeatedly as the ice grows. These difficulties become particularly important in multilayer simulations, where the geometry must be updated repeatedly as the ice grows. Each update may degrade the mesh quality or generate invalid surface and volume elements (Fig. 1.4). As a result, a simulation that has already required substantial computational effort may fail before reaching the final accretion time.

This robustness constraint partly explains why single-step ice accretion strategies remain

widely used in practical applications. They are simpler to execute and reduce the risk of mesh failure, but they also limit the ability of the solver to account for the feedback between the evolving ice shape and the surrounding flow and droplet fields. In the Ice Prediction Workshop (IPW)[3], many submitted ice-shape predictions relied on such single-step approaches. While effective from a computational standpoint, these methods generally showed reduced ability to reproduce complex accretion features, especially when compared with multilayer simulations [26,27]. This highlights the need for numerical strategies capable of preserving the robustness of single-step approaches while retaining the improved physical fidelity of multilayer ice-growth modeling.

1.5.1 Motivation for the Present Work

The preceding discussion identifies the repeated update of the iced geometry as a major limitation of 3D multilayer ice accretion simulations. Although the flow, droplet, and thermodynamic solutions must be recomputed consistently as the ice shape evolves, the ability to reach the final accretion time often depends on the robustness of the geometry-update procedure itself.

This motivates the use of geometry representations that are less sensitive to local surface irregularities. Previous studies [2, 28, 29] have demonstrated the potential of level-set descriptions for representing and evolving iced geometries in 2D configurations. In a level-set formulation, the interface is not tracked explicitly by the surface mesh. Instead, it is represented implicitly as the zero level of a scalar function ϕ. The sign of ϕ distinguishes the ice and air regions, while the interface is recovered from the condition $\phi = 0$.

Building on these advances, the present work extends this idea to 3D multilayer ice accretion simulations, where geometry updates become more challenging due to spanwise variations and increasingly complex ice features. By representing the iced surface implicitly, the geometry can be displaced and reconstructed in a more systematic manner, while preserving the clean portions of the geometry that are not modified by ice accretion. This reconstruction strategy reduces the sensitivity of the workflow to local geometric defects and provides greater control over the quality of the reconstructed surface mesh.

Since geometry evolution is only one component of the icing simulation framework, the preceding physical models must also provide reliable inputs to the accretion process. In particular, droplet impingement predictions determine the local water mass flux supplied to the thermodynamic model and directly influence the ice-growth rate. For this reason,

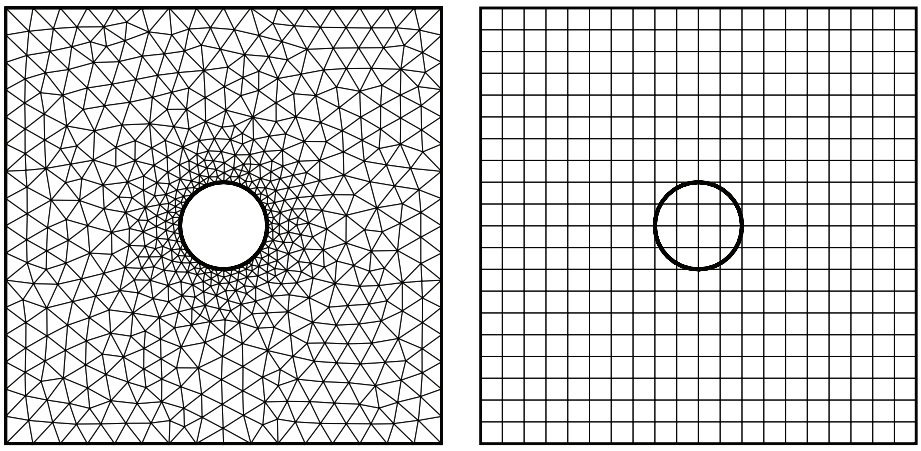
[3]The Ice Prediction Workshops (IPW) [26,27] are an AIAA international benchmark initiative for assessing state-of-the-art in numerical aircraft icing tools against experimental data.

Eulerian and Lagrangian droplet formulations, including their extension to Supercooled Large Droplet conditions, are also examined in this thesis.

Finally, to reduce the mesh-generation cost associated with successive geometry updates in an icing framework, growing interest has been directed toward immersed-boundary methods (IBMs) [29–31].

In a body-fitted approach, the computational mesh conforms to the solid surface, as shown in Figure 1.5a. As the ice geometry evolves, maintaining this conformity generally requires deformation or regeneration of the surrounding volume mesh. In contrast, an immersed-boundary method embeds the geometry within a non-conformal background mesh, as shown in Figure 1.5b. The geometry is therefore represented independently of the mesh, which can simplify mesh generation and geometry handling for complex or evolving iced surfaces.

This decoupling between the geometry and the computational mesh makes immersed-boundary methods particularly attractive for multilayer ice accretion, where the surface evolves repeatedly throughout the simulation. In this context, their integration within a complete icing framework provides a potential means of reducing the meshing constraints associated with conventional body-fitted approaches.



(a) Body-fitted mesh. (b) Immersed-boundary mesh.

Figure 1.5 Mesh representation around a cylinder using (a) a body-fitted mesh and (b) an immersed-boundary mesh.

1.5.2 Objectives of the Present Study

The primary objective of this thesis is to improve the robustness of 3D multilayer ice accretion simulations through level-set-based geometry evolution. The proposed developments aim to

reduce geometric failures during the accretion process, improve the quality of reconstructed iced surfaces, and maintain reliable aerodynamic, droplet-impingement, thermodynamic, and ice-shape predictions throughout the icing simulation framework.

More specifically, this work develops and assesses numerical methods required to support robust multilayer ice accretion simulations, including Lagrangian particle tracking for droplet impingement prediction, Supercooled Large Droplet effects in Eulerian and Lagrangian formulations, level-set-based geometry update and surface reconstruction, and an immersed-boundary extension of the existing icing framework.

To accomplish this objective, the following sub-objectives are defined.

O.1: Develop, Accelerate, and Assess a Lagrangian Droplet Impingement Model and Extend Droplet Impingement Modeling to Supercooled Large Droplet Conditions

The first objective is to develop and evaluate Lagrangian particle-tracking methods for predicting droplet trajectories and collection efficiency.

The method is assessed in terms of impingement limits, collection-efficiency distributions, computational cost, and parallel performance, and is compared with the existing Eulerian droplet model.

This objective also extends both the Eulerian and Lagrangian impingement models to account for Supercooled Large Droplet effects. These developments are used to assess the influence of large-droplet effects on droplet collection and impingement limits.

O.2: Develop and Assess a Robust Level-Set Methodology for 3D Multilayer Ice-Shape Evolution

The second objective is to develop a robust methodology for multilayer ice-shape evolution using a level-set description of the evolving geometry. The method is intended to support geometry displacement, surface reconstruction, and preservation of the clean portions of the geometry, while reducing geometric failures during multilayer simulations.

The methodology is then applied and assessed on representative 2D and 3D icing configurations. These cases are used to demonstrate its applicability to multilayer ice accretion and to examine its behavior for increasingly complex geometries.

O.3: Develop, Integrate, and Assess an Immersed-Boundary Framework for Ice Accretion Simulations

The third objective is to extend an existing immersed-boundary flow solver by integrating Eulerian and Lagrangian droplet impingement, thermodynamic ice accretion, and geometry-evolution capabilities within a unified ice-accretion framework. The objective is to demonstrate the feasibility of performing the complete ice-accretion sequence without repeated body-fitted volume remeshing and to establish a numerical foundation for the further development of immersed-boundary icing methods.

The resulting immersed-boundary framework is then assessed against established body-fitted formulations using selected verification and validation cases. Comparisons of droplet collection efficiencies and final ice shapes are used to evaluate the agreement between the two approaches and identify areas requiring further development.

1.6 Thesis Outline

This thesis is organized in a hybrid format, combining conventional chapters with one article-based chapter. Chapter 2 first presents the state of the art in numerical ice accretion modeling. It reviews the main components of icing simulation workflows, including aerodynamics, droplet impingement, SLD effects, thermodynamics, geometry evolution strategies, and immersed-boundary methods.

Chapter 3 focuses on Lagrangian particle tracking for droplet-impingement prediction under Supercooled Large Droplet conditions. Particular attention is given to the treatment of large-droplet effects and the accuracy of the predicted collection-efficiency distributions, as well as the computational efficiency and parallel scalability of the method for large-scale simulations.

Chapter 4, presented as the submitted journal article, addresses the problem of robust geometry evolution in 3D multilayer ice accretion simulations. A level-set-based approach is developed and assessed as a means of updating complex ice shapes while reducing the risk of mesh degradation and invalid geometries during the accretion process.

Chapter 5 extends the developments introduced in the preceding chapters to an immersed-boundary framework for ice-accretion simulations. This chapter investigates the use of immersed-boundary methods as an alternative to body-fitted approaches for aerodynamic and droplet-impingement calculations.

Chapter 6 provides a general discussion comparing the main numerical methodologies investigated throughout the thesis, including Eulerian and Lagrangian approaches for droplet

impingement, algebraic and level-set methods for geometry evolution, and body-fitted and immersed-boundary approaches for ice-accretion simulations. Finally, Chapter 7 presents a synthesis of the main contributions, followed by the limitations of the proposed methods and recommendations for future work.

CHAPTER 2 LITERATURE REVIEW

This chapter reviews the principal numerical ingredients involved in aircraft ice-accretion simulations. It first presents the conventional quasi-steady framework adopted by most icing solvers before reviewing the aerodynamic, dispersed-phase, thermodynamic, geometry-evolution, and geometry-representation models and techniques reported in the literature. The chapter concludes with a synthesis of the current state of the art and identifies the remaining challenges that motivate the developments presented in the following chapters.

2.1 Overview of Numerical Ice Accretion Modeling

Aircraft icing is the accumulation of ice on aircraft surfaces caused by the impingement and freezing of supercooled water droplets. From a numerical point of view, the problem involves the coupling of an external aerodynamic flow, a dispersed liquid phase, a surface thermodynamic balance, and an evolving fluid-solid interface. These different components are characterized by distinct physical time scales, which motivates the modular formulation used in most ice-accretion solvers [24]. Certification icing conditions generally involve exposure over several minutes, resulting in an ice-accretion time scale that is several orders of magnitude longer than the characteristic flow and droplet time scales [24]. Owing to this disparity, the transient ice-accretion process may be approximated as a succession of quasi-steady states.

In addition to this separation of time scales, the low concentration of liquid water in icing clouds permits a further simplification in the treatment of the dispersed phase. Certification icing envelopes involve standard supercooled droplets (SSD) with characteristic diameters ranging from approximately 10 to 50 μm, while SLD conditions can extend to diameters approaching 1000 μm. For a typical liquid water content (LWC) of 1 g/m^3, the corresponding liquid volume fraction is of the order of 10^{-6} [32,33]. This small volume fraction justifies the common assumption of one-way coupling between the air and droplet phases.

These assumptions provide the foundation for the conventional quasi-steady ice-accretion framework (Fig. 2.1). In its simplest form, the aerodynamic flow is first computed around the clean geometry. The resulting velocity field is then used to determine the transport of the dispersed phase and the local droplet impingement rate on the exposed surface. The latter is then provided to a surface thermodynamic model, which evaluates the local mass and energy balances and returns the ice-growth rate. Finally, the geometry is updated

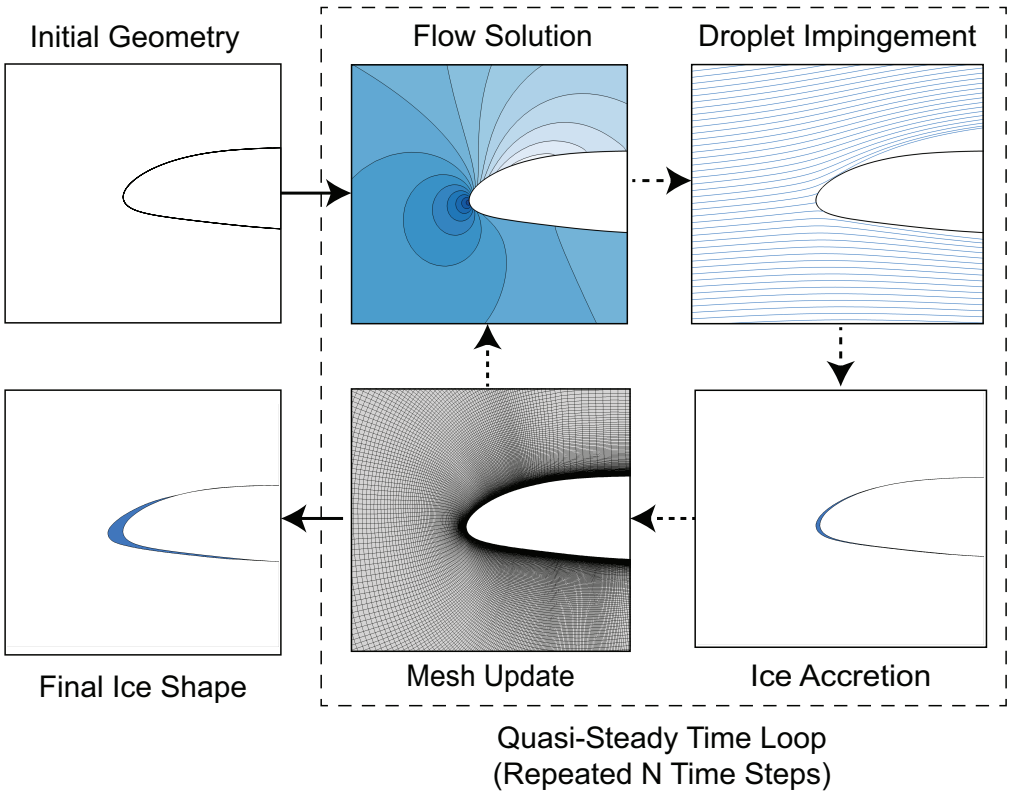


Figure 2.1 Schematic of the multilayer ice accretion framework with representative solutions at each stage.

according to the predicted ice thickness. Representative aerodynamic, droplet, geometry-evolution, and mesh-update solutions associated with this sequence are shown in Fig. 2.1. When this sequence is performed only once over the full exposure time, the approach is often referred to as a single-step or single-shot accretion simulation. When the total icing time is divided into several shorter intervals, the same sequence is repeated on the evolving geometry, leading to a multilayer formulation. Intermediate strategies, such as predictor–corrector formulations [34, 35], have also been proposed to partially account for the retroaction of the growing ice shape on the flow and droplet fields. This limits the additional computational cost to a small number of surface-solution correction cycles.

Although the quasi-steady multilayer approach remains the standard strategy in engineering ice-accretion simulations, the physical process is inherently unsteady. Fully time-accurate formulations have been reported in the literature [36]. More recently, unsteady couplings involving scale-resolving flow simulations, such as LES, have also been explored [37]. These approaches provide a more direct representation of the continuous interaction between the flow and the evolving ice surface, but their computational cost remains prohibitive for routine

industrial icing applications, especially in 3D configurations.

To provide a broader perspective on existing numerical approaches, Table 2.1 summarizes several state-of-the-art ice-accretion solvers reported in the literature [26, 27]. The table highlights the main numerical ingredients used in each framework. The remainder of this chapter then examines these components in more detail.

Table 2.1 Comparison of representative numerical aircraft icing solvers.

Org.	Solver*	Flow	Droplet	Thermo.	Geo.	IBM	Multishot	Ref.
ANSYS Canada	FENSAP-ICE	RANS	Eul.	Messinger	Algebraic + Wrapping	✗	✓	[38]
CIRA	SIMBA	RANS	Eul.	Messinger	Algebraic	✓	✓	[30]
NASA	GlennICE	RANS/Euler	Lag.	Messinger	Algebraic	✗	✓	[39]
ONERA	IGLOO3D	RANS/Euler	Eul.	Messinger	Algebraic/LS	✓	✓	[29, 31, 34]
Politecnico di Milano	PoliMIce	RANS	Lag.	Mod. Myers	Algebraic + LS	✗	✓	[40]
Polytechnique Montréal**	CHAMPS	RANS/Euler/Potential	Eul.	Messinger/SWIM	Algebraic + Hyp.	✗	✓	[2, 5]
Uni. of Strasbourg	NSMB	RANS	Eul.	SWIM	LS	✓	✓	[41]

Abbreviations: Eul.: Eulerian; Lag.: Lagrangian; LS: Level Set; Hyp.: Hyperbolic; Mod.: Modified; IBM: Immersed-Boundary Method.
Notes:
* The solvers listed here refer to the icing frameworks. The aerodynamic flow solution may be provided by separate CFD solvers; therefore, the Flow column indicates the types of flow formulations compatible with each icing framework rather than a specific flow solver.
** Prior to the developments presented in this thesis, multilayer ice-accretion simulations in CHAMPS were limited to 2D/2.5D configurations and exhibited robustness limitations, while 3D applications were restricted to single-step accretion.

2.2 Aerodynamics

Early theoretical developments in aircraft icing were established before the availability of modern computational resources. Foundational contributions by Langmuir and Blodgett, Hardy, and Messinger provided the theoritical basis for ice accretion modeling [42]. However, these early models were restricted to simple configurations, such as cylinders and spheres, or relied on empirical corrections when analytical solutions were not available. The extension of icing analysis to more representative aerodynamic geometries became possible only with the development of computational methods.

The first generation of numerical icing tools emerged from research efforts led mainly by NASA Lewis Research Center in the United States, Polytechnique Montréal in Canada, and ONERA in France [42]. Early codes such as LEWICE [43] and CANICE [44] relied primarily on inviscid aerodynamic descriptions, typically based on potential-flow or panel methods, coupled with boundary-layer corrections to estimate the near-wall quantities required by the icing model. Similarly, ONERA developed icing tools based on Euler flow solutions combined with boundary-layer calculations [45]. These approaches provided an efficient framework for ice-accretion prediction and formed the basis of several widely used engineering tools. The capabilities of these early numerical icing tools were subsequently assessed through international benchmarking efforts, notably the NATO/RTO exercise on “Assessment of Icing Code Prediction Capabilities,” which culminated in a workshop held at CIRA in Capua, Italy, in 2000. By comparing blind numerical predictions against common experimental icing

cases, this initiative provided a coordinated international assessment of existing icing codes and helped identify limitations and directions for their further development [46].

With the maturation of computational fluid dynamics (CFD), especially from the 1970s onward, Reynolds-averaged Navier–Stokes (RANS) solvers progressively became the dominant aerodynamic model for practical icing simulations [47]. RANS solvers provide a more complete description of viscous effects and near-wall flow features, which are important for the ice shape prediction. This transition is reflected in recent IPW, where the majority of reported simulations rely on RANS-based aerodynamic solutions, most often using turbulence closures such as the Spalart–Allmaras or k–ω models [48, 49]. Although RANS calculations are more expensive than inviscid approaches, they remain affordable for industrial applications and provide a practical compromise between physical fidelity and computational cost.

At the same time, recent work has renewed interest in lower-cost inviscid formulations strongly coupled with boundary-layer models [29, 31, 50]. These approaches aim to recover part of the accuracy of RANS-based icing simulations [51, 52] while retaining the computational efficiency of Euler or potential-flow solvers. Such strategies are particularly attractive for design studies, parametric analyses, and multilayer ice-accretion calculations. The current literature therefore reflects two complementary trends: the continued use of RANS solvers as the standard aerodynamic model for engineering icing simulations, and the renewed interest in revisiting earlier inviscid and low- to medium-fidelity approaches using the knowledge and reference data gained from high-fidelity computational tools.

An important output of the aerodynamic solution for ice-accretion calculations is the convective heat-transfer coefficient (HTC). Its accurate evaluation is particularly important because convection generally represents one of the dominant contributions to the surface energy balance. The convective heat flux may be expressed as

$$\dot{Q}_c = HTC\,(T_s - T_{rec}) \tag{2.1}$$

where T_s is the surface temperature and T_{rec} the recovery temperature. The HTC may be estimated from empirical correlations, such as those based on the Reynolds analogy and skin-friction coefficient [53]. Alternatively, it may be obtained directly from the aerodynamic solution by imposing an isothermal wall condition and evaluating the resulting wall heat flux. In this case, the recovery temperature can be approximated as

$$T_{rec} = T_e \left(1 + r\frac{\gamma - 1}{2} M_e^2\right) \tag{2.2}$$

where T_e and M_e are the temperature and Mach number at the edge of the boundary layer, respectively, with M_e obtained from the local flow conditions using isentropic relations [34], and r is the recovery factor. Montreuil et al. [54] use $r = Pr^{1/3} \approx 0.9$ for turbulent flows and $r = 0.85$ for laminar flows.

To avoid relying on an approximate recovery factor, Montreuil et al. [54] also proposed evaluating the HTC from two RANS solutions performed with different thermal wall conditions. For two prescribed wall temperatures $T_{w,0}$ and $T_{w,1}$, the corresponding wall heat fluxes satisfy

$$\dot{Q}_{w,0} = HTC\,(T_{w,0} - T_{rec}) \qquad \dot{Q}_{w,1} = HTC\,(T_{w,1} - T_{rec}) \tag{2.3}$$

Assuming that the HTC and recovery temperature remain unchanged between the two solutions, subtraction eliminates T_{rec}, giving

$$HTC = \frac{\dot{Q}_{w,1} - \dot{Q}_{w,0}}{T_{w,1} - T_{w,0}} \tag{2.4}$$

One of the solutions may alternatively be performed with an adiabatic wall, for which $\dot{Q}_{w,0} = 0$ and the resulting wall temperature directly provides T_{rec}. This two-solution procedure therefore allows both the HTC and recovery temperature to be obtained directly from the CFD solution without relying on empirical recovery-temperature correlations. However, it approximately doubles the aerodynamic computational cost, which limits its practicality in repeated ice-accretion calculations. Consequently, most icing solvers rely on a modeled recovery temperature together with a single thermal flow solution.

2.3 Dispersed Phase Modeling

Droplet-impingement prediction provides the local impingement rate required by the thermodynamic stage. Before the local ice-growth rate can be evaluated, the amount of supercooled water reaching the surface must first be determined. This quantity is commonly expressed through the collection efficiency, denoted by β, a dimensionless variable that represents the ratio of the local impinging water mass flux to the freestream water mass flux. The dimensional impinging water mass flux can therefore be written as

$$\dot{m}_{\text{imp}} = \beta\,\text{LWC}\,U_\infty \tag{2.5}$$

where U_∞ is the freestream velocity and LWC is the Liquid Water Content. The collection

efficiency directly determines the water mass available for freezing or runback and therefore strongly influences the predicted ice shape.

The definition of β depends on the droplet formulation. In an Eulerian framework, the droplet phase is treated as a continuous medium. The local collection efficiency is then obtained from the normal component of the droplet mass flux at the wall [55]. In nondimensional form, it can be expressed as

$$\beta = -\alpha \, \boldsymbol{u}_d \cdot \boldsymbol{n} \tag{2.6}$$

where α is the nondimensional water concentration, $\boldsymbol{u}_d$ is the droplet velocity, and $\boldsymbol{n}$ is the outward unit normal to the surface. In a Lagrangian framework, the collection efficiency is instead determined by tracking the deformation of a droplet streamtube initialized in the freestream and tracked until it intersects the surface. As illustrated in Fig. 2.2, β may be interpreted, for a given surface element, as the ratio between the upstream seeding area associated with a set of neighboring droplet trajectories and the corresponding impingement area on the wall [56]:

$$\beta = \frac{A_{\text{upstream}}}{A_{\text{impingement}}} \tag{2.7}$$

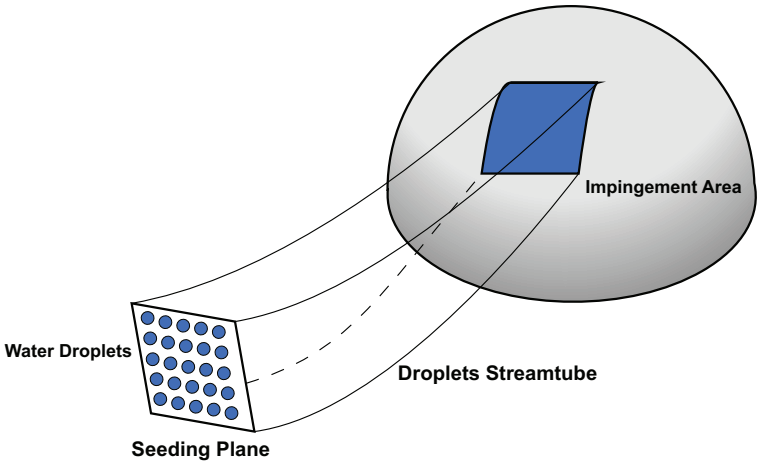


Figure 2.2 Schematic representation of the Lagrangian droplet-seeding procedure. Droplets are released from an upstream plane and tracked until they impinge on the surface.

Historically, Lagrangian droplet tracking was the only approach in early icing codes [43, 44]. In solvers based on potential-flow or panel methods, the airflow field was inexpensive to evaluate and droplet trajectories could be integrated directly through the computed velocity

field. This made the Lagrangian formulation attractive because it provided a clear physical description of droplet motion. However, as ice-accretion solvers evolved toward Euler and RANS aerodynamic models, the computational cost of Lagrangian tracking became more restrictive. Accurate impingement maps require a large number of droplets, particularly in 3D configurations. In addition, each droplet trajectory requires repeated interpolation of the carrier-flow solution, localization of the droplet within the computational mesh, and detection of surface impacts. These operations can become expensive and difficult to parallelize efficiently on complex unstructured grids. These limitations motivated the development of Eulerian droplet formulations [55], in which the dispersed phase is solved as a continuum using partial differential equations (PDE).

2.3.1 Eulerian Droplet Formulation

Eulerian dispersed-phase models describe the droplet field through continuous variables defined over the computational domain [55]. The main unknowns are the nondimensional water concentration α and the droplet velocity $\boldsymbol{u}_d$. A common formulation consists of a conservation equation for the droplet concentration

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \boldsymbol{u}_d) = 0 \tag{2.8}$$

combined with a momentum equation for the droplet phase

$$\frac{\partial (\alpha \boldsymbol{u}_d)}{\partial t} + \nabla \cdot (\alpha \boldsymbol{u}_d \otimes \boldsymbol{u}_d) = \frac{C_D Re_D}{24 S_t} \alpha (\boldsymbol{u}_a - \boldsymbol{u}_d) + \alpha \left(1 - \frac{\rho_a}{\rho_d}\right) \frac{\boldsymbol{g}}{Fr^2} \tag{2.9}$$

where $\boldsymbol{u}_a$ is the airflow velocity, C_D is the droplet drag coefficient, Re_D is the droplet Reynolds number, S_t is the droplet Stokes number, ρ_a and ρ_d are the air and droplet densities, respectively, $\boldsymbol{g}$ is the gravitational acceleration vector, and Fr is the Froude number. The nondimensional parameters are defined as

$$\mathrm{Re}_D = \frac{\rho_a |\boldsymbol{u}_a - \boldsymbol{u}_d| D}{\mu_a}, \tag{2.10}$$

$$S_t = \frac{\tau_d}{\tau_f} = \frac{\rho_d D^2 U_\infty}{18 \mu_a c_{ref}}, \tag{2.11}$$

and

$$\mathrm{Fr} = \frac{U_\infty}{\sqrt{Lg}}, \tag{2.12}$$

where D is the droplet diameter, μ_a is the dynamic viscosity of air, U_∞ is the freestream velocity, c_{ref} is the reference length, $\tau_d = \rho_d D^2/(18\mu_a)$ is the characteristic droplet response time, $\tau_f = c_{ref}/U_\infty$ is the characteristic flow timescale, and $g = |\boldsymbol{g}|$ is the magnitude of the gravitational acceleration.

The Eulerian approach is well suited to modern finite-volume CFD solvers. Since the droplet equations are written in conservative form, they can be discretized using numerical techniques similar to those used for the airflow equations. This makes the approach relatively efficient, naturally compatible with parallel computing, and convenient for 3D simulations on complex grids.

Nevertheless, the Eulerian formulation has limitations. Since the droplet phase is averaged within each computational cell, individual droplet trajectories, impact velocities, and impact angles are not directly available. Moreover, numerical diffusion may smear the impingement limits, especially near shadow zones. Numerical oscillations may also appear in low-concentration regions and can lead to nonphysical values of α. Additional numerical treatments are therefore often required to preserve positivity, reduce artificial diffusion, and improve conservation properties [57, 58].

2.3.2 Lagrangian Droplet Formulation

In a Lagrangian formulation, droplets are tracked individually by integrating their equations of motion through the carrier-flow field. The droplet velocity $\boldsymbol{u}_d$ and position $\boldsymbol{x}_d$ satisfy

$$\frac{d\boldsymbol{u}_d}{dt} = \frac{\boldsymbol{f}}{m} + \boldsymbol{g} = F\left(\boldsymbol{u}_a - \boldsymbol{u}_d\right) + \left(1 - \frac{\rho_a}{\rho_d}\right)\boldsymbol{g} \tag{2.13}$$

and

$$\frac{d\boldsymbol{x}_d}{dt} = \boldsymbol{u}_d \tag{2.14}$$

where $\boldsymbol{f}$ is the aerodynamic force acting on the droplet, and m is the droplet mass. The scalar drag coefficient F is given by

$$F = \frac{3C_D\rho_a\left|\boldsymbol{u}_a - \boldsymbol{u}_d\right|}{4\rho_d D} \tag{2.15}$$

The airflow velocity $\boldsymbol{u}_a$ is evaluated at the instantaneous droplet position.

Although Lagrangian methods are considered expensive for high-fidelity icing simulations, they have recently gained renewed interest [59–63]. Modern high-performance computing resources, improved particle-localization algorithms, and the need for detailed impact information have made Lagrangian approaches increasingly attractive. This is particularly true for SLD conditions, where splashing, bouncing, and the motion of ejected droplets resulting from splashing are more naturally described within the Lagrangian framework.

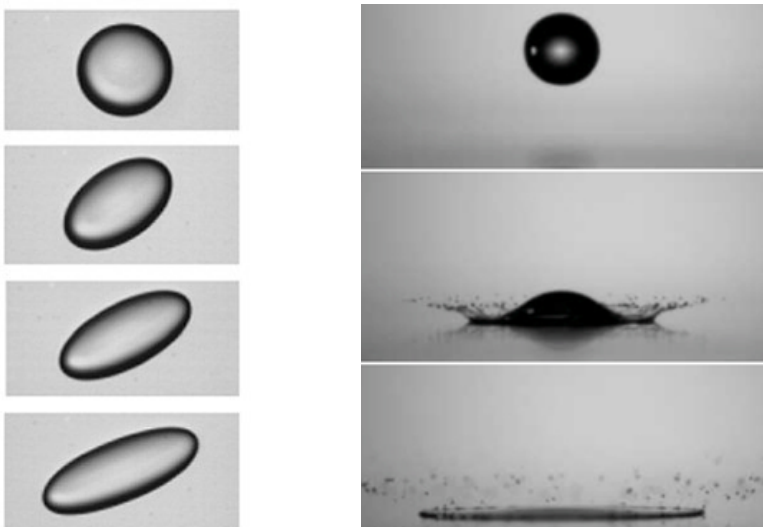
2.3.3 Supercooled Large Droplet Modeling

Supercooled Large Droplet (SLD) conditions are characterized by droplet sizes substantially larger than those typically encountered under Standard Supercooled Droplet (SSD) conditions. As indicated by the Stokes number defined in Eq. (2.11), the response of a droplet to changes in the surrounding airflow depends strongly on its diameter. Larger droplets have higher Stokes numbers and therefore respond more slowly to changes in the airflow, exhibiting a greater tendency to maintain their trajectories [24]. SLDs may consequently impinge farther downstream and potentially outside regions protected by conventional Ice Protection Systems (IPS), particularly when splashing occurs and ejected secondary droplets re-enter the airflow and re-impinge farther downstream.

The additional phenomena associated with SLDs may be separated into effects occurring before and during impact. In-flight deformation and breakup (Fig. 2.3a) can be regarded as second-order effects because they indirectly influence impingement by modifying the droplet shape, aerodynamic drag, and trajectory before contact with the surface. By contrast, splashing, bouncing, and secondary-droplet ejection (Fig. 2.3b) constitute first-order wall-interaction effects because they directly determine the fraction of the impinging water mass that remains on the surface [64].

As an SLD travels through a nonuniform flow field, variations in the aerodynamic pressure and shear stresses acting over its surface may deform it from its initially spherical shape. The resistance of the droplet to this deformation is governed primarily by surface-tension and viscous forces. When the aerodynamic forces become sufficiently large relative to these restoring forces, the spherical-droplet assumption is no longer valid, and the droplet may adopt an oblate or more complex nonspherical shape. Under more severe conditions, continued deformation may ultimately lead to breakup into smaller droplets [67].

Droplet deformation can affect impingement by modifying the aerodynamic drag and, consequently, the droplet response time and trajectory. Breakup may additionally alter the



(a) Droplet deformation [65]. (b) Droplet splashing [66].

Figure 2.3 Examples of droplet deformation and splashing phenomena relevant to SLD conditions.

droplet-size distribution and produce smaller fragments that interact more strongly with the airflow. Nevertheless, the influence of deformation and breakup on the resulting impingement distribution has generally been found to remain limited for many icing conditions, with potentially greater importance under freezing-drizzle conditions [64, 68]. For this reason, these mechanisms are often neglected or represented through simplified corrections rather than modeled explicitly.

A commonly used approximation accounts for deformation by modifying the drag coefficient without explicitly resolving the evolving droplet shape. The effective drag coefficient may be obtained by blending correlations for spherical droplets and oblate disks as a function of the droplet Reynolds and Weber numbers [69]:

$$C_D = (1 - f)\, C_{D,\text{sphere}} + f C_{D,\text{disk}} \tag{2.16}$$

with

$$f = 1 - \left(1 + 0.07\sqrt{\text{We}}\right)^{-6} \tag{2.17}$$

where the Weber number is defined as

$$\mathrm{We} = \frac{\rho_a \, D \, |u_a - u_d|^2}{\sigma} \tag{2.18}$$

where σ is the surface tension of water. $C_{D,\mathrm{sphere}}$ and $C_{D,\mathrm{disk}}$ depend on the droplet Reynolds number.

The most consequential SLD effects generally arise during droplet-wall interaction. At relatively low impact energy, an impinging droplet may stick to or spread over the surface and contribute directly to the surface water film. Under other conditions, it may bounce, splash, or break into secondary droplets [70, 71]. These secondary droplets can be ejected from the surface and may subsequently reimpinge farther downstream, thereby extending the effective impingement region. The amount of water deposited on the surface is therefore determined not only by the collection efficiency, but also by the outcome of the impact process.

Because resolving individual droplet impacts with interface-capturing methods would be prohibitively expensive at aircraft scale [72], these first-order SLD wall-interaction effects are generally represented through semi-empirical models. Such models are commonly formulated in terms of nondimensional impact parameters, including the Reynolds, Weber, and Ohnesorge numbers, and may also account for the impact angle and impact velocity. A common approach consists in correcting the collection efficiency using a sticking rate ϵ, or equivalently a mass-loss coefficient f_m:

$$\beta_{\mathrm{final}} = \epsilon \beta_0 = (1 - f_m) \, \beta_0 \tag{2.19}$$

where β_0 is the collection efficiency obtained before accounting for splashing or bouncing. This correction represents the fraction of the impinging water mass that remains on the surface after impact.

More advanced SLD models also account for the generation and reimpingement of secondary droplets. In a Lagrangian framework, this can be handled by reinjecting secondary droplets with prescribed diameters, velocities, and ejection directions obtained from semi-empirical correlations [62]. In an Eulerian framework, secondary reimpingement is more difficult to represent directly because the dispersed phase is described as a continuous field. One strategy consists in modifying the boundary conditions on splashing surfaces so that the secondary droplets are reemitted into the computational domain and transported by the Eulerian equations [73, 74]. The resulting collection efficiency may then include both the retained contribution and the mass associated with secondary reimpingement:

$$\beta_{\text{final}} = \epsilon\beta_0 + \sum \beta_{\text{reimp}} \tag{2.20}$$

where β_{reimp} denotes the collection-efficiency contribution associated with reimpinging secondary droplets. The modeling of SLD effects therefore remains closely linked to the selected droplet formulation. Eulerian methods provide an efficient and robust framework for impingement calculations, but require additional boundary-condition treatments to represent secondary droplets. Lagrangian methods are generally more expensive, but naturally provide the impact quantities required by splashing and bouncing correlations and allow secondary droplets to be tracked explicitly.

Several semi-empirical models have been proposed to describe droplet impact and splashing under SLD conditions [75–77]. Among these, the formulations of Wright et al. [64] and Trontin et al. [68] have been widely adopted in aircraft-icing applications due to their ability to provide reliable predictions against experimental data. Wright et al. [64] incorporated the impact angle into the splashing criterion and calibrated the model using experimental water-collection data. Trontin et al. [68] proposed an alternative formulation in which the mass loss is described through separate dependencies on the impact angle and kinetic energy. The principal characteristics of these two formulations are summarized in Table 2.2, where the splashing criterion defines the condition under which splashing is taken into account.

Table 2.2 SLD splashing models.

Source	**Mass Loss Coefficient**	**Splashing Criterion**
Wright et al. [64]	$0.7(1-\sin\theta_0)\left[1-\exp\left(-0.0092(K_M-200)\right)\right]$	$\dfrac{0.859\,K_M^{1/2}\left(\dfrac{\rho_w}{LWC}\right)^{1/8}}{\sin^{5/4}\theta_0} > 200, \qquad \theta < \dfrac{\pi}{2}$
Trontin et al. [68]	$1-g\left(\dfrac{\theta}{\theta_c}\right) f\left(\dfrac{K_c-K_0}{K_0}\right)$	$K_c = We_n\,Oh^{-2/5} > 657$

2.4 Thermodynamic Modeling

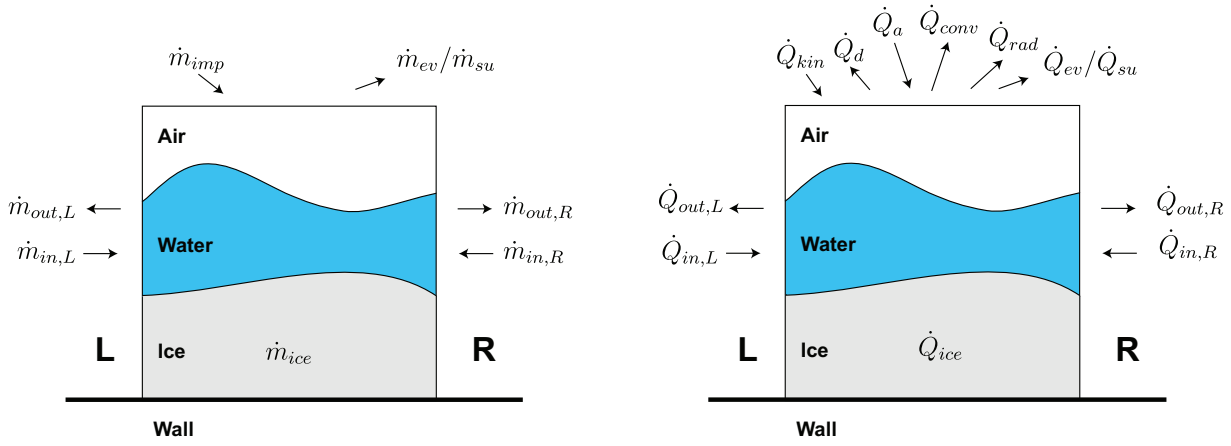
Once the airflow and droplet-impingement solutions have been obtained, a thermodynamic analysis is performed over the exposed surface to determine the local ice-accretion rate. The aerodynamic solution provides the convective heat-transfer coefficient, wall shear stress, and surface-pressure distribution, whereas the dispersed-phase solution provides the local impinging-water mass flux and droplet kinetic energy at impact. These quantities are introduced into surface mass- and energy-balance equations describing the freezing, evaporation, and transport of water over the surface (Fig. 2.4). These balances are written as

$$\dot{m}_{imp} + \sum \dot{m}_{in} = \sum \dot{m}_{out} + \dot{m}_{ev/su} + \dot{m}_{ice} \tag{2.21}$$

and

$$\dot{Q}_{kin} + \dot{Q}_d + \dot{Q}_a + \sum \dot{Q}_{in} = \dot{Q}_{conv} + \dot{Q}_{rad} + \dot{Q}_{ev/su} + \sum \dot{Q}_{out} + \dot{Q}_{ice} \tag{2.22}$$

A detailed description of the individual mass- and energy-balance terms and their corresponding expressions is provided in Appendix A.



(a) Mass balance. (b) Energy balance.

Figure 2.4 Control volume diagrams for mass and energy balance equations. Adapted from [2].

The thermodynamic problem involves three unknowns: the local ice-accretion rate, $\dot{m}_{\text{ice}}$; the outgoing runback-water flow rate, $\dot{m}_{\text{out}}$; and the surface temperature, T_s. The mass and energy conservation equations alone are insufficient to determine these quantities simultaneously. The system is therefore closed by introducing phase-state conditions derived from the classical Messinger model [78]:

- **Dry regime:** a dry element containing no liquid water;
- **Rime or fully freezing regime:** an element for which all incoming water freezes and $\dot{m}_{\text{out}} = 0$;
- **Glaze or partially freezing regime:** an element covered by a liquid-water film, for which only part of the incoming water freezes and $T_s = T_{\text{freeze}}$;
- **Fully liquid regime:** an element for which no ice is formed and $\dot{m}_{\text{ice}} = 0$.

Although the model remains the basis of many contemporary ice-accretion solvers, its original implementation relied on a surface-marching procedure initiated near the stagnation point. This dependence on a clearly identified stagnation location complicates its application to configurations involving multiple stagnation regions, separated flows, or general 3D geometries.

Several extensions have consequently been proposed to improve either the physical description or the numerical robustness of the thermodynamic calculation. Myers et al. [79], for example, extended the classical formulation by accounting for heat conduction through the growing ice layer. Although this modification provides a more detailed representation of the temperature distribution within the ice, it does not remove the dependence of the solution on the treatment of the stagnation point.

A different strategy was introduced by Zhu et al. [80] through an iterative implementation of the Messinger model. Rather than relying exclusively on a predefined marching direction originating from a stagnation point, the surface-water distribution and thermodynamic state are updated iteratively. The transport direction of the runback water is determined from the local wall-shear-stress direction, with the surface-pressure gradient providing an additional contribution. This formulation is better suited to complex and 3D geometries and has consequently been adopted by several modern ice-prediction tools, as reflected in the methods reported during recent ice-prediction workshops [26, 27].

Bourgault et al. [81] proposed an alternative formulation known as the Shallow-Water Icing Model (SWIM). In this approach, the liquid-water film is represented through surface PDEs describing its transport and evolution. In contrast to the iterative Messinger formulation, SWIM retains an explicitly unsteady description of the water film. It can therefore represent transient film-transport phenomena more naturally, but generally requires a more involved numerical treatment and introduces an additional computational cost. For most ice-accretion conditions, the resulting ice shapes have nevertheless been found to remain broadly comparable to those obtained using iterative Messinger formulations [24]. A more comprehensive assessment and comparison of the classical Messinger model, its extensions, and SWIM formulations is provided by Lavoie et al. [82]. In that work, it was established that, for time scales beyond the transient film response, SWIM recovers the Messinger solution by construction [82].

2.5 Geometry Evolution

Once the local ice-accretion rate has been determined by the thermodynamic solver, the solid surface must be updated to represent the newly accumulated ice. The updated surface then becomes the aerodynamic geometry for the subsequent accretion layer. Geometry-evolution techniques can generally be divided into two broad categories: interface-tracking methods, in which the ice–air interface is represented explicitly, and interface-capturing methods, in which the interface is embedded implicitly within a field defined over a computational domain.

2.5.1 Interface Tracking

Interface-tracking methods represent the ice–air interface using marker points located directly on the surface or within the surrounding control volumes. In ice-accretion simulations, the markers generally correspond to the nodes of the surface mesh. The position of the interface is therefore known explicitly throughout the geometry-update procedure. The most commonly used approach is the algebraic geometry-evolution method, also referred to as Lagrangian node displacement [83]. In its simplest form, each surface node is moved along the local normal by the predicted ice thickness (Fig. 2.5):

$$\boldsymbol{x}_{\text{new}} = \boldsymbol{x}_{\text{old}} + h_{\text{ice}}\boldsymbol{n} \tag{2.23}$$

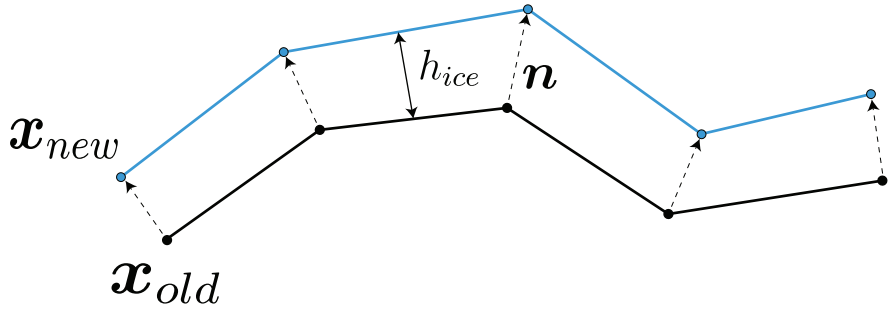


Figure 2.5 Schematic representation of surface-node displacement along the local normal according to the predicted ice thickness.

where $\boldsymbol{x}_{\text{old}}$ and $\boldsymbol{x}_{\text{new}}$ denote the initial and updated node positions, respectively, h_{ice} is the local accumulated ice thickness, and $\boldsymbol{n}$ is the outward surface-normal vector. Several extensions have been introduced to improve the robustness and accuracy of this displacement procedure. One approach consists of dividing the prescribed ice thickness into a sequence of smaller increments and recomputing the surface normals after each intermediate displacement [84].

This results in smoother surface normals and decreases the likelihood of surface-mesh entanglement during geometry evolution. Another formulation that has been proposed consists of deriving the displacement direction from a normal-vector field rather than directly from the discrete surface normals [31, 34]. Such fields are typically obtained from the gradient of a wall-distance field [85]. Since the distance field is computed over the surrounding domain, neighboring displacement directions become naturally aligned and vary smoothly across the surface.

Bourgault-Côté et al. [2] proposed a different interface-tracking approach based on constrained hyperbolic mesh generation. In this method, successive mesh layers are generated outward from the initial surface until the prescribed ice thickness is reached. An advantage of this formulation is that the newly accreted region is directly represented by a volume mesh, allowing the added ice volume to be evaluated without requiring a separate reconstruction procedure. Its application is nevertheless restricted to structured surface meshes, which limits its generality for complex 3D configurations.

The principal advantage of interface-tracking methods is that the interface position is explicitly available, allowing fine geometric structures to be represented accurately when the surface discretization is sufficiently resolved. However, large node displacements may lead to surface-element distortion, self-intersections, and a loss of mesh quality. Topological changes, such as the merging of neighboring ice structures or the formation of highly concave regions, are also difficult to handle robustly using an explicit surface representation. Although several successful treatments have been proposed in the literature [86], maintaining a valid surface and volume mesh remains a challenging aspect of multilayer simulations involving complex ice shapes often requiring additional corrective procedures, such as mesh smoothing, to maintain acceptable mesh quality [6].

2.5.2 Interface Capturing

Interface-capturing methods represent the interface implicitly through a field variable defined over the surrounding computational domain. Among these methods, the level-set approach embeds an n-dimensional interface within an (n+1)-dimensional space and tracked through the evolution of a scalar function ϕ defined in that higher-dimensional domain [87]. The ice–air interface is then recovered as the zero level set of ϕ:

$$\Gamma = \{\boldsymbol{x} \,|\, \phi(\boldsymbol{x}) = 0\}\,. \tag{2.24}$$

As illustrated in Figure 2.6, the level-set function is commonly defined as positive in the fluid

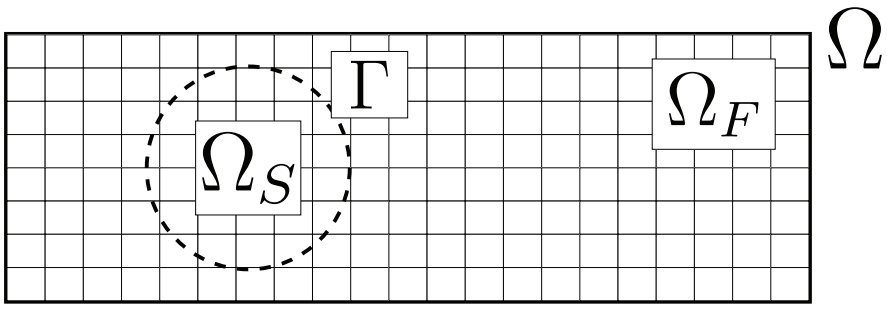


Figure 2.6 Schematic representation of the level-set function. The ice–air interface Γ is implicitly defined by the zero level set ($\phi = 0$), separating the solid domain Ω_S ($\phi < 0$) from the surrounding air domain Ω_F ($\phi > 0$).

region, negative within the solid, and zero at the interface. Its evolution is governed by an advection equation of the form

$$\frac{\partial \phi}{\partial t} + \boldsymbol{V} \cdot \nabla \phi = 0 \tag{2.25}$$

where $\boldsymbol{V}$ is an extension of the interface-growth velocity to the computational region surrounding the interface. When ϕ is maintained as a signed-distance function, the interface-normal vector $\boldsymbol{n}$ and curvature κ can be evaluated as

$$\boldsymbol{n} = \frac{\nabla \phi}{|\nabla \phi|} \tag{2.26}$$

and

$$\kappa = \nabla \cdot \left(\frac{\nabla \phi}{|\nabla \phi|} \right) \tag{2.27}$$

Several attempts have been made to introduce level-set representations into aircraft ice-accretion simulations. Peña et al. [28] presented one of the first applications of a level-set method to the evolution of an ice shape. Their work demonstrated the feasibility of representing ice growth implicitly, but the application was limited to single-step accretion. Beaugendre et al. [88] employed a level-set representation in a different icing context, namely to describe the evolving geometry and trajectories of ice fragments during shedding simulations. In this case, the implicit representation facilitated the treatment of moving solid interfaces and their interaction with the surrounding flow.

Bourgault-Côté et al. [2] and Lavoie et al. [29] subsequently extended the use of level-set methods to 2D multilayer ice-accretion simulations. Their studies considered both body-fitted and immersed-boundary frameworks.

An alternative formulation was proposed by Donizetti et al. [89], who introduced a discrete level-set representation constructed from a Lagrangian description of the surface. In this approach, the interface remains explicitly represented, while the level-set values at surrounding mesh points are recovered geometrically from the distance to the Lagrangian description of the surface. The level-set field is therefore used primarily as an implicit geometric description rather than as the solution of a continuous advection equation.

Level-set methods offer several advantages for ice-accretion simulations. They can accommodate large interface displacements and topological changes without requiring the connectivity of an explicit surface mesh to remain unchanged. These properties make them attractive for the development of robust 3D multilayer geometry-evolution procedures. However, the signed-distance property

$$|\nabla\phi| = 1 \tag{2.28}$$

is generally not preserved as the interface evolves, particularly when the interface undergoes large displacements. A reinitialization or redistancing procedure may therefore be required to recover a suitable distance field [90]. The frequency at which this procedure should be applied depends on the numerical formulation and the evolution of the interface. Furthermore, reinitialization must be performed carefully, since it may displace the interface, alter the enclosed ice volume, or smooth sharp geometric features [41]. The updated surface must then be extracted from the implicit level-set representation before the surrounding volume mesh can be regenerated. Particular care is required during this surface-reconstruction and volume-remeshing process to preserve distinctive features of the original clean geometry, especially in regions unaffected by ice accretion, while accurately representing the newly formed ice structures. In addition, numerical difficulties associated with explicit time integration of the level-set equation have been reported, with stability constraints potentially imposing restrictive time-step requirements during interface evolution [2].

2.6 Geometry Representation

The robustness of aircraft icing simulations is strongly influenced by the geometry-evolution and mesh-update procedures. In conventional body-fitted approaches, the computational mesh must conform to the aircraft surface and must therefore be deformed or regenerated after each accretion layer. As the ice shape develops, the surface may exhibit regions of high curvature, horns, scallops, and other localized irregularities that make it increasingly difficult to preserve high-quality near-wall elements and avoid mesh degeneration or failure.

Immersed-boundary methods provide an alternative framework in which the solid geometry is represented independently of the underlying computational mesh. The origins of the immersed-boundary approach are commonly traced to the pioneering work of Charles Peskin [91], who introduced forcing terms into the Navier–Stokes equations to represent the interaction between blood flow and the moving boundaries of heart valves. Since this early contribution, a wide range of immersed-boundary formulations has been developed.

In an immersed-boundary framework, the physical boundary is generally allowed to intersect the computational cells rather than coincide with mesh faces. The boundary conditions may then be enforced through modifications of the governing equations, the numerical fluxes, or the discrete solution near the interface. Because the mesh does not need to conform explicitly to the surface, these methods can substantially simplify grid generation and geometry updates.

Immersed-boundary methods have gained considerable attention and have been applied to a broad range of problems, including inviscid flows [92, 93], turbulent flows [94, 95], multiphase flows [96, 97], and fluid–structure interaction [98–100]. More recently, they have also been introduced into aircraft ice-accretion simulations [29, 31, 101, 101].

Their use in aircraft icing, however, remains comparatively limited. In particular, immersed-boundary approaches were reported by only one participant in each of the first two Ice Prediction Workshops, indicating that their application to complete ice-accretion frameworks is still at an early stage of development [26, 27]. Additional work is therefore required to assess their accuracy, robustness, and compatibility within a coupled multilayer ice-accretion framework.

Following the classification proposed by Sotiropoulos and Yang [102], immersed-boundary methods may be divided into three broad families: geometrical methods, continuous-forcing methods, and discrete-forcing methods. Geometrical methods modify the computational mesh, control volumes, and numerical integration procedure in the vicinity of the immersed interface. Continuous-forcing methods introduce forcing or source terms into the governing equations to represent the effect of the solid boundary. Discrete-forcing methods instead impose the boundary conditions directly within the discretized equations, typically by reconstructing or modifying the solution near the interface. The following sections summarize the main characteristics of these three approaches.

2.6.1 Geometrical Methods

The principal geometrical approach is the cut-cell method. In this formulation, the solid interface is represented by geometrically modifying the cells of a background mesh that it intersects. The portions of these cells located inside the solid are removed, producing control volumes that conform locally to the immersed boundary while leaving the remainder of the mesh unchanged. One of the earliest cut-cell formulations was introduced by Clarke et al. [3] for Euler-flow simulations around multi-element airfoils. A general illustration of the cut-cell approach is provided in Figure2.7.

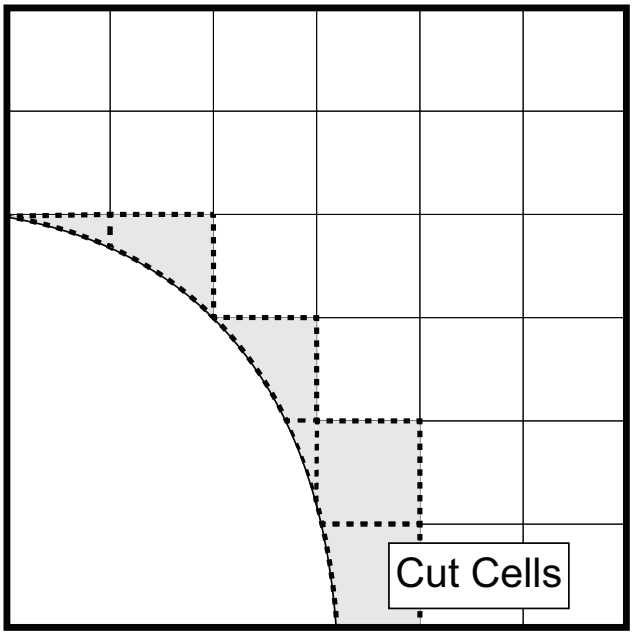


Figure 2.7 Illustration of the boundary cut-cell method [3].

Because the interface is represented through an actual modification of the computational cells, some authors treat cut-cell methods separately and do not classify them strictly as immersed-boundary methods.

Cut-cell methods provide a sharp representation of the interface and allow boundary conditions and numerical fluxes to be imposed directly on the reconstructed faces. They are also naturally suited to conservative finite-volume formulations, although the solver must support general polygonal or polyhedral control volumes.

Their main limitations are the complexity of the geometric preprocessing, particularly in 3D, and the formation of very small cut cells when the interface passes close to a mesh node or face. These cells may impose severe time-step restrictions in explicit schemes and degrade numerical stability. To alleviate these problems, several remedies have been proposed in the

literature, including cell merging [103], cell linking [104], and conservative cell mixing [105].

2.6.2 Continuous-Forcing Methods

Continuous-forcing methods impose the immersed-boundary conditions by introducing a distributed force directly into the continuous governing equations. For example, the incompressible Navier–Stokes equations may be written as

$$\nabla \cdot \boldsymbol{u} = 0 \tag{2.29a}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \boldsymbol{u} + \boldsymbol{f} \tag{2.29b}$$

where $\boldsymbol{u}$ is the fluid velocity, ρ is the density, p is the pressure, ν is the kinematic viscosity, and $\boldsymbol{f}$ is the forcing term used to represent the immersed boundary. The expression of $\boldsymbol{f}$ distinguishes the different continuous-forcing formulations.

The original immersed-boundary method was introduced by Peskin for the simulation of blood flow interacting with elastic heart structures [91]. The fluid equations are solved on an Eulerian background mesh, whereas the immersed boundary is represented by massless Lagrangian markers with coordinates $\boldsymbol{X}(s,t)$, where s denotes the curvilinear coordinate along the immersed boundary and t is time. The force exerted by the boundary on the fluid is distributed to the surrounding Eulerian points through a regularized Dirac function:

$$\boldsymbol{f}(\boldsymbol{x},t) = \int_\Gamma \boldsymbol{F}(s,t)\delta(\boldsymbol{x} - \boldsymbol{X}(s,t))\ \mathrm{d}s \tag{2.30}$$

where $\boldsymbol{F}$ is the force defined on the Lagrangian boundary Γ. Conversely, the velocity of the markers is interpolated from the Eulerian velocity field according to

$$\frac{\partial \boldsymbol{X}(s,t)}{\partial t} = \int_\Omega \boldsymbol{u}(\boldsymbol{x},t)\delta(\boldsymbol{x} - \boldsymbol{X}(s,t))\ \mathrm{d}\boldsymbol{x} \tag{2.31}$$

This two-way transfer allows the fluid to displace the immersed boundary while the boundary exerts a force on the fluid. The regularized Dirac function spreads this interaction over several grid points, resulting in a diffuse rather than sharp representation of the interface. Although this formulation is well suited to deformable interfaces, its extension to rigid bodies is more problematic. As the boundary stiffness is increased to suppress deformation, the constitutive laws used to determine the restoring force become increasingly stiff and may become ill posed

in the rigid-body limit [106].

These limitations motivated the development of penalization methods for rigid boundaries. Initially introduced by Arquis and Caltagirone to model flow through porous media [107], the method was subsequently extended to represent solid obstacles as regions of very low permeability [108]. A mask function (Fig. 2.8) distinguishes the solid and fluid domains:

$$\chi(\boldsymbol{x}, t) = \begin{cases} 1, & \boldsymbol{x} \in \Omega_s \\ 0, & \boldsymbol{x} \in \Omega_f \end{cases} \tag{2.32}$$

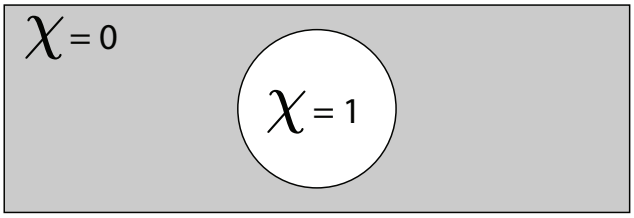


Figure 2.8 Schematic representation of the solid and fluid domains using the mask function χ.

For the incompressible Navier–Stokes equations, penalization is commonly applied to the momentum equation to enforce the no-slip boundary condition. This is achieved by introducing the forcing term

$$\boldsymbol{f} = \frac{\chi}{\eta} (\boldsymbol{u}_s - \boldsymbol{u}) \tag{2.33}$$

where $\boldsymbol{u}_s$ is the prescribed solid velocity and η is a permeability or penalization parameter. In the fluid region, $\chi = 0$ and the original equations are recovered. Within the solid, $\chi = 1$, and a sufficiently small value of η ($\approx 10^{-8}$ [109]) drives the fluid velocity toward the prescribed solid velocity.

Penalization methods require only limited modifications to an existing solver, with the main preprocessing operation consisting of identifying the solid region and constructing the mask function. They also possess a well-established mathematical foundation and can be combined with a wide range of spatial discretizations [110]. However, the boundary is generally represented diffusely, which commonly limits the classical formulation to first-order spatial accuracy near the interface. Higher-order corrections have nevertheless been proposed [111]. In the context of aircraft icing, Lavoie et al. [29] successfully extended the penalization approach to the Eulerian droplet equations, demonstrating its applicability to droplet-impingement

prediction on immersed geometries.

Finally, the penalization parameter must be selected carefully: Decreasing η improves the enforcement of the boundary condition but increases the stiffness of the governing equations and may therefore require careful numerical treatment. Although the example presented above involves a single penalization term, the enforcement of other boundary conditions may require several distinct penalization parameters. Their simultaneous selection can introduce additional case-dependent tuning and may require manual adjustment [112].

2.6.3 Discrete-Forcing Methods

Discrete-forcing methods impose the immersed-boundary conditions directly within the discretized governing equations. Their development is commonly traced to the work of Mohd-Yusof [113], who modified the discrete momentum equations near the immersed interface to enforce a prescribed velocity. This idea was subsequently formalized and extended by Fadlun et al. [114]. For illustration, the incompressible momentum equation may be written as

$$\frac{\partial \boldsymbol{u}}{\partial t} = \boldsymbol{R} + \boldsymbol{f} \tag{2.34}$$

where $\boldsymbol{R}$ contains the convective, pressure, and viscous contributions

$$\boldsymbol{R} = -\boldsymbol{u} \cdot \nabla \boldsymbol{u} - \nabla p + \frac{1}{Re} \nabla^2 \boldsymbol{u} \tag{2.35}$$

and $\boldsymbol{f}$ denotes the discrete forcing. Using a forward-Euler time discretization gives

$$\frac{\boldsymbol{u}^{n+1} - \boldsymbol{u}^n}{\Delta t} = \boldsymbol{R}^n + \boldsymbol{f}^n \tag{2.36}$$

Imposing a target velocity $\boldsymbol{u}_{\text{target}}$ at the forcing point leads to

$$\boldsymbol{f}^n = -\boldsymbol{R}^n + \frac{\boldsymbol{u}_{\text{target}} - \boldsymbol{u}^n}{\Delta t} \tag{2.37}$$

Although this formulation is expressed through a forcing term, it does not generally need to be evaluated as an independent physical quantity. Instead, the velocity update is directly constructed so that the desired target value is obtained.

The accuracy of the method depends primarily on the definition of $\boldsymbol{u}_{\text{target}}$. The boundary value may be imposed directly from the nearest computational point, resulting in a staircase representation of the interface. Alternatively, the target value may be weighted using the local

solid volume fraction to obtain a smoother transition. More accurate approaches reconstruct the target value from neighboring fluid data through interpolation [115]. Unlike continuous-forcing methods, discrete forcing does not introduce a large source term into the continuous equations and therefore avoids the associated stiffness. However, its implementation depends on the spatial and temporal discretizations and generally requires geometrical preprocessing near the interface.

A widely used form of discrete forcing is the ghost-cell immersed-boundary method, developed by Tseng and Ferziger [4]. It generalizes the ghost-cell treatment of body-fitted boundaries to interfaces that do not coincide with the computational mesh. Cells located inside the solid but required to complete the stencil of neighboring fluid cells are identified as ghost cells. For each ghost cell, a wall point is determined on the immersed interface and an image point is constructed in the fluid, commonly along the wall-normal direction (Fig. 2.9).

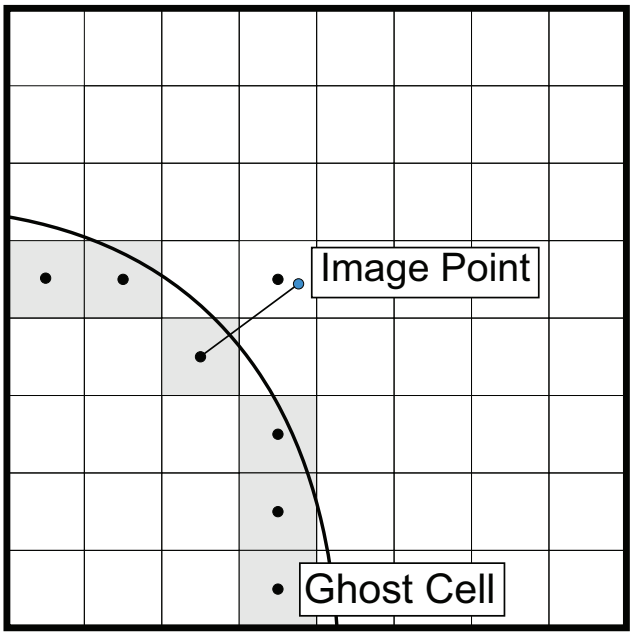


Figure 2.9 Image-point and ghost-cell construction near an immersed boundary [4].

Depending on the interpolation procedure, ghost-cell methods can retain the order of accuracy of the underlying numerical scheme and provide a sharp representation of the immersed interface [4, 116]. The governing equations and flux calculations in the fluid domain remain largely unchanged, with the principal additions consisting of the geometrical preprocessing and the interpolation used to update the image and ghost values. These operations must be performed efficiently to limit the computational overhead.

The main difficulties arise from the construction of suitable ghost, wall, and image points in

regions containing high curvature, sharp features, or closely spaced boundaries. The image point may fall inside the solid or may not have a sufficient number of neighboring fluid cells for the required interpolation stencil. The number and placement of ghost-cell layers also depend on the numerical scheme and the width of its computational stencil. These issues become particularly important for complex iced geometries.

In addition to ghost-cell methods, other discrete-forcing formulations have been developed. Capizzano et al. [117, 118] proposed a face-forcing approach in which the boundary condition is imposed through the fluxes of the cell faces intersected by the immersed interface rather than through reconstructed cell values. The procedure remains similar to the ghost-cell approach: a wall-normal direction is constructed through the intersected face, an image point is defined in the fluid, and the state at this point is interpolated from neighboring cells. The reconstructed state is then used to determine the flux at the immersed-boundary face. This approach has been applied to Euler flows and RANS simulations [117, 118].

Face forcing avoids some of the difficulties associated with placing several rows of ghost cells near complex interfaces. However, it requires a dedicated treatment of the numerical fluxes on immersed-boundary faces and therefore introduces more extensive modifications to the flow solver than conventional ghost-cell methods.

2.7 Synthesis and Positioning of the Present Work

Despite the continued improvement of the individual modules involved in numerical ice-accretion prediction, the automation and robustness of complete multilayer frameworks remain limited by three closely connected aspects: droplet-impingement modeling, geometry evolution, and the representation of the evolving geometry within the computational mesh. These components are particularly important in 3D simulations, where accurately representing the droplet trajectories and their interaction with increasingly complex iced geometries becomes more challenging, while maintaining a valid surface and a high-quality volume mesh becomes progressively more difficult as the ice shape develops. At the same time, the large meshes required for 3D configurations and the repeated solution of the coupled physical models over multiple accretion steps impose substantial computational requirements. Consequently, the numerical methods employed within each module must not only remain robust as the geometry evolves, but also be compatible with efficient parallel execution on modern high-performance computing architectures.

Regarding droplet impingement, the literature does not identify a universally superior formulation. Eulerian and Lagrangian approaches provide comparable access to the collection-

efficiency distribution, but each presents distinct advantages and limitations. Eulerian methods integrate naturally within finite-volume CFD solvers, are readily parallelized, and remain attractive for large 3D calculations. Their continuum description, however, introduces numerical diffusion near impingement limits and does not directly provide individual impact quantities. Lagrangian methods offer a more direct physical representation of droplet motion and naturally provide impact locations, velocities, and angles. These properties are particularly valuable for SLD modeling and for tracking droplets generated by splashing. Their main limitation is computational cost, especially since a large number of trajectories must be tracked to obtain sufficiently resolved impingement maps. For large-scale 3D applications, efficient particle localization, trajectory integration, and parallel workload distribution therefore become important considerations for maintaining computational performance. Consequently, the choice between Eulerian and Lagrangian formulations remains dependent on the intended application rather than on a clear overall advantage of either approach.

The literature on geometry evolution reveals a growing interest in interface-capturing methods. Classical interface-tracking approaches remain widely used because of their simplicity and because the evolving surface is available explicitly. Nevertheless, their robustness deteriorates when large displacements, concave regions, sharp features, or surface self-intersections develop. These difficulties become increasingly restrictive in 3D multilayer simulations, where maintaining a valid surface connectivity is considerably more involved than in 2D. Interface-capturing techniques, particularly level-set methods, provide a more flexible alternative because they represent the interface implicitly and can accommodate large deformations and topological changes without explicitly preserving the original surface connectivity. They are therefore well suited to the formation of complex ice morphologies. Their benefits are accompanied by challenges related to signed-distance preservation, preservation of sharp features, and the extraction of an explicit surface required for the subsequent accretion layer.

Geometry representation and volume-mesh generation constitute the third central difficulty. In conventional body-fitted frameworks, the volume mesh must be regenerated or deformed after each accretion step. Although this treatment provides a direct and accurate application of the wall boundary conditions, its automation becomes increasingly difficult as the ice shape develops sharp horns, scallops, and highly concave regions. Immersed-boundary methods have therefore gained considerable attention because they decouple the evolving surface from the underlying volume mesh and can reduce, or in some cases eliminate, repeated body-fitted remeshing.

No single immersed-boundary formulation is equally suited to all the equations involved in an ice-accretion framework. Penalization methods require relatively limited modifications to an

existing solver and have proved particularly attractive for Eulerian droplet transport [31,109]. Their extension to aerodynamic simulations involving high-Reynolds-number turbulent flows is less straightforward.

Discrete approaches, such as ghost-cell immersed-boundary methods, provide a sharper representation of the interface and can retain the formal accuracy of the underlying numerical scheme. They are therefore generally better suited to aerodynamic-flow calculations, including compressible and turbulent formulations [31]. Their implementation, however, requires more elaborate geometrical preprocessing and robust interpolation procedures. In particular, the construction of ghost, wall, and image points becomes challenging near sharp ice features, highly curved surfaces, and closely spaced portions of the interface.

The preceding review therefore suggests that the most appropriate methodology for a complete ice-accretion framework may involve different numerical treatments for its individual modules rather than the systematic use of a single formulation. Moreover, these treatments must remain computationally tractable and scalable as simulations progress toward increasingly complex 3D configurations on modern HPC systems. The present thesis follows this direction by focusing on complementary developments for droplet impingement, geometry evolution, and geometry representation. First, a Lagrangian droplet-tracking framework is investigated as an alternative to the established Eulerian formulation, with particular attention to its computational efficiency, parallel scalability, collection-efficiency prediction, and ability to represent SLD wall interactions. Second, a level-set method is developed for robust 3D multilayer geometry evolution, with the objective of reducing the dependence of the update procedure on the connectivity of an explicit surface mesh. Finally, the integration of the evolving geometry within an immersed-boundary framework is examined to assess the extent to which repeated body-fitted remeshing can be reduced.

CHAPTER 3 A Ray-Tracing-Enhanced Lagrangian Particle Tracking Framework for Ice Accretion Applications

This chapter presents the development of a Lagrangian particle-tracking solver for aircraft ice accretion simulations on distributed-memory systems. The primary aim of this work is to enhance the robustness, accuracy, and computational efficiency of Lagrangian droplet tracking. To this end, a face-to-face tracking scheme is employed for robust boundary-intersection detection, combined with a ray-tracing-enhanced bilinear-patch intersection technique to improve computational performance and scalability. Improvements to the droplet-seeding strategy are also introduced to facilitate the initialization and refinement of the seeding grid. Supercooled large droplet (SLD) effects are incorporated to improve agreement with experimental collection-efficiency data. Results demonstrate good consistency with the Eulerian formulation and agreement with available experimental data.

3.1 Introduction

Particle-laden flows have long been a subject of scientific investigation due to their prevalence in numerous applications ranging from climate prediction [119] to combustion systems [120] and, in the context of the present work, aircraft ice accretion [60, 121–123]. A key challenge lies in the large number of particles required for high-fidelity simulations, which imposes significant computational demands. With recent advances in high-performance computing (HPC) on distributed systems, there is a growing need to optimize computational frameworks for tracking large particle populations.

Recent Ice Prediction Workshops [26, 27] have shown an increasing use of Eulerian droplet-tracking methods [124–128]. These methods adopt the same computational framework as the flow solver, providing advantages in terms of implementation and computational efficiency. However, Eulerian formulations can present challenges when incorporating complex droplet phenomena such as splashing and secondary-droplet re-impingement. Lagrangian formulations, on the other hand, provide direct tracking of individual droplet trajectories but can become computationally expensive when a large number of droplets is required. Advances in high-performance computing therefore provide an opportunity to improve the computational efficiency of Lagrangian approaches for aircraft icing applications.

Significant progress has been made in the development of Lagrangian particle-tracking methods for ice accretion, with recent studies proposing different strategies to address their

computational challenges. For instance, [122] adapts the grid-partitioning scheme based on particle trajectories, while [123] employs a dynamic load-balancing approach in which a dedicated computational rank redistributes droplets across the remaining ranks. Further developments are nevertheless required to reduce oscillations in surface collection-efficiency distributions [129] and to automate the generation and refinement of the droplet-seeding grid. The present work addresses these aspects by introducing improvements throughout the Lagrangian tracking process, from droplet seeding to trajectory integration, geometric intersection, surface impingement, and parallel execution. The chapter is structured as follows: Section 3.2 reviews droplet-tracking techniques for ice accretion applications and introduces the Lagrangian formulation used in this work. Section 3.3 presents the proposed solver enhancements. Section 3.4 presents and discusses the benchmark cases and corresponding results. Finally, Section 3.5 summarizes the main findings.

3.2 Lagrangian Droplet Tracking for Ice Accretion

Water droplet tracking is a key component of ice accretion simulations, as it determines both impingement locations and the amount of water collected on solid surfaces. Numerous approaches have been proposed in the literature, all aiming to balance robustness, accuracy, and computational efficiency.

Historically, early ice accretion solvers such as LEWICE [130] relied on Lagrangian tracking, in which individual droplet trajectories are computed by solving their equations of motion. While well suited to panel-based potential-flow solvers, the extension of this approach to modern RANS-based frameworks introduces additional complexity due to the need for repeated interpolation of flow quantities at droplet locations.

To address these limitations, Eulerian approaches [55] were introduced, modeling droplets as a continuous dispersed phase governed by transport equations for momentum and volume fraction α (Fig. 3.1). This formulation integrates naturally within flow solvers and offers excellent computational efficiency and scalability. However, accurately representing complex droplet physics, such as splashing, remains challenging and often computationally expensive. These limitations motivate the consideration of Lagrangian particle tracking, in which individual droplet trajectories and post-impact behavior can be treated more directly.

Lagrangian droplet tracking is based on integrating the equation of motion from Newton's second law (Eq. 3.1) for each droplet individually. In the present formulation, the forces considered are the drag force resulting from the relative motion between the air and the droplet, and the gravitational force. The droplet velocity $\boldsymbol{u_d}$ is obtained from Eq. 3.1, where

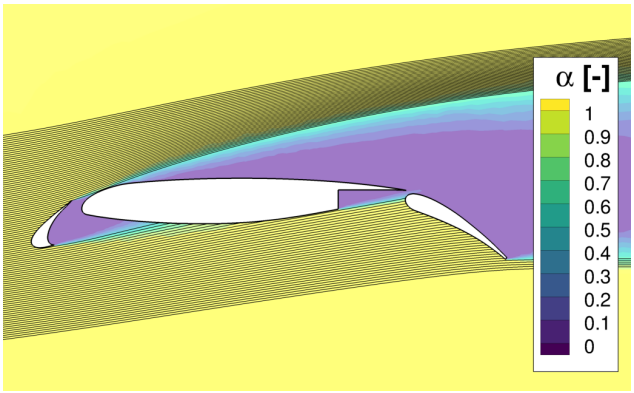


Figure 3.1 Example of Eulerian solver results : Countour of water volume fraction α with droplet trajectories streamlines.

$\boldsymbol{u_a}$ is the air velocity. The droplet position $\boldsymbol{x_d}$ is subsequently obtained from Eq. 3.2. The force coefficient F is defined by Eq. 3.3 as a function of the droplet drag coefficient C_D, the relative velocity $\|\boldsymbol{u_a} - \boldsymbol{u_d}\|$, the air and droplet densities ρ_a and ρ_d, and the droplet diameter D.

$$\frac{d\boldsymbol{u_d}}{dt} = \frac{\mathbf{f}}{m} + \mathbf{g} = F(\boldsymbol{u_a} - \boldsymbol{u_d}) + \mathbf{g} \tag{3.1}$$

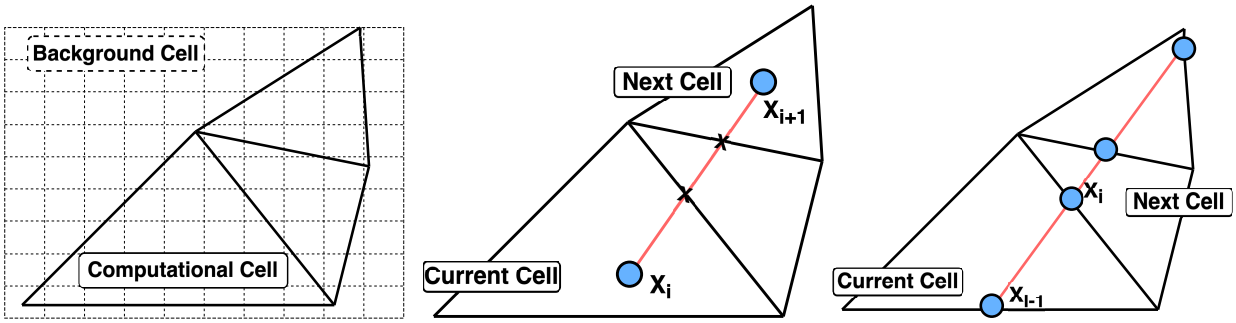
$$\frac{d\boldsymbol{x_d}}{dt} = \boldsymbol{u_d} \tag{3.2}$$

$$F = \frac{3C_D\rho_a\|\boldsymbol{u_a} - \boldsymbol{u_d}\|}{4\rho_d D} \tag{3.3}$$

The main assumptions of the formulation are the following:

- The droplet mass remains constant prior to impact;
- Droplets are independent from one another, such that collisions and coalescence are neglected;
- The droplet concentration does not affect the fluid flow;
- The effects of turbulence on the droplets are neglected;
- Droplet rotation and lift are neglected;
- Phase change within the droplets is not considered, as the solver is limited to computing the deposited liquid water mass under the quasi-steady icing hypothesis [131].

A central difficulty in Lagrangian particle tracking on unstructured meshes is that droplet trajectories do not generally follow the computational grid. The integration of Eq. 3.1 requires the interpolation of the flow velocity along the trajectory and the identification of the cells traversed by the droplet. Various approaches have been developed for this purpose. Tree-based search techniques [132] and auxiliary-grid approaches [133], illustrated in Fig. 3.2a, rely on spatial-search structures to identify the host cell. Alternatively, face-to-face tracking techniques use the connectivity of the computational mesh to follow the droplet from one cell to the next.



(a) Auxiliary grid tracking [133].

(b) face-to-face tracking [60] (In Cell Positions).

(c) face-to-face tracking [121] (On Face Positions).

Figure 3.2 Droplet tracking techniques.

The face-to-face approach of [60], illustrated in Fig. 3.2b, uses the previous droplet position and velocity together with the mesh connectivity to identify the successive cells traversed by the trajectory. This approach requires particular care near solid boundaries, where strong flow gradients and small cells can affect the trajectory integration and impact detection.

The face-to-face tracking method presented by [121], illustrated in Fig. 3.2c, instead advances the droplet from one mesh face to the next using edge-to-cell connectivity in 2D or face-to-cell connectivity in 3D. A ray is cast based on the droplet velocity, and the nearest face intersection determines the next cell along the trajectory. The corresponding residence time within the cell defines the local time step, which therefore adapts to the local mesh resolution. The host cell is obtained directly from the mesh connectivity without requiring an additional spatial search. The method also operates directly on the same unstructured mesh used by the flow solver. For these reasons, this face-to-face formulation is adopted in the present work.

The integration of Eq. 3.1 is achieved using a second order backward difference scheme [121]:

$$a\boldsymbol{u_d}^n + b\boldsymbol{u_d}^{n-1} + c\boldsymbol{u_d}^{n-2} = F(\boldsymbol{u_a} - \boldsymbol{u_d}) + \mathbf{g} \tag{3.4}$$

$$\boldsymbol{u_d}^n = \frac{F\boldsymbol{u_a} - b\boldsymbol{u_d}^{n-1} - c\boldsymbol{u_d}^{n-2} + \mathbf{g}}{a + F} \tag{3.5}$$

$$a = \frac{2\Delta t_c + \Delta t_{c-1}}{(\Delta t_c + \Delta t_{c-1})\Delta t_c} \tag{3.6}$$

$$b = \frac{-(\Delta t_c + \Delta t_{c-1})}{\Delta t_c \Delta t_{c-1}} \tag{3.7}$$

$$c = \frac{\Delta t_c}{(\Delta t_c + \Delta t_{c-1})\Delta t_{c-1}} \tag{3.8}$$

where Δt_c and Δt_{c-1} are the current and previous local time steps, respectively, and the superscripts n, $n-1$, and $n-2$ denote successive integration levels. Since the local time step depends on the next face intersection, the droplet velocity and intersection position are determined iteratively. The flow velocity at the face $\boldsymbol{u_a}^n$ is first evaluated by averaging the flow velocity between the current and next cells:

$$\boldsymbol{u_a}^n = \frac{1}{2}(\boldsymbol{u_a}_c + \boldsymbol{u_a}_{c+1}) \tag{3.9}$$

Since the next cell depends on the trajectory itself, an iterative procedure is employed. Initially, $\boldsymbol{u_d}^n$ is assumed equal to $\boldsymbol{u_d}^{n-1}$ and the nearest face intersection is determined. The corresponding local time step and the coefficients in Eqs. 3.6–3.8 are then calculated. Equation 3.5 provides an updated value of $\boldsymbol{u_d}^n$, and the procedure is repeated until the prescribed convergence criterion is satisfied. The face intersection is determined using the intermediate droplet velocity

$$\boldsymbol{u_d}^{n-\frac{1}{2}} = \frac{1}{2}(\boldsymbol{u_d}^n + \boldsymbol{u_d}^{n-1}) \tag{3.10}$$

In regions of strong flow gradients, particularly near solid surfaces, the method can locally reduce to first-order accuracy [121]. To improve the evaluation of the flow velocity at the crossed face, the cell-centered velocity is reconstructed at the facet-intersection position using the local velocity gradient:

$$\boldsymbol{u_{a,face}} = \boldsymbol{u_{a,c}} + \nabla(\boldsymbol{u_a})_c \cdot \boldsymbol{d} \tag{3.11}$$

where $\boldsymbol{d}$ is the displacement vector from the cell center to the facet-intersection position.

3.3 Improvements in Lagrangian Particle Tracking

Several improvements are introduced to enhance the robustness, accuracy, and computational efficiency of the Lagrangian particle-tracking methodology, including trajectory intersection, adaptive seeding, collection-efficiency evaluation, SLD treatment, and parallel workload distribution.

3.3.1 Ray–Quadrilateral Intersections in CFD Applications

Hexahedral elements are commonly used in CFD meshes because they can provide high-quality and structured resolution in regions of interest. This is illustrated in Fig. 3.3 using the Common Research Model (CRM) wing–body configuration from the Fifth AIAA High-Lift Prediction Workshop (HLPW5), where successive levels of hexahedral refinement are used around the aircraft geometry. For Lagrangian particle tracking on such meshes, the trajectory calculation repeatedly requires the intersection of the droplet path with the quadrilateral faces of the hexahedral elements.

Unlike triangular faces, quadrilateral faces are not necessarily planar. In a general unstructured hexahedral mesh, the four vertices defining a face may therefore not lie on a single plane, making a planar ray–face intersection ambiguous. A common approach is to split the quadrilateral into two planar triangles and perform two independent ray–triangle intersection tests following the method of [134]. Although straightforward, this introduces an artificial diagonal and treats the quadrilateral as two separate planar surfaces.

An alternative is to represent the quadrilateral directly as a bilinear patch, parameterized by the local coordinates $(u, v) \in [0, 1]^2$. The particle trajectory is represented by the ray $\boldsymbol{r}(t)$,

$$\boldsymbol{r}(t) = \boldsymbol{O} + t\boldsymbol{D}, \quad t \geq 0 \tag{3.12}$$

while the quadrilateral is represented by the bilinear patch

$$\boldsymbol{p}(u, v) = (1-u)(1-v)\boldsymbol{p}_{00} + (1-u)v\boldsymbol{p}_{01} + u(1-v)\boldsymbol{p}_{10} + uv\boldsymbol{p}_{11} \tag{3.13}$$

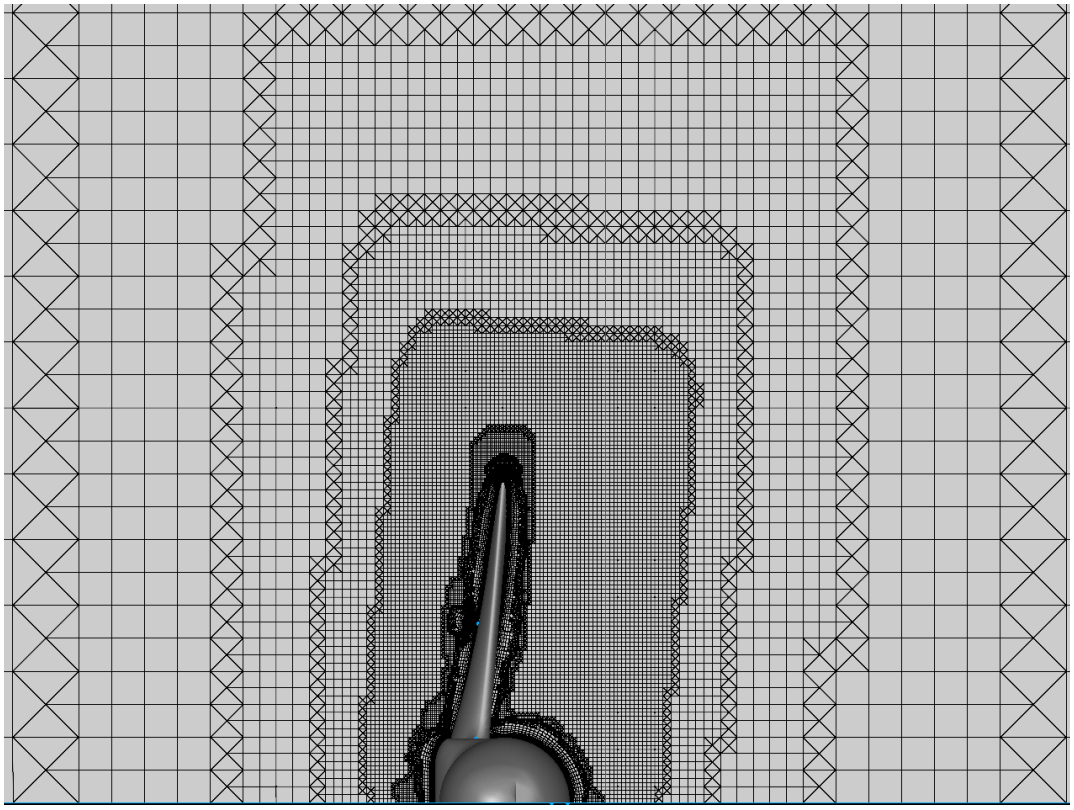
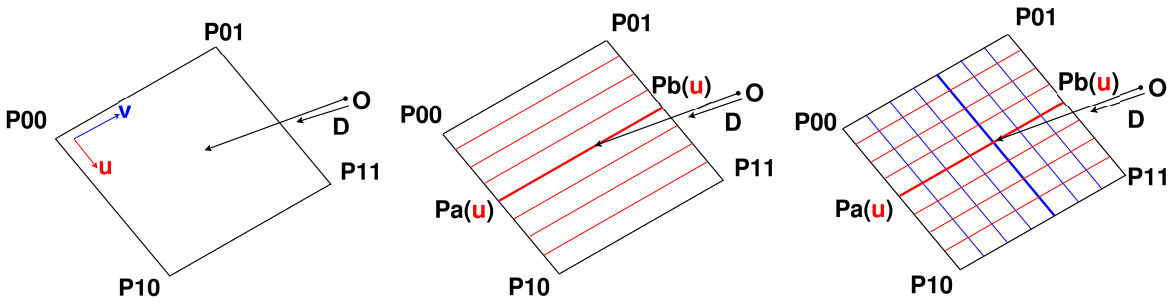


Figure 3.3 Cut through the computational mesh of the CRM wing–body configuration from HLPW5, illustrating the hexahedral refinement around the aircraft geometry.

Here, $\boldsymbol{O}$ and $\boldsymbol{D}$ denote the ray origin and direction, respectively, and $\boldsymbol{p}_{ij}$ are the four vertices of the quadrilateral. This representation defines a continuous surface passing through the four vertices without requiring them to be coplanar. The ray–quadrilateral intersection problem therefore consists of determining the parameters (t, u, v) for which $\boldsymbol{r}(t) = \boldsymbol{p}(u, v)$.

Several algorithms have been proposed for computing ray–bilinear patch intersections. Equating Eqs. 3.12 and 3.13 yields a system of three equations that can be solved analytically [135]. This algebraic formulation has been used in Lagrangian particle-tracking applications [136, 137]. The more recent GARP (Geometric Approach to Ray/bilinear Patch intersections) method of [138] instead uses the geometric properties of the bilinear patch to determine the intersection.

The GARP method interprets the bilinear patch as a ruled surface obtained by linear interpolation between two opposite edges. The intersection procedure is carried out in two parametric stages, as illustrated in Fig. 3.4b–3.4c. For a given value of u, the corresponding ruling segment is defined by the two interpolated points



(a) Step 0: Definition of the ray and bilinear patch geometry.

(b) Step 1: Determination of the intersection u-parameter along the bilinear patch.

(c) Step 2: Determination of the intersection v-parameter along the bilinear patch.

Figure 3.4 Steps for ray–bilinear patch intersection.

$$\boldsymbol{P}_a(u) = (1-u)\boldsymbol{p}_{00} + u\boldsymbol{p}_{10}, \qquad \boldsymbol{P}_b(u) = (1-u)\boldsymbol{p}_{01} + u\boldsymbol{p}_{11} \tag{3.14}$$

The first stage determines the values of u for which this segment is coplanar with the ray. This condition is expressed by setting the scalar triple product of the vectors $\boldsymbol{OP}_a$, $\boldsymbol{P}_a\boldsymbol{P}_b$, and $\boldsymbol{D}$ to zero, leading to a polynomial equation of degree at most two in u. Only solutions satisfying $u \in [0,1]$ are retained.

For each retained value of u, the ruling segment $\boldsymbol{P}_a\boldsymbol{P}_b$ is therefore fixed. The second stage determines v and t by finding the closest points between the line supporting this segment and the line supporting the ray. A valid intersection requires $v \in [0,1]$ and $t \geq 0$. In the presence of multiple valid solutions, the intersection corresponding to the smallest admissible value of t is selected.

Whereas Ramsey et al. [135] determine the intersection by algebraically manipulating the ray and bilinear-patch equations (Eqs. (**??**) and (**??**)), resulting in relatively complex intermediate expressions. GARP instead exploits the geometric nature of the problem to obtain a coplanarity condition for one patch parameter. Both approaches lead to a polynomial equation of degree at most two, but differ in how the intersection parameters are determined. The geometric formulation of GARP reduces the number of arithmetic operations required to determine the intersection [138].

To evaluate computational performance, one million random rays and quadrilateral patches were generated for intersection testing using the GARP and Ramsey et al. [135] formulations, together with the traditional two-triangle approach. Both ray–patch methods reduce the computational cost relative to the two-triangle approach, with GARP providing an additional

improvement over the formulation of Ramsey et al. [135]. The implications of the intersection algorithm on the overall solver scalability are discussed in Section 3.4.

Table 3.1 Performance comparison of ray–quadrilateral intersection methods.

Method	ETR* vs. Two-Triangles [134]	ETR* vs. Ramsey et al. [135]
Ramsey et al. [135]	51.3%	-
GARP [138]	58.2%	14.2%

* ETR: execution-time reduction.

3.3.2 Adaptive Seeding

The definition of the seeding region is an important step in Lagrangian droplet tracking, as the initial droplet seeding distribution must adequately cover the portion of the incoming cloud that can impinge on the geometry. A first estimate can be obtained by projecting the geometry onto an upstream seeding plane along the freestream direction. An initial set of trajectories can then be computed, and the droplets that effectively impact the geometry are used to adjust the dimensions of the seeding region.

A more systematic estimation of the seeding region can be obtained through trajectory backtracking. Direct backtracking from the aerodynamic RANS solution is not possible because the no-slip boundary condition imposes a zero fluid velocity at the wall. The Eulerian droplet formulation provides a complementary solution to this problem, since the computed droplet velocity remains defined at the impingement location. Building on the 2D hybrid approach of [5], the Eulerian droplet velocity field is therefore used to trace trajectories backward from the surface toward a user-defined upstream seeding plane, as illustrated in Fig. 3.5. The resulting intersections with the seeding plane provide an estimate of the region from which impacting droplets originate and are used to initialize the Lagrangian seeding grid.

Once the initial seeding region has been established, the spatial resolution of the seeding grid becomes the main factor controlling convergence of the Lagrangian impingement solution. Rather than prescribing a uniformly fine distribution of droplets, an adaptive refinement procedure is employed. An initial structured seeding grid is generated and the corresponding droplet trajectories are computed. Cells of the seeding grid associated with droplets that impact the geometry are subsequently refined, thereby progressively increasing the number of trajectories in regions contributing to the impingement solution while avoiding unnecessary refinement elsewhere.

Figure 3.6 illustrates this process. Starting from a relatively coarse initial distribution, successive refinement levels concentrate the seeding resolution within the region associated with

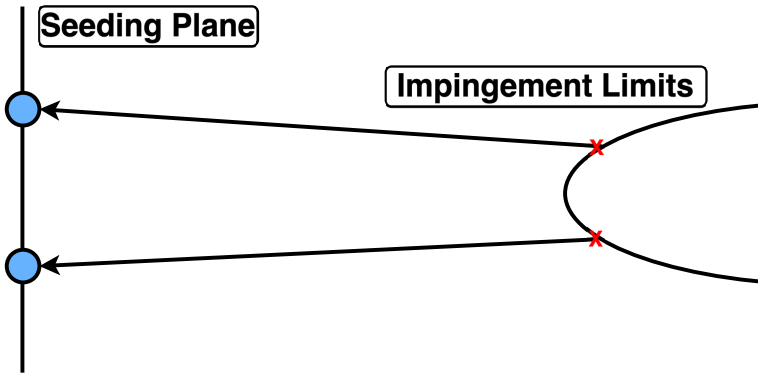


Figure 3.5 Hybrid Eulerian–Lagrangian backtracking for initialization of the Lagrangian seeding region [5].

impacting trajectories. Previously computed droplets are retained between refinement passes, such that only the newly introduced seeding points require additional trajectory calculations. In practice, approximately six to eight refinement passes are typically sufficient, although the refinement can instead be continued until the variation in the total collection efficiency criterion between successive refinement passes falls below a prescribed tolerance of $\epsilon = 10^{-5}$. Another advantage of performing the refinement over multiple passes is that the workload distribution can be progressively improved from one pass to the next. The computational time of previously computed trajectories is retained and used to estimate the workload associated with newly generated droplets during subsequent refinement passes.

3.3.3 Collection Efficiency Computations and SLD Effects

The collection efficiency associated with a Lagrangian trajectory is commonly evaluated by constructing a local droplet streamtube around the injected droplet, as illustrated in Fig. 3.7a. In principle, this droplet streamtube can be constructed directly from neighboring droplets of the seeding grid. However, there is no guarantee that neighboring droplets upstream will impact the same region of the geometry. This becomes particularly problematic for complex configurations, where neighboring trajectories may separate and impact different geometric components, as illustrated in Fig. 3.7b. A sufficiently small spacing between droplets can reduce the occurrence of such situations, but applying this resolution to the complete seeding grid can lead to a prohibitively large number of trajectories.

To avoid this requirement, the droplets originating from the adaptive seeding grid are treated as primary droplets, while additional auxiliary droplets are introduced locally for the sole purpose of constructing the streamtube and evaluating the collection efficiency [139]. For

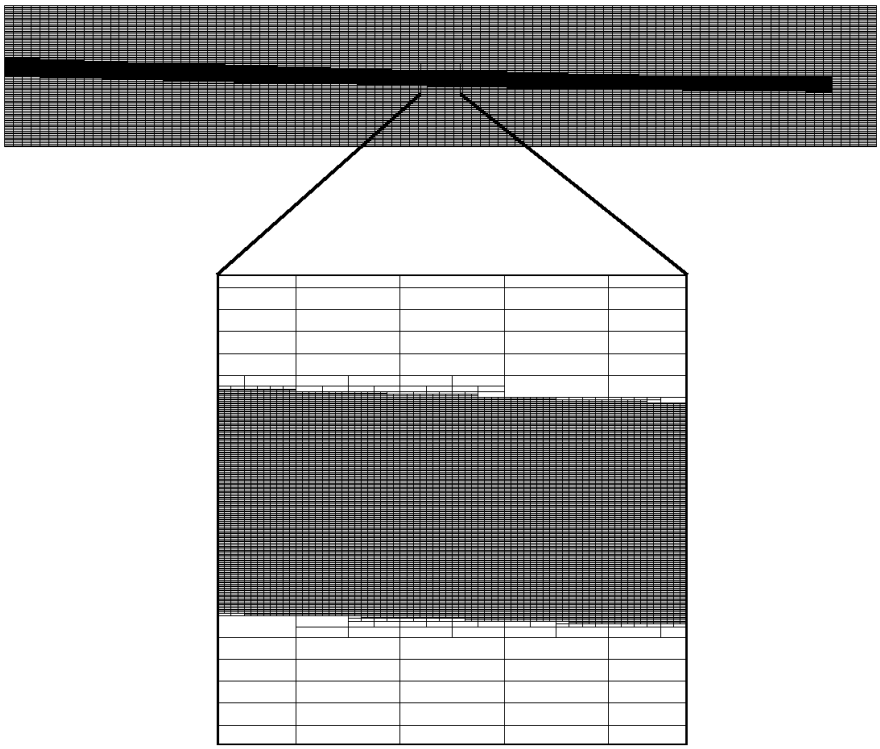
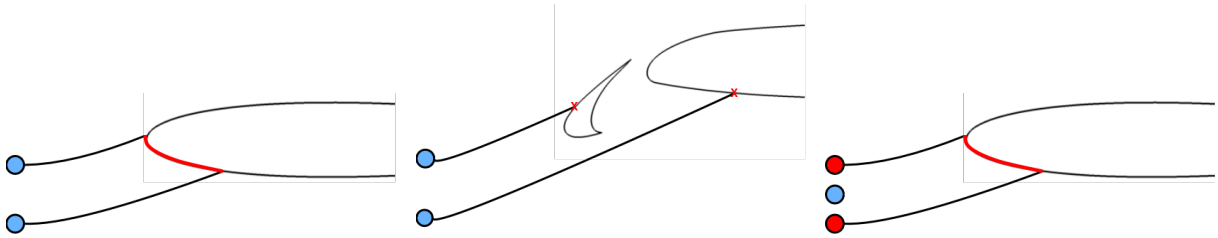


Figure 3.6 Example of adaptive refinement of the Lagrangian seeding grid around the region associated with impacting droplet trajectories.

each primary droplet, two auxiliary droplets are released in its immediate vicinity in 2D, while three or four auxiliary droplets are used in 3D depending on the topology of the local streamtube. The auxiliary droplets are initialized with a spacing of $dy \approx O(10^{-8}c_{\text{ref}})$ from the primary droplet, where c_{ref} is the reference length of the geometry, as illustrated in Fig. 3.7c. The resulting local streamtube is then used to determine the collection efficiency associated with the primary impact as

$$\beta = \frac{A_\infty}{A_s} \tag{3.15}$$

where A_∞ is the area of the local streamtube at the upstream seeding location and A_s is the corresponding area at the surface impact location. The auxiliary droplets are used only for this calculation and are discarded afterward.



(a) Droplet streamtube. (b) Pathological case for the collection efficiency. (c) Local droplet streamtube.

Figure 3.7 Collection-efficiency computations. Primary and auxiliary droplets are represented in blue and red, respectively.

Once the collection efficiency has been evaluated at each primary impact location, the Lagrangian information is transferred to the cell-centered surface representation used in CHAMPS. Since multiple primary droplets may contribute to the same surface face, a distance-weighted average is applied to obtain a single value β_i for each impacted surface cell:

$$\beta_i = \frac{\sum \beta_{i,d} \cdot \omega_{i,d}}{\sum \omega_{i,d}}, \tag{3.16}$$

where the weight $\omega_{i,d}$ is based on the inverse square of the distance between the primary-droplet impact position and the center of the corresponding surface face. To improve the surface coverage without increasing the number of primary trajectories, the influence of each impact is extended to neighboring boundary faces identified through the existing surface connectivity. This treatment extends the interpolation of the available Lagrangian information to neighboring faces without requiring additional trajectory calculations.

Experimentally, an icing cloud does not consist of droplets of a single diameter, but rather of a distribution of droplet sizes, each representing a fraction of the total liquid water content. To better represent these experimental conditions, a polydisperse cloud is discretized into a number of droplet-size bins, and the preceding procedure is performed independently for each bin. Each bin n is assigned a weighting factor γ_n corresponding to its fraction of the distribution, and the resulting collection efficiencies are combined to obtain the total collection efficiency on each surface cell:

$$\beta_{i,total} = \sum_{n=1}^{N_{bins}} \gamma_n \beta_{i,n}, \tag{3.17}$$

where $\beta_{i,n}$ is the collection efficiency on surface cell i for droplet-size bin n. This formulation allows the contribution of the complete droplet-size distribution to be represented within the final surface collection-efficiency.

For SLD conditions, the models introduced in Section 2.3.3 are subsequently applied to account for the fraction of the impinging water that is not retained following impact. The mass-loss coefficient f_m, or equivalently the sticking rate $\epsilon = 1 - f_m$, modifies the collection efficiency associated with the primary impact according to

$$\beta_{i,\,final} = (1 - f_m)\,\beta_{i,\,0} = \epsilon\,\beta_{i,\,0}, \tag{3.18}$$

where $\beta_{i,\,0}$ denotes the collection efficiency prior to the splashing correction. In addition to this correction, the model of Wright et al. [64] provides the post-impact properties required to track the secondary droplets generated by splashing. In the present implementation, as a simplification, the ejected mass associated with each impact is represented by a single secondary-droplet parcel rather than by individual secondary droplets. The post-impact normal and tangential velocities and the secondary-droplet diameter are given by

$$\begin{aligned} u_{n,\,post\,impact} &= u_{n,\,impingement}(1.075 - 0.0025\,\alpha_{impingement}) \\ u_{t,\,post\,impact} &= u_{t,\,impingement}(0.300 - 0.0020\,\alpha_{impingement}) \end{aligned} \tag{3.19}$$

$$\frac{D_{new}}{D} = exp(-0.0281\,K_M). \tag{3.20}$$

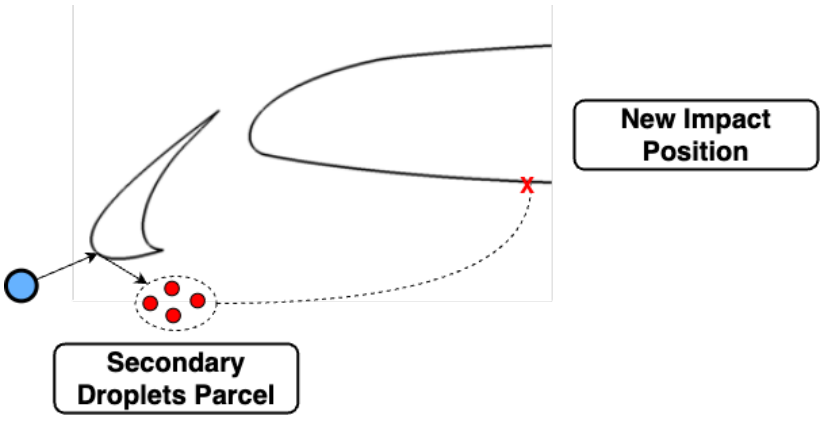


Figure 3.8 Schematic representation of secondary-droplet parcel tracking and re-impingement following splashing.

The secondary-droplet parcel is then tracked from the primary impact location until it either leaves the computational domain or encounters the geometry again, as illustrated in Fig. 3.8. In the case of re-impingement, its contribution to the local collection efficiency is normalized with respect to the incoming water flux according to

$$\beta_{Re-Impingement} = \frac{m_{sd} \cdot (\boldsymbol{u_{sd}} \cdot \boldsymbol{n})/A_s}{m_\infty \cdot u_\infty / A_\infty} \tag{3.21}$$

where m_{sd} is the mass contained in the secondary-droplet parcel, $\boldsymbol{u_{sd}}$ is its velocity at the re-impingement location, A_s is the area of the impacted surface face, A_∞ is the area of the upstream seeding grid, and m_∞ is the initially injected droplet mass. This contribution is added to the collection efficiency of the corresponding boundary face to account for secondary-droplet re-impingement.

3.3.4 Parallelism

The Lagrangian droplet solver is implemented in Chapel within the distributed-memory framework of CHAMPS [140]. The aerodynamic mesh remains spatially decomposed among the available computational resources, whereas the adaptive droplet cloud is generated and scheduled independently of this decomposition. This separation is particularly useful for Lagrangian tracking, since the computational cost of a trajectory is not necessarily correlated with the spatial partition in which its seeding point is located. To minimize inter-node communication during trajectory integration, the flow and mesh data required for droplet tracking are replicated across the computational locales. This allows trajectories to access the required aerodynamic and geometric information locally, avoiding repeated remote data accesses as droplets travel across the computational domain. Each computational resource is therefore assigned a subset of the droplet population and independently tracks these droplets through the domain until impact or trajectory termination.

The adaptive seeding procedure progressively introduces new trajectories during refinement, each of which is assigned an estimated computational cost for workload distribution. During the initial refinement stage, all trajectories are assigned the same cost. In subsequent stages, the cost of each new trajectory is estimated from the measured cost of previously computed trajectories within its parent cell, progressively improving the workload estimate throughout the refinement process.

The resulting trajectories are distributed using a longest-processing-time (LPT) scheduling strategy. Trajectories are first sorted according to their predicted computational cost and are successively assigned to the computational resource with the lowest accumulated predicted

workload. After integration, their measured computational costs are retained and used to improve the workload estimates during subsequent refinement stages. Previously computed trajectories are also cached and reused, such that only newly introduced droplets contribute to the computational workload during refinement.

Auxiliary-droplet calculations associated with wall impacts are balanced separately from the primary trajectories. Following the primary-droplet calculation, impacting droplets are redistributed using a second cost-based assignment before auxiliary-droplet integration. The measured cost of each primary trajectory, together with the number of auxiliary droplets required to construct the local streamtube, is used to estimate this additional workload. This two-stage scheduling procedure allows both primary and auxiliary trajectory calculations to be distributed according to their expected computational cost rather than solely according to the number of droplets.

3.4 Results

The results are first presented through a parallel scalability study to evaluate the computational performance of the Lagrangian solver on distributed-memory systems. The IPW2 CRM Hybrid configuration is used for this scalability assessment. The droplet-tracking methodology and SLD implementation are then assessed using the two IPW1 configurations summarized in Table 3.2, for which collection-efficiency results are compared with available numerical and experimental data. Finally, the HLPW5 CRM wing–body configuration is considered as a large 3D application of the Lagrangian methodology.

Table 3.2 Summary of the droplet-impingement benchmark cases.

Parameter	IPW2-1.3	IPW1-112	IPW1-122	HLPW5-2.1
Geometry	CRM Hybrid	NACA 64A008 H-tail	Three-element airfoil	CRM wing–body
MVD $[\mu m]$	25	92	92	FR*
AOA [°]	3.7	6.0	4.0	6.0
Ma [-]	0.21	0.23	0.23	0.20
C_{ref} $[inch]$	39.37	37.65	36.0	275.8
T $[K]$	247.15	280.0	278.0	288.15
Re [-]	$8.1 \cdot 10^6$	$5.0 \cdot 10^6$	$4.8 \cdot 10^6$	$5.4 \cdot 10^6$

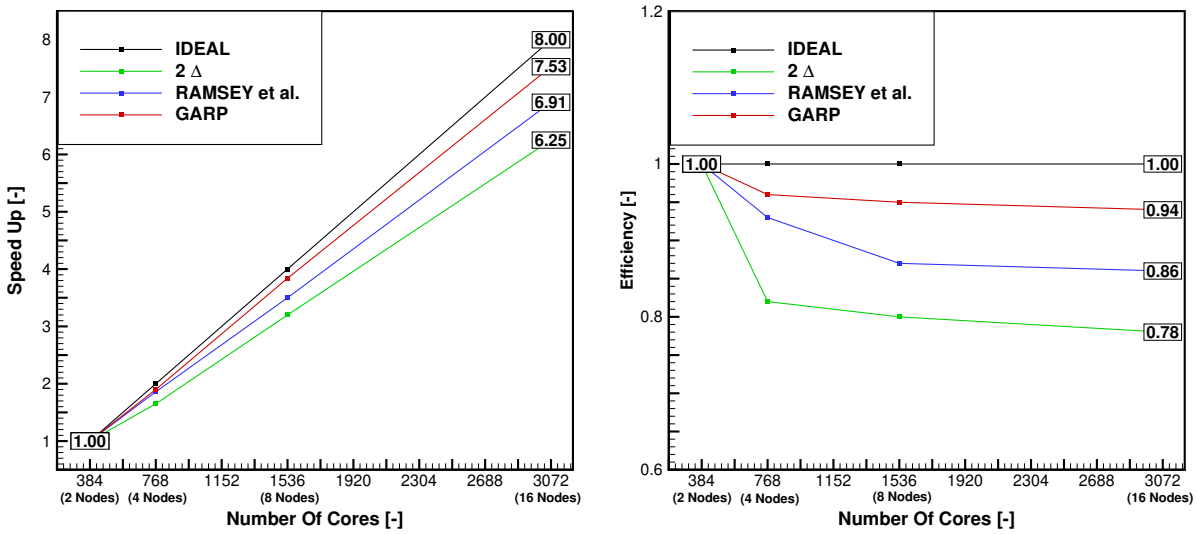
Note: *FR: Freezing Rain.

3.4.1 Scalability

To evaluate the parallel scalability of the Lagrangian solver, Case IPW2 - 1.3 is selected as the benchmark. All simulations are conducted on the Digital Research Alliance of Canada Trillium cluster. Each compute node contains two AMD EPYC 9655 processors, providing a total of 192 cores per node. Simulations are performed using between 2 and 16 nodes, with the number of nodes doubled at each measurement point.

The flow conditions from the original workshop case are adopted. A single droplet diameter corresponding to the median volumetric diameter (MVD) of 25 μm is considered. A total of 12.8 million droplets are seeded for the benchmark, providing a sufficiently large and fixed workload for the scalability assessment.

Unlike Eulerian simulations, for which the computational cost can conveniently be characterized by the time per solver iteration, individual Lagrangian trajectories may require substantially different numbers of integration steps before impacting the geometry or leaving the computational domain. Consequently, the total computation time is used as the performance metric for the scalability study.



(a) Solver speedup.

(b) Solver efficiency.

Figure 3.9 Parallel scalability of the Lagrangian solver: speedup and parallel efficiency relative to the two-node baseline.

The computational speedup is determined using Eq. 3.22, and the parallel efficiency is computed using Eq. 3.23. Since the two-node calculation is used as the baseline, a factor of 2 is included in the efficiency definition. The results are presented in Fig. 3.9. With the GARP technique, the solver retains approximately 94% parallel efficiency at 16 nodes relative to the

two-node baseline. Over the considered range, GARP also exhibits better scalability than the two other approaches.

The remaining loss in parallel efficiency can be associated with several factors inherent to Lagrangian particle tracking. Although the flow and mesh data required for trajectory integration are replicated to minimize communication between computational resources, droplet trajectories access these data according to their paths through the domain, resulting in irregular memory-access patterns. In addition, the high-level parallel abstractions provided by Chapel facilitate the implementation of distributed-memory algorithms but provide less direct control over some architecture-specific, low-level optimizations. These effects become increasingly relevant as the computational workload is distributed over a larger number of resources and may contribute to the departure from ideal parallel scaling.

$$Speed\,Up = \frac{Simulation\,Time\,2\,Nodes}{Simulation\,Time\,N\,Nodes} \tag{3.22}$$

$$\eta = 2 \cdot \frac{Speed\,Up}{N\,Nodes} \tag{3.23}$$

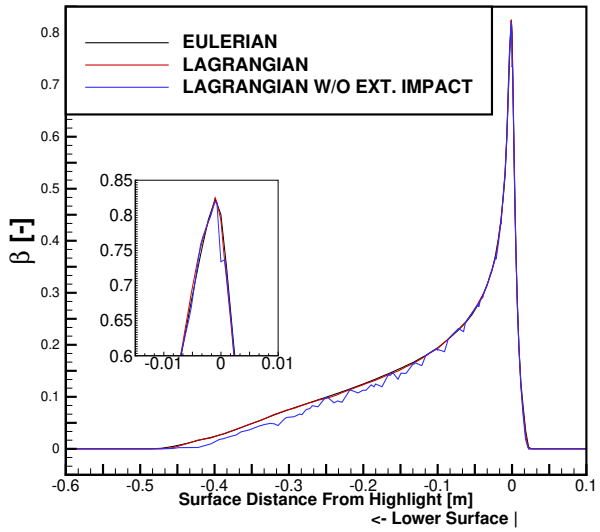
3.4.2 Case IPW1 - 112: NACA64A008 Horizontal Tail

Case IPW1-112 corresponds to a NACA64A008 horizontal-tail configuration considered during IPW1, for which experimental collection-efficiency measurements are available under SLD conditions. Figure 3.10 presents the corresponding collection-efficiency results. The effect of extending the influence of each droplet impact to neighboring surface faces is first illustrated in Fig. 3.10a. Without this treatment, the Lagrangian solution exhibits local oscillations in the collection-efficiency distribution. Extending the interpolation to neighboring faces produces a smoother distribution and improves the consistency with the Eulerian prediction.

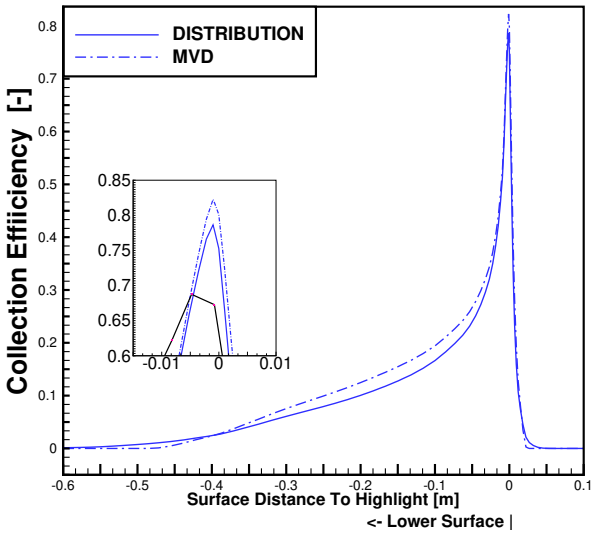
The influence of the droplet-size distribution is then shown in Fig. 3.10b. Compared with the single-diameter calculation using the MVD of 92μm, the polydisperse case modifies the collection-efficiency distribution. The different droplet diameters respond differently to the aerodynamic flow field, resulting in broader impingement limits.

Figure 3.10c compares the Eulerian and Lagrangian formulations using the droplet-size distribution prescribed by the workshop. Close agreement is obtained between the two formulations over the complete impingement region.

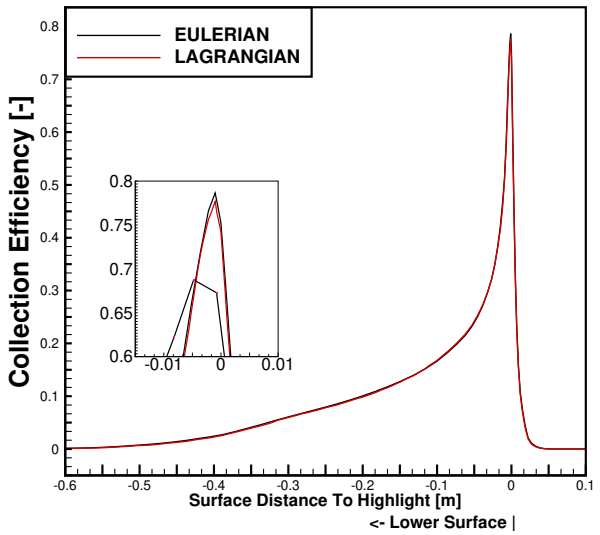
Finally, the influence of SLD effects is evaluated in Fig. 3.10d through comparison with the experimental collection-efficiency data of [141]. Without SLD effects, the predicted collection



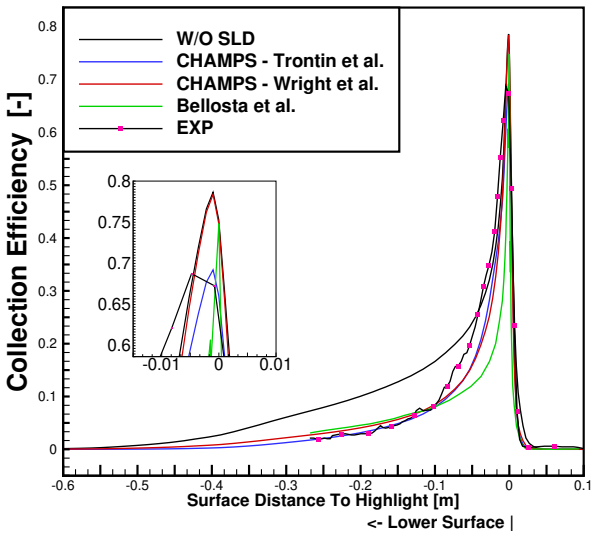
(a) Effect of droplet impact zone.



(b) Effect of droplet distribution on collection efficiency.



(c) Impingement results.



(d) SLD results.

Figure 3.10 Collection-efficiency results for Case IPW1-112: impact-zone treatment, droplet-size distribution, Eulerian–Lagrangian comparison, and SLD-model assessment.

efficiency is substantially higher than the experimental measurements. Accounting for post-impact mass loss reduces the retained water and improves agreement with the experimental distribution.

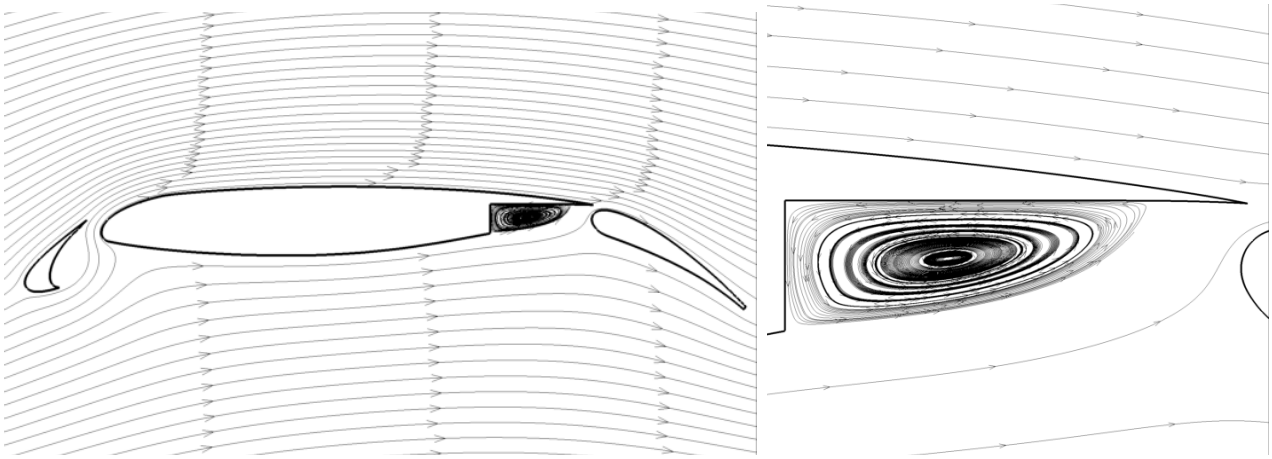
The model of Wright et al. [64] predicts little mass loss near the stagnation region, where the droplets impact the surface nearly normally. Consequently, the collection efficiency remains close to the solution without SLD effects near the stagnation point. As the impact becomes increasingly tangential farther downstream, the predicted mass loss increases and progressively reduces the collection efficiency. In contrast, the model of Trontin et al. [68] predicts mass loss near the stagnation region as a function of the impact conditions, resulting in a lower peak collection efficiency and closer agreement with the experimental measurements. Both models progressively reduce the collection efficiency away from the stagnation region and improve agreement with the experimental distribution.

The CHAMPS predictions are also compared with the results reported by [60], which employ the SLD model of Wright et al. [64]. For this case, CHAMPS provides closer agreement with the experimental collection-efficiency measurements. An important methodological difference lies in the evaluation of the collection efficiency from the computed trajectories. The local streamtube approach employed in the present methodology may contribute to the improved agreement obtained with CHAMPS.

3.4.3 Case IPW1 - 122: Three Element Airfoil

Case IPW1-122 also corresponds to a configuration for which experimental collection-efficiency measurements are available under SLD conditions. The three-element airfoil presents a particularly challenging impingement problem because of the recirculation region in the cove behind the main element, illustrated in Fig. 3.11. Disparities in the predicted collection efficiency were observed among the IPW1 participants, particularly on the main and flap components (Fig. 3.12). Figures 3.13a–3.13c show the CHAMPS Eulerian and Lagrangian results using the droplet-size distribution prescribed by the workshop. Overall, the two formulations exhibit good agreement. The Eulerian formulation predicts some impingement toward the end of the main element on both sides of the geometry. This behavior is associated with non-physical values of the droplet volume fraction resulting from dispersion-induced oscillations. The slat collects the largest fraction of the incoming droplets because of its upstream position. This effect is particularly pronounced on the pressure side at the prescribed angle of attack of 4°. Consequently, substantially less impingement is observed on the suction sides of the main and flap elements. The remaining differences between the Eulerian and Lagrangian predictions may be related to numerical dissipation in the Eulerian formulation,

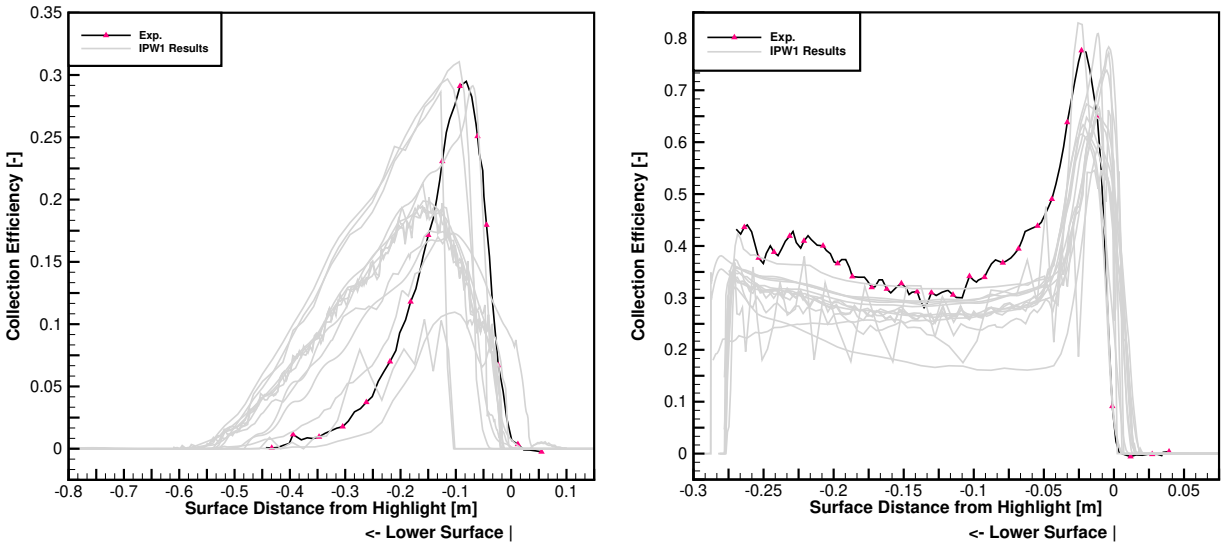
particularly near the cove behind the main element, as discussed in [58]. These differences are expected to decrease with mesh refinement, as illustrated in Appendix B.1.



(a) Three Element Airfoil Geometry.

(b) Recirculation zone in the cove of the main element.

Figure 3.11 Description of Case IPW1 - 122.



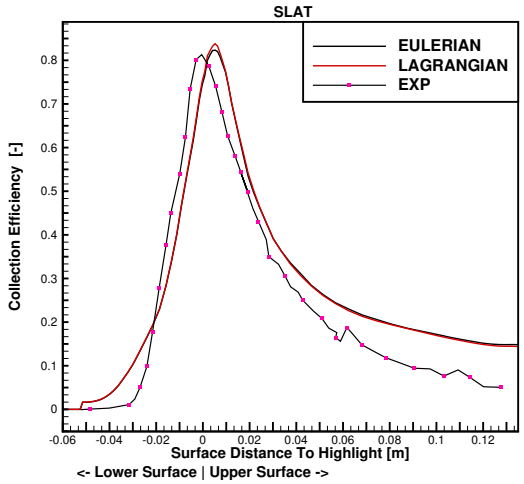
(a) Main component results.

(b) Flap component results.

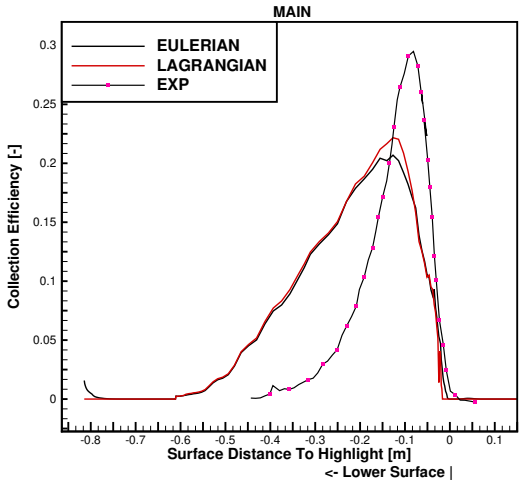
Figure 3.12 Aggregated impingement results of all IPW1 participants for Case IPW1-122.

The SLD results are presented in Fig. 3.13d–3.13f. On the slat, the predicted collection efficiency agrees well with the experimental measurements in terms of both the peak value and the impingement limits. On the main element, however, the collection efficiency is significantly underestimated. Better agreement is obtained on the flap, where the impingement

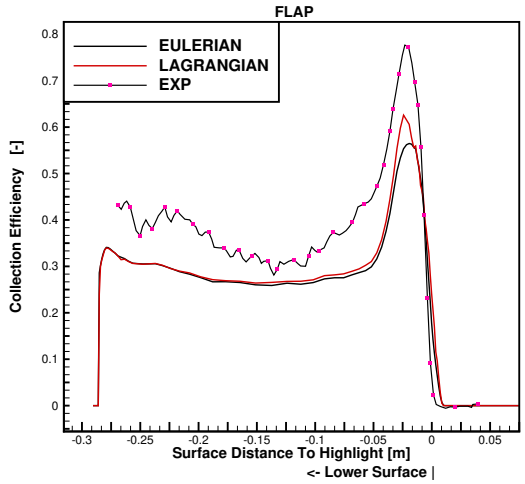
limits are well predicted, although the peak collection efficiency remains underestimated. When secondary-droplet re-impingement is included using the methodology presented in Section 3.3.3, the peak collection efficiency increases on both the main and flap elements, resulting in slightly closer agreement with the experimental measurements. Similar discrepancies between numerical predictions and the experimental measurements have also been reported for this case in the literature [60, 142], particularly on the downstream elements.



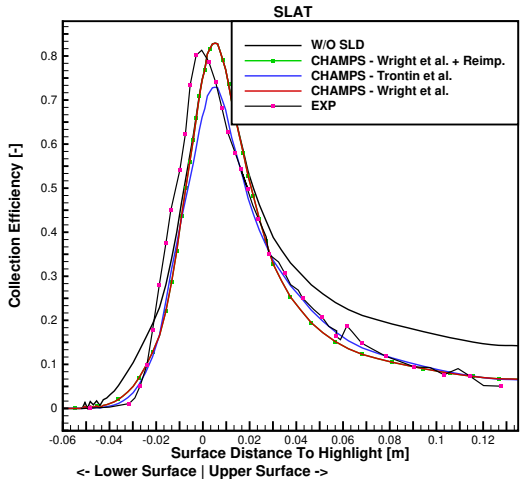
(a) Impingement results on the Slat.



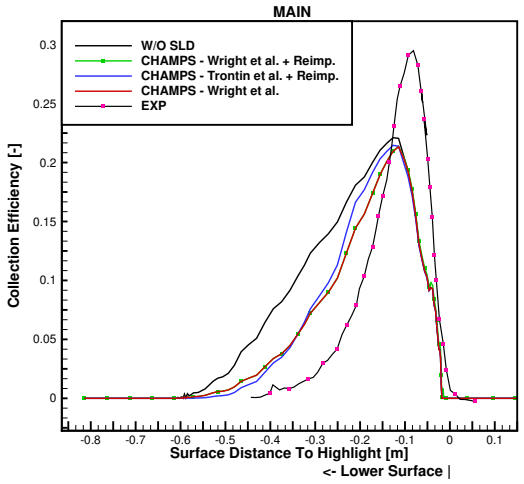
(b) Impingement results on the Main.



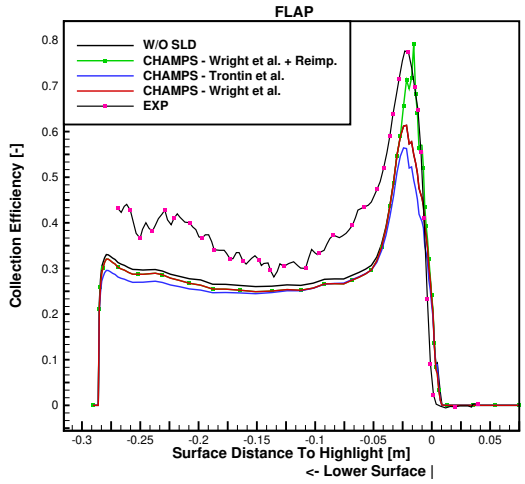
(c) Impingement results on the Flap.



(d) SLD results on the Slat.



(e) SLD results on the Main.



(f) SLD results on the Flap.

Figure 3.13 Collection-efficiency results for Case IPW1-122. The upper row compares Eulerian and Lagrangian impingement predictions; the lower row assesses the SLD models and secondary-droplet re-impingement.

3.4.4 Case HLPW5 - 2.1: CRM Wing-Body

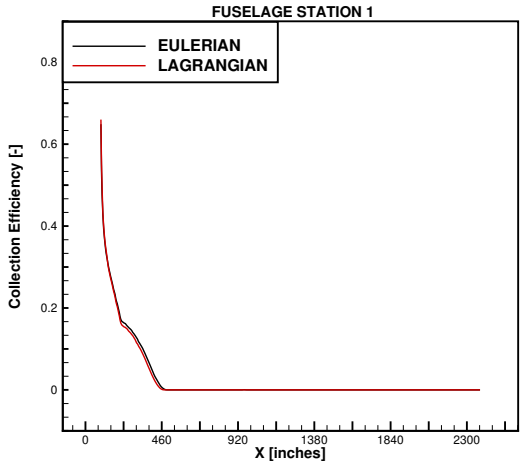
The final case considers the Common Research Model (CRM) wing–body configuration to assess the Lagrangian methodology on a larger and more complex 3D geometry. The collection efficiency is extracted along five spanwise wing cuts and one fuselage cut, whose locations are shown in Fig. 3.14. The corresponding results for Case HLPW5 - 2.1 are presented in Fig. 3.15, where the Lagrangian predictions are compared with the Eulerian formulation.

For the fuselage, as shown in Fig. 3.15a, the peak collection efficiency is obtained near the nose and progressively decreases along the surface. The collection-efficiency distributions along the wing are presented in Figs. 3.15b–3.15f. The distributions are asymmetric between the pressure and suction sides because of the prescribed angle of attack of 6°. Higher values of β are also observed near the wing–fuselage junction, indicating increased droplet impingement in this region. The distribution evolves along the span because of the swept-wing geometry and the corresponding variation in the local flow field. In particular, while impingement is predominantly observed on the pressure side at the inboard sections, the relative distribution between the pressure and suction sides changes farther outboard.

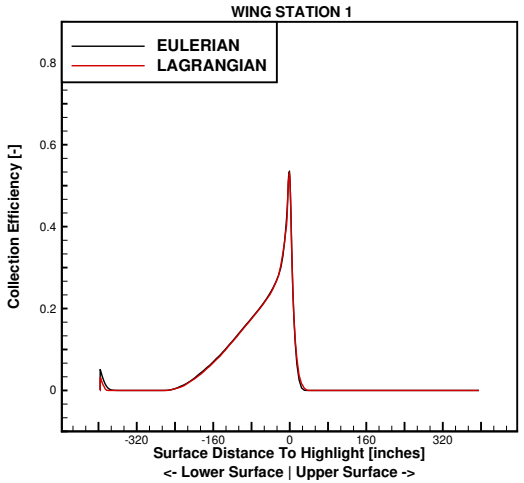
Overall, the Lagrangian and Eulerian formulations provide consistent collection-efficiency distributions across the fuselage and the five wing cuts. Good agreement is obtained in both the magnitude and spatial distribution of the collection efficiency, demonstrating the consistency of the two droplet formulations when applied to a large 3D configuration.

(a) Collection-efficiency cuts along the wing span. (b) Collection-efficiency cut on the fuselage.

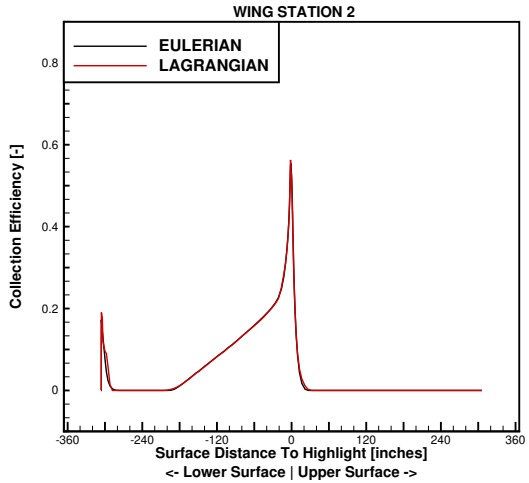
Figure 3.14 Locations of the collection-efficiency cuts on the CRM wing and fuselage.



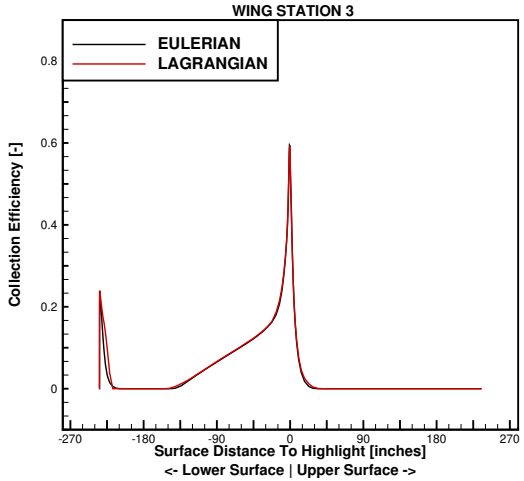
(a) Impingement results on the Fuselage.



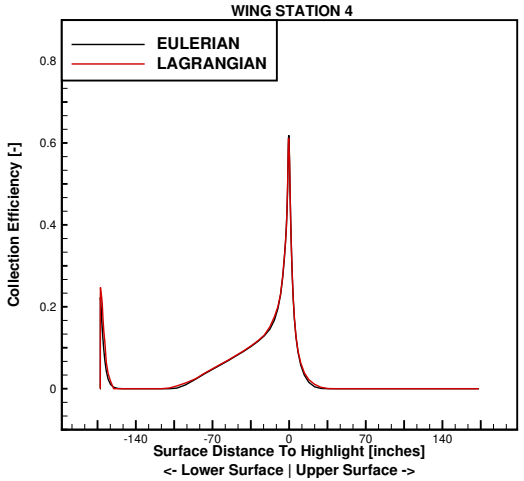
(b) Impingement results on the Wing-1.



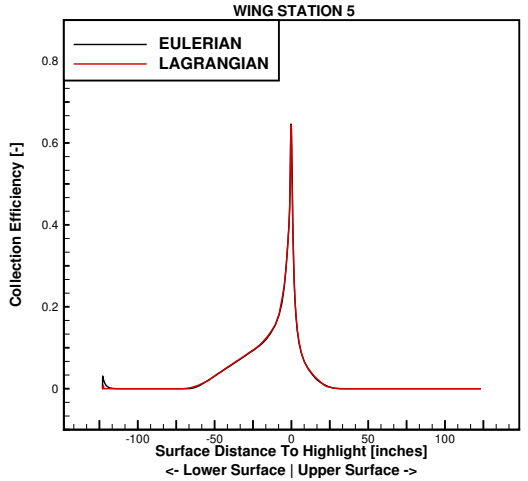
(c) Impingement results on the Wing-2.



(d) Impingement results on the Wing-3.



(e) Impingement results on the Wing-4.



(f) Impingement results on the Wing-5.

Figure 3.15 Eulerian and Lagrangian collection-efficiency results for Case HLPW5-2.1 at the fuselage and five spanwise wing cuts.

3.5 Conclusion

This chapter presents the development of a Lagrangian particle-tracking methodology for aircraft ice accretion simulations within the CHAMPS framework. The methodology combines face-to-face trajectory integration with a ray-tracing-enhanced intersection technique for robust and efficient tracking on unstructured meshes. An adaptive seeding strategy is introduced to progressively refine the droplet cloud in regions associated with surface impingement while reusing previously computed trajectories. The resulting workload is distributed according to estimates of the computational cost of individual trajectories to improve parallel execution on distributed-memory systems. For the evaluation of the collection efficiency, auxiliary trajectories are introduced locally around each seeding droplet to construct a local streamtube, while a neighboring-face interpolation treatment improves the coverage of the resulting surface distribution without requiring additional trajectories. SLD effects are also incorporated through mass-loss models and the tracking of secondary droplets generated by splashing.

The methodology is assessed using configurations from the AIAA Ice Prediction Workshops and the Fifth AIAA High-Lift Prediction Workshop. The parallel scalability study demonstrates a parallel efficiency of approximately 94% on 16 nodes relative to the two-node baseline. For the IPW1 configurations, the Lagrangian and Eulerian formulations provide consistent collection-efficiency distributions, while the inclusion of SLD effects improves agreement with the available experimental measurements. Finally, application to the CRM wing–body configuration demonstrates consistent Eulerian and Lagrangian predictions across the fuselage and multiple spanwise wing sections. Overall, the results demonstrate the capability of the developed methodology to provide robust and computationally efficient Lagrangian droplet-impingement predictions for complex 3D configurations.

CHAPTER 4 Article : Three-Dimensional Multilayer Ice Accretion Using a Level-Set Method

This chapter presents a continuous level-set framework for simulating 3D ice accretion within a quasi-steady, multilayer approach. In contrast to traditional Lagrangian surface-node displacement methods, which may suffer from mesh distortion and topological limitations, the proposed method relies on an implicit interface-capturing formulation that naturally represents complex geometric features and topological changes. The continuous level-set equation is solved on adapted meshes using a conservative formulation combined with a semi-implicit Rosenbrock-Wanner time integration scheme. A local interface-reconstruction procedure is introduced to monitor and control the accreted ice volume without requiring full surface extraction. The zero level set is then extracted using an external library (MMG) to generate a conformal surface mesh. To reduce numerical oscillations while preserving the overall ice shape and volume, a smoothing filter is applied to the extracted surface. The proposed methodology is assessed on representative icing configurations, including a manufactured test case and experimental cases from the first Ice Prediction Workshop (IPW1). The results demonstrate good agreement with reference and experimental data.

4.1 Introduction

Ice accretion on aircraft components remains a critical challenge for the aerospace industry. The resulting geometrical changes on lifting surfaces can significantly alter the aerodynamic characteristics of the aircraft. In particular, ice accretion leads to a reduction in aerodynamic efficiency, characterized by a decrease in the maximum lift coefficient ($C_{L,\max}$) and an increase in drag [143]. In more severe situations, however, ice accumulation can compromise aircraft controllability and potentially endanger flight safety.

To better understand and control these phenomena, numerical tools are increasingly employed. Advances in computational methods and high-performance computing (HPC) have significantly improved the capability of simulation frameworks to capture the complex multiphysics processes involved in ice accretion. As a result, numerical modeling has become an essential component of the aircraft design and certification process, enabling more reliable predictions of ice formation and its aerodynamic consequences [144].

Ice accretion is an unsteady and intricate multiphysics phenomenon involving the interaction of aerodynamic flow, droplet dynamics, thermodynamic exchanges, and geometric evolution.

However, the characteristic time scales of ice growth are significantly larger than those of the underlying physical processes (e.g. flow fluctuations). As a result, the accretion process can be modeled using a quasi-steady framework in which the total icing exposure time is subdivided into N successive discrete steps [131]. At each step, the governing equations are solved assuming steady-state conditions, after which the geometry is updated. This procedure corresponds to the commonly adopted multilayer approach for ice accretion simulations. Figure 4.1 illustrates a typical icing simulation workflow. The simulation begins with a clean geometry, which can either be meshed to generate a body-fitted computational grid or embedded within a pre-existing background mesh in the case of immersed boundary approaches [31]. A flow solver is first used to compute the airflow around the geometry, often using the Reynolds-Averaged Navier-Stokes equations (RANS). This is followed by droplet tracking to determine the distribution of liquid water on the wetted surfaces. Subsequently, thermodynamic exchanges are evaluated to estimate the local ice mass accretion rate. Finally, the geometry is advanced using a time-accurate update algorithm to account for the newly formed ice shape. The resulting modified surface is then used to update the surrounding volume mesh accordingly.

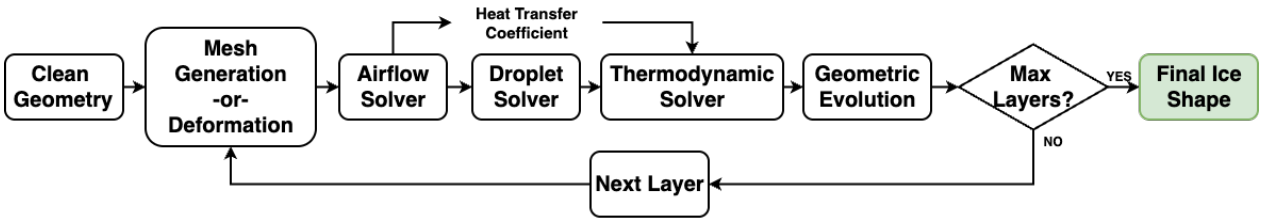


Figure 4.1 Schematic of the multilayer ice accretion framework.

Although all these steps remain active areas of research, the present work focuses on the last stage of the framework: the geometric evolution required to represent the accreted ice shape. This aspect has received considerable attention in recent years due to the inherent topological challenges associated with ice growth. Early approaches relied on Lagrangian displacement of surface nodes along their local normals [145], accompanied by a volume propagation of the deformation. While these methods are computationally efficient and relatively straightforward to implement, they encounter significant difficulties when complex ice shapes develop, particularly in the presence of concave and convex regions where mesh entanglement can occur. Such methods do not inherently resolve these topological issues. Although heuristic strategies can sometimes mitigate these problems in 2D configurations by removing problematic nodes or locally adjusting the geometry, extending these techniques to

3D becomes considerably more challenging due to the complexity of stitching and merging surface regions [86]. More recently, surface-wrapping techniques have been used to circumvent some of these topological limitations in complex ice-accretion simulations [38].

An increasingly popular alternative is to represent the evolving ice geometry implicitly using interface-capturing methods such as level-set formulations [146]. Level-set methods have been widely used to model moving-interface problems across a broad range of applications, including multiphase flows [147, 148], shape optimization [149], and forest-fire propagation [150], to name a few. In this approach, the interface is described by a level-set function ϕ, whose evolution is governed by an advection equation driven by a velocity field. In the context of aircraft icing, an early use of the level-set method was reported by [88], where it was employed to track ice-shedding trajectories rather than to model the accretion process itself. Subsequent developments extended the use of level-set methods to ice accretion. One of the earliest formulations in this direction was proposed by [28], and later extended by [2], who modified the icing velocity field while still relying on explicit Lagrangian tracking of surface nodes to extract the ice shape. These approaches were applied on RANS meshes, where strict mesh-resolution requirements and time-step constraints can limit the effectiveness of level-set evolution. More recently, [89] introduced a discrete formulation of the level-set method based on a Lagrangian prediction of the ice shape. While this approach proved effective, conservation of the accreted ice mass remained dependent on the quality of the underlying Lagrangian displacement, which can still suffer from mesh distortion and entanglement.

The present work addresses this gap in the literature by adopting a framework for solving the level-set equations on adapted meshes, thereby alleviating constraints imposed by the computational mesh. In particular, the level-set formulation is solved in its continuous form within a multilayer accretion framework. To the best of the authors' knowledge, this work represents one of the first extensions of a partial-differential-equation-based level-set formulation to 3D multilayer ice accretion. In addition, a robust and efficient time integration strategy, namely the Rosenbrock-Wanner scheme, is employed to accelerate the solution process while maintaining numerical stability. The proposed methodology enables consistent monitoring and control of the accreted ice volume through a local interface-reconstruction procedure, while simultaneously generating a conformal geometry suitable for subsequent simulation layers. Overall, the approach provides a robust, consistent, and scalable framework for 3D ice accretion simulations.

The remainder of this paper is organized as follows. Section 4.2 presents the CHAMPS icing framework and the overall simulation strategy. Section 4.3 introduces the level-set formulation, including the governing equations, velocity models, and time integration approach.

Section 4.4 describes the geometry extraction and smoothing procedures used to obtain conformal ice shapes. The proposed methodology is then assessed and discussed in Section 4.5 through representative test cases. Finally, conclusions and perspectives are provided in Section 4.6.

4.2 CHAMPS Icing Framework

The present work is implemented within CHAMPS [151], an in-house 3D unstructured cell-centered finite-volume solver developed at Polytechnique Montréal. Under the quasi-steady icing assumption, the total icing exposure time is discretized into N discrete layers (Fig. 4.1). For each layer, the flow field is computed using a RANS solver with Roe-based convective fluxes, a weighted least-squares gradient reconstruction, and a Venkatakrishnan limiter.

The aerodynamic solution is followed by Lagrangian droplet tracking [152] in order to determine the water catch distribution β on the wetted surfaces. Water droplets are released from an upstream seeding plane located in an unperturbed region of the flow ($u_{\text{local}}/u_\infty \approx 0.9$) and tracked until impact with the surface. Rather than assuming a single droplet diameter, a polydispersed droplet cloud is considered to better reproduce the experimental conditions. Multiple droplet size bins are simulated successively and then combined through a weighted procedure to obtain the final impingement distribution β.

Ice accretion is then computed using an iterative Messinger model [153] that resolves the thermodynamic exchanges at the surface. This model relies on mass and energy balances applied locally at each surface cell and embedded within an iterative loop to determine the incoming and outgoing water fluxes. The resulting solution yields the local ice mass accretion rate, $\dot{m}_{\text{ice}}$, which drives the geometric evolution of the ice layer.

The geometric update can be performed either through a Lagrangian surface displacement approach [145], in which the computed ice accretion rate $\dot{m}_{\text{ice}}$ is used to determine the displacement velocity of surface nodes, or through the level-set method [146] considered in the present work. In the latter case, the ice interface is tracked implicitly within an unsteady Eulerian framework and subsequently extracted to obtain a conformal surface mesh representing the updated ice geometry.

Finally, to complete the accretion layer, a remeshing stage is performed in order to generate a new volume grid for the subsequent icing step. In 2D, this remeshing stage relies on an in-house hyperbolic mesh generator [154]. For 3D simulations, the volume mesh is generated using Pointwise scripts [155].

4.3 Level-Set Ice Accretion Formulation

4.3.1 Level-Set Governing Equations and Initialization

The level-set method, introduced by [156], provides a robust framework for tracking moving interfaces through a Hamilton–Jacobi formulation. An illustration of the level-set field is given in Fig. 4.2. Its primary objective is to represent evolving boundaries implicitly within an unsteady Eulerian framework. Following [157], the level-set equation can be written in conservative form for the signed distance function ϕ (Eq. 4.1):

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \boldsymbol{V}) - \phi \nabla \cdot \boldsymbol{V} = 0,$$

$$\phi = \begin{cases} < 0 & \text{in the solid (ice) region,} \\ = 0 & \text{at the interface,} \\ > 0 & \text{in the fluid (air) domain.} \end{cases} \tag{4.1}$$

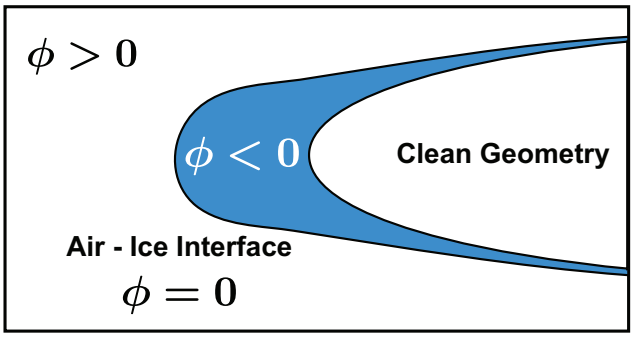


Figure 4.2 Representation of the level-set function ϕ. The ice region is defined by $\phi < 0$ (blue), while the surrounding air corresponds to $\phi > 0$. The interface is implicitly represented by the zero level set $\phi = 0$.

In Eq. 4.1, t denotes the physical time and $\boldsymbol{V}$ represents the interface velocity field. The equation is solved in an unsteady manner. A linear extrapolation boundary condition is applied at all domain boundaries along the level-set unit normal direction, defined as $\boldsymbol{n}_\phi = \frac{\nabla \phi}{\|\nabla \phi\|}$. The extrapolated value is obtained using the distance vector between the ghost-cell center and the adjacent interior cell center.

The initial level-set field is defined as a wall-distance field. A natural approach to compute this field consists in solving the steady form of the level-set equation [85], commonly referred to as the Eikonal equation (Eq. 4.2). This equation is solved using a pseudo-time τ scheme,

which vanishes upon convergence.

$$\frac{\partial \phi}{\partial \tau} + \boldsymbol{n}_{\phi} \cdot \nabla \phi = 1 \tag{4.2}$$

Owing to its unit source term, the Eikonal equation enforces a unit norm of the level-set gradient, thereby ensuring the signed-distance property. In this formulation, the extrapolation boundary condition is replaced by a Dirichlet condition imposed at the wall, $\phi_{\text{wall}} = 0$. Figure 4.3 shows the Eikonal contours around a NACA 23012 airfoil, which represent the resulting wall-distance field.

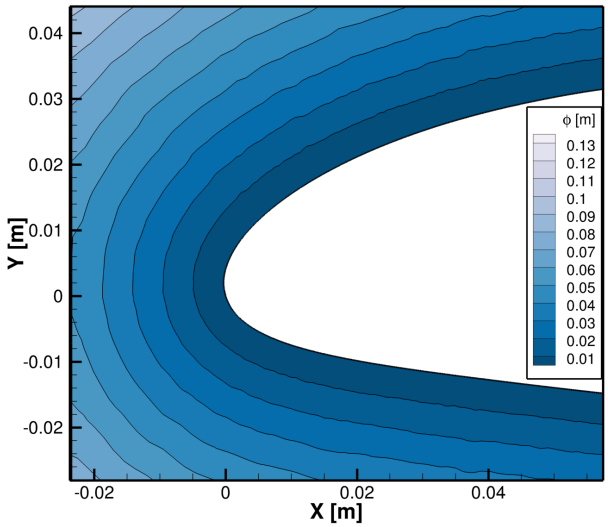


Figure 4.3 Contours of the level-set function ϕ obtained from the Eikonal equation, representing the wall-distance field.

4.3.2 Level-Set Advection: Velocity Formulation, Surface-Quantity Propagation, and Time Integration

The velocity field governing interface motion must be related to the ice accretion variables. Three main approaches have been proposed in the literature and are summarized in Table 4.1. The work of [28] first introduced a surface-based icing-velocity formulation (Eq. 4.3), in which the interface velocity depends exclusively on surface quantities. In this formulation, the three velocity components computed at the surface must be propagated throughout the level-set computational domain. In Eq. 4.3, $\dot{m}_{ice}$ denotes the ice mass accretion rate and ρ_{ice} the ice density.

$$\boldsymbol{V} = \frac{\dot{m}_{ice}}{\rho_{ice} A_{wall}} \boldsymbol{n}_{wall} \tag{4.3}$$

The approach proposed by [2] later pointed out that this formulation introduces an implicit dependency on the local surface area, which should evolve with the deformation of the interface. A flux-based velocity formulation (Eq. 4.4) was therefore proposed, in which the ice accretion rate is first propagated within the volume and then locally adjusted according to a characteristic area. This formulation is intended to improve local volume conservation.

$$\boldsymbol{V} = \frac{\dot{m}_{ice}}{\rho_{ice}\, \Delta A_{local}} \boldsymbol{n}_{\phi} \tag{4.4}$$

where the projected local area ΔA_{local} is defined as

$$\Delta A_{local} = \sum_{facets} |\boldsymbol{n}_f \cdot \boldsymbol{n}_{\phi}|\, \Delta A_{facet} \tag{4.5}$$

In Eq. 4.5, $\boldsymbol{n}_f$ denotes the facet normal and ΔA_{facet} the corresponding facet area.

The formulation introduced by [29] constitutes a further improvement over that of [28]. In this approach, the velocity magnitude remains constant and is propagated along the ϕ-normals, while its direction is computed locally from the initial ϕ-normals. This feature is particularly attractive in the present context, as it helps preserve the signed-distance property of the level-set field and thereby limits the need for re-initialization [157]. For this reason, this formulation is adopted in the present work.

$$\begin{aligned} \boldsymbol{V} &= \frac{h_{ice}}{\Delta t_{icing}} \boldsymbol{n}_{\phi} = \frac{\dot{m}_{ice}}{\rho_{ice} A_{wall}} \boldsymbol{n}_{\phi}, \\ h_{ice} &= \frac{\Delta t_{icing}\, \dot{m}_{ice}}{\rho_{ice}\, A_{wall}}. \end{aligned} \tag{4.6}$$

Table 4.1 Icing Velocity Formulations

Icing Velocity	Norm	Direction
Pena et al. [28]	$\dot{m}_{ice}/(\rho_{ice} A_{wall})$	$\boldsymbol{n}_{wall}$
Bourgeault-Côté et al. [154]	$\dot{m}_{ice}/(\rho_{ice}\, \Delta A_{local})$	$\boldsymbol{n}_{\phi}$
Lavoie et al. [29]	$h_{ice}/\Delta t_{icing}$	$\boldsymbol{n}_{\phi}$

Since the chosen interface velocity is defined from surface quantities, such as $\dot{m}_{ice}$ and h_{ice}, these must be extended into the volume. This is achieved by solving a convection equation [154] that propagates the variable along the ϕ-normals:

$$\frac{\partial \varphi}{\partial \tau} + \nabla \cdot (\varphi \boldsymbol{n}_\phi) - \varphi \nabla \cdot \boldsymbol{n}_\phi = 0 \tag{4.7}$$

Here, φ denotes the propagated quantity. A Dirichlet boundary condition is imposed at the wall, while a zero-gradient Neumann condition is prescribed at the far field. An example of the resulting propagated field is shown in Fig. 4.4. It should be noted that the Eikonal and propagation equations described above are not the only possible approaches for initializing the level-set field and extending surface information into the volume. In the present study, a geometric wall-distance method based on [158] was also tested. With this approach, the wall distance is computed in the volume, while the propagated quantity is assigned from the closest wall cell. Both approaches yielded similar results.

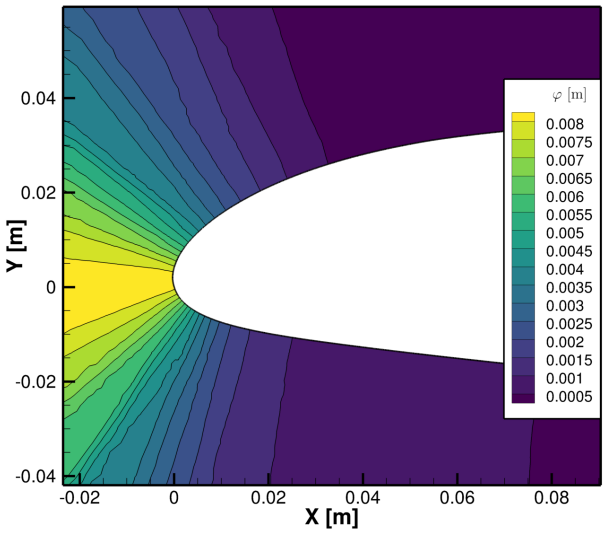


Figure 4.4 Contours of the propagated scalar field φ, here representing the ice thickness extended from the clean surface into the volume. The field is transported along the ϕ-normal direction.

Beyond the choice of interface-velocity formulation, the time-integration strategy also has a direct impact on robustness and computational efficiency. Most existing level-set-based icing approaches rely on explicit time integration schemes [28, 29], which impose restrictive stability constraints on the time step, or on dual time stepping methods [2], which require additional pseudo-time iterations and thereby increase computational cost.

In contrast, the present work employs a Rosenbrock–Wanner (ROW) semi-implicit time integration scheme [159] recently introduced in CHAMPS and presented at the Cambridge Unsteady Flow Symposium (CUFS 2026) [160]. This approach improves stability and allows for larger time steps. In the present context, this feature is particularly advantageous since

the multilayer ice accretion framework involves only small surface deformations at each accretion step. As a result, the signed-distance property of the level-set field is better preserved, which significantly reduces the need for re-initialization in the accretion cases considered here. The two-stage scheme is given in Eq. 4.8:

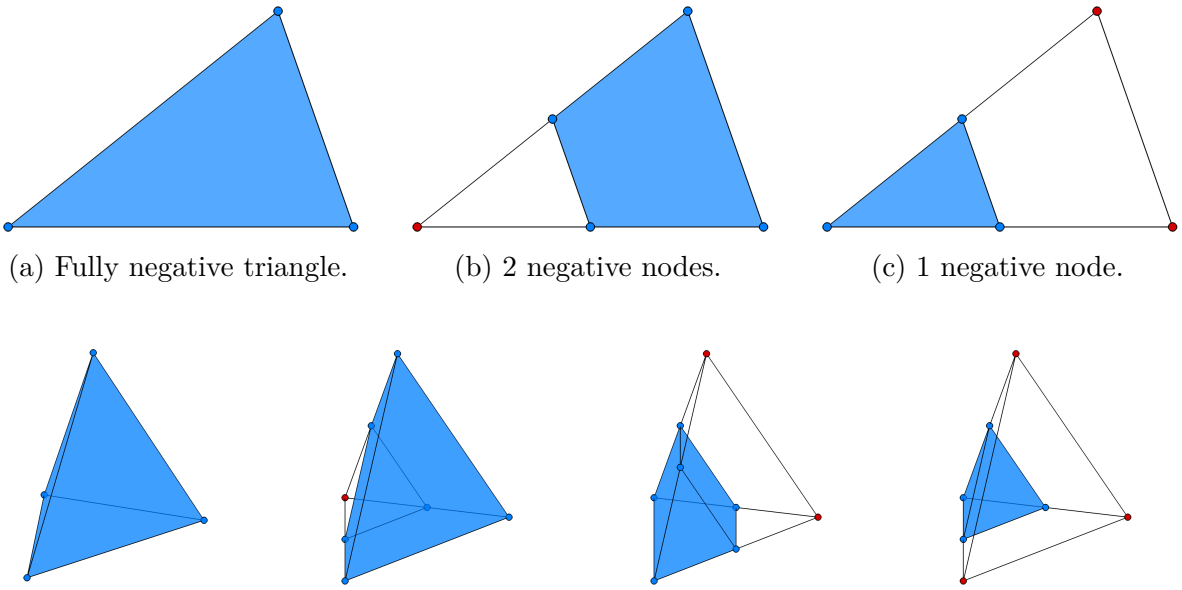
$$\begin{gathered}
\phi^{n+1} = \phi^n + \sum_{j=1}^{s} m_j S_j \\
\left(\frac{\boldsymbol{I}}{\gamma \Delta t} + \boldsymbol{J}\right)^n S_i = -R\left(\phi^n + \sum_{j=1}^{i-1} a_{ij} S_j\right) \\
\gamma = 1 - \frac{1}{\sqrt{2}} \\
\boldsymbol{A} = \begin{pmatrix} 0 & 0 \\ 4 - 8\gamma & 0 \end{pmatrix}, \qquad \boldsymbol{m}^T = \begin{pmatrix} \frac{1}{\gamma}\left(1 - \frac{1}{8\gamma}\right) \\ \frac{1}{8\gamma^2} \end{pmatrix}.
\end{gathered} \tag{4.8}$$

where $R(\phi)$ denotes the spatial residual associated with the semi-discrete level-set equation, $\boldsymbol{J} = \partial R/\partial \phi$ is the corresponding Jacobian matrix, S_i are the stage increments. This formulation resembles a two-stage Runge–Kutta scheme, with the distinction that the Jacobian matrix appears on the left-hand side, leading to the solution of a linear system at each stage. In the present work, this system is solved using the Generalized Minimal Residual (GMRES) method [161].

4.3.3 Ice Mass Computation and Local Interface Reconstruction

The global ice mass is evaluated at each time step by integrating the volume of the region where $\phi < 0$, allowing the simulation to terminate at the prescribed total ice mass. In the present work, variable ice density is not currently considered; therefore, the mass is assumed to be directly proportional to the accumulated volume. Cells are first classified according to the sign of ϕ at their nodes: cells located entirely within the ice region contribute fully to the volume, whereas intersected cells require a partial volume evaluation. In the latter case, a local interface reconstruction is performed only within the intersected cells, following the different intersection configurations illustrated in Fig. 4.5, thereby avoiding the cost of a full surface extraction. This local treatment is sufficient for volume estimation and remains inexpensive to evaluate at each time step. The reconstruction is carried out on simplicial elements. In 2D, interface intersections within triangles are determined by linear interpolation along the edges, forming a polygon whose area is then evaluated. In 3D, a

similar procedure is applied to tetrahedra, where the intersection points define a clipped polyhedron whose volume is computed. The use of simplicial elements, as imposed by the MMG extraction library [162], also proves advantageous by limiting the number of possible intersection configurations to be considered.



(a) Fully negative triangle. (b) 2 negative nodes. (c) 1 negative node.

(d) Fully negative tetrahedron. (e) 3 negative nodes. (f) 2 negative nodes. (g) 1 negative node.

Figure 4.5 Illustration of local interface reconstruction within simplicial elements based on the sign of the level-set function ϕ. Red nodes denote positive values ($\phi > 0$), whereas blue nodes denote negative values ($\phi < 0$). The interface $\phi = 0$ is reconstructed by linear interpolation along the intersected edges. In 2D, the negative subregion is either the full triangle, a quadrilateral, or a triangle. In 3D, the negative subregion is either the full tetrahedron, a truncated tetrahedron, a clipped polyhedron, or a smaller tetrahedral subregion. These clipped regions are used to evaluate the local ice area or volume contribution without requiring a full surface extraction.

4.4 Geometry Extraction and Smoothing

One of the main limitations reported in [154] was related to the RANS mesh, which constituted a restrictive aspect of the level-set simulation framework. Although the mesh was sufficiently refined to satisfy the $y^+ < 1$ requirement of the Spalart–Allmaras turbulence model, it was not well suited for the level-set formulation. In particular, the very fine near-wall cells imposed highly restrictive time-step constraints for the level-set advection, making the evolution of the interface computationally demanding. Furthermore, the geometry tracking

was performed using a Lagrangian approach, which required heuristic procedures to handle topological events such as merging or breaking of the interface.

To address these limitations, the present study adopts a framework that generates adapted meshes for solving the level-set formulation. The constructed mesh accounts for both the geometry extents and the maximum ice thickness predicted by the thermodynamic model, thereby limiting the overall element count. In order to improve the accuracy of the continuous level-set tracking and extraction, the mesh is refined in regions where ice accretion occurs to reach a user-prescribed target element size. These regions are identified based on the condition $\dot{m}_{ice} > 0.0$. In addition, a set of transition layers is introduced to ensure a smooth transition between iced and clean regions of the geometry. The auxiliary grid is generated using MMG in 2D cases and through a POINTWISE script [155] in 3D cases. An example of this adapted mesh is shown in Figure 4.6.

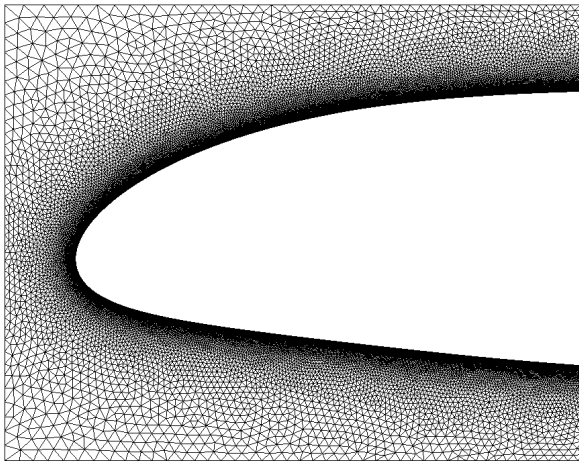


Figure 4.6 Adapted unstructured mesh around the iced surface of the geometry, showing local refinement in order to accurately capture the interface in the level-set framework.

An important aspect of the procedure is the preservation of the original clean geometry. Based on the previously defined accretion criterion, nodes and edges belonging to regions without ice accumulation are tagged as required entities so that they remain unaltered during the MMG adaptation/extraction process. Moreover, as the adapted surface mesh does not share the same discretization as the original surface in the iced regions, an interpolation step is required. This is performed using a linear interpolation by locating the closest element on the original surface mesh.

Once this mapping step is completed, the level-set framework is initiated by computing the

eikonal ϕ and normals, propagating surface variables into the volume along these normals, and subsequently advecting the unsteady level-set field. Prior to extracting the geometry, the level-set values must be defined at mesh nodes since MMG operates on nodal values. Therefore, a final interpolation step is performed to transfer the cell-centered solution of the CHAMPS solver to nodal values. The zero level set is then extracted to obtain the updated surface geometry, which is subsequently used to generate the volume grid for the next accretion layer.

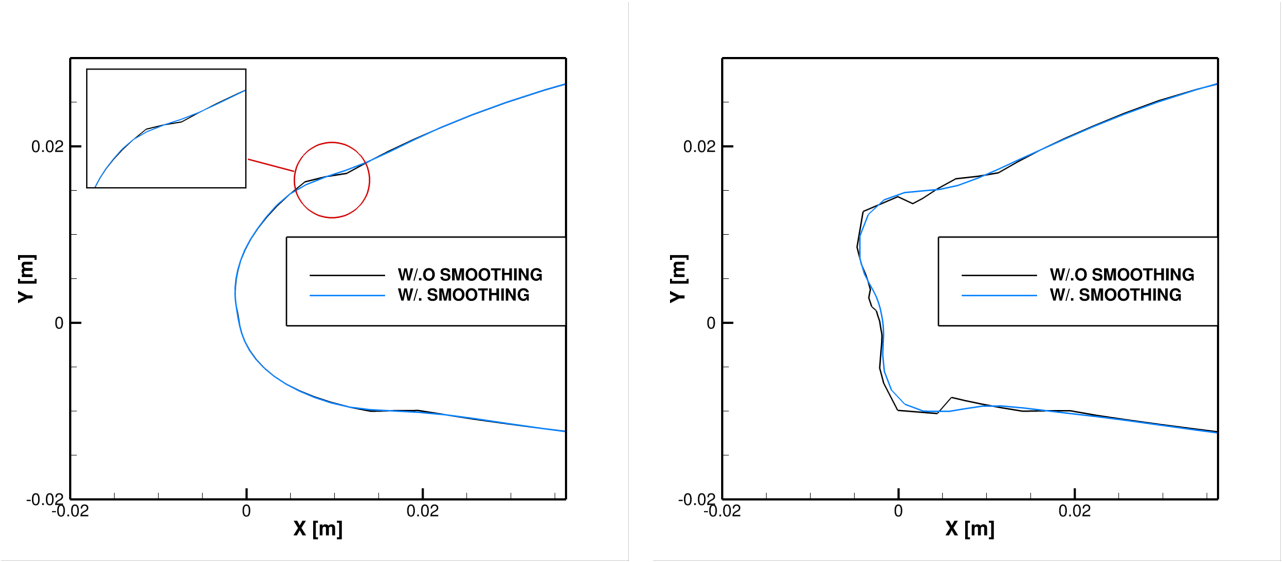
Surface perturbations may arise during several stages of the ice accretion simulation process, including the level-set solution, the geometry extraction procedure, and even during the computation of the droplet collection efficiency. As reported in [163], these perturbations can introduce small geometric oscillations on the extracted interface. If left untreated, such oscillations may amplify during subsequent accretion layers and eventually lead to numerical instabilities or undesirable topological artifacts in the evolving ice geometry.

To mitigate these issues, a smoothing step is applied to regularize the surface while preserving the physically relevant geometric features of the ice shape. In the case of surface variables such as the collection efficiency, the smoothing procedure must also preserve the overall distribution and magnitude of the quantities involved. An effective approach reported in [163] is the smoothing filter proposed in [6], which is well suited for this purpose. Unlike classical Gaussian smoothing, which tends to introduce geometric shrinkage and consequently alters the predicted ice volume, the filter reported in [6] was specifically designed to minimize such shrinkage effects while still removing high-frequency surface noise.

This smoothing method can be interpreted as a low-pass filter applied to the mesh geometry. It attenuates high-frequency perturbations while preserving the large-scale structure of the surface. The algorithm operates through two successive passes of Laplacian smoothing. In the first pass, node positions are updated using a positive scaling factor λ, which produces a smoothing effect similar to standard Laplacian smoothing. In the second pass, a corrective update is applied using a negative factor μ, which counteracts the shrinkage introduced during the first step. The combined effect of these two passes suppresses high-frequency oscillations while maintaining the global shape and volume of the surface. The pass-band frequency controlling the filter behavior is defined as $k_{pb} = \frac{1}{\lambda} + \frac{1}{\mu}$. The same filtering procedure can also be applied to surface fields when regularization of field variables is required. In this work, the pass-band frequency was set to $k_{pb} = 0.01$, with $\lambda = 0.33$. The number of smoothing iterations ranged from 2 to 50.

Through this two-stage filtering process, this smoothing filter effectively regularizes the extracted surface while preserving the geometric integrity of the predicted ice shape, making

it particularly suitable for multilayer ice accretion simulations where accumulated numerical noise could otherwise degrade the solution quality. An example is presented in Figure 4.7.



(a) Smoothing effects on the first ice layer.

(b) Smoothing effects on the final ice shape.

Figure 4.7 Comparison of the effect of [6] smoothing on the extracted ice surface. Smoothing improves interface regularity in the first layer and prevents the accumulation of geometric distortions in the final ice shape.

4.5 Results

A primary objective of the present study is to assess the proposed level-set-based framework for ice accretion prediction on representative benchmark configurations, while also highlighting its advantages over a Lagrangian node-displacement approach in the presence of complex geometric features. To this end, a manufactured three-horn test case from [29] is first considered to illustrate the ability of the level-set formulation to handle intricate interface evolutions and topological complexity that are more difficult to capture with explicit surface-node displacements. The numerical results are then evaluated on four cases drawn from the first Ice Prediction Workshop (IPW1, [26]), namely Cases 241, 242, 361, and 362, as summarized in Table 4.2. The Ice Prediction Workshop (IPW) [26, 27] is a major international initiative organized by the American Institute of Aeronautics and Astronautics (AIAA) to evaluate and compare numerical tools for aircraft icing prediction against experimental data. It provides well-documented benchmark cases covering both 2D and 3D geometries under a variety of icing conditions.

Cases 241 and 242 correspond to 2D icing configurations on a NACA 23012 airfoil, whereas Cases 361 and 362 extend the analysis to a 3D configuration involving a swept NACA 0012 wing with a sweep angle of 30°. Together, these benchmark cases make it possible to evaluate the robustness of the proposed method on both 2D and 3D geometries, as well as its ability to capture a range of ice-shape features under different accretion conditions. Whenever available, the numerical results are compared with the experimental data and reference results provided within the IPW1 database. The experimental results are presented in the form of a maximum combined cross section (MCCS), obtained by extracting streamwise slices from the 3D scanned ice shape at multiple spanwise locations and combining these cross-sections into a single representation [26]. Both the 3D scan and the corresponding MCCS data are available on the workshop website.

Table 4.2 Summary of the IPW1 benchmark cases.

Parameter	241	242	361	362
Geometry	2D NACA 23012	2D NACA 23012	3D Swept NACA 0012	3D Swept NACA 0012
Icing Time $[min]$	5.0	5.0	20.0	20.0
LWC $[g/m^3]$	0.42	0.81	0.5	0.5
MVD $[\mu m]$	30.0	15.0	34.7	34.7
Ma [-]	0.32	0.31	0.32	0.31
AOA [°]	2.0	2.0	0.0	0.0
C_{ref} $[inch]$	18	18	36	36
T_{static} $[K]$	250.15	266.05	257.0	266.0
Re [-]	$3.8 \cdot 10^6$	$3.4 \cdot 10^6$	$7.2 \cdot 10^6$	$6.7 \cdot 10^6$

4.5.1 Three-Horns Ice Accretion Case

In order to highlight the capabilities of the level-set method, the first test case considered is the three-horns configuration initially introduced in [82]. This manufactured case features three distinct ice horns, producing highly concave and convex regions that challenge interface tracking methods. An ice thickness of 0.02 m is uniformly imposed on all surface nodes with $x < 0.05$ m.

Figure 4.8 compares the interface obtained from level-set contour extraction with that resulting from a Lagrangian node displacement approach. The latter exhibits clear limitations when handling the geometric complexity of the case. In particular, mesh entanglement is observed in regions of strong curvature, leading to local inconsistencies and ice mass loss, which explains the mismatch between the two predicted fronts. In contrast, the level-set method

produces a smooth and conformal surface representation that remains robust in highly distorted regions. This allows for a consistent geometry extraction suitable for subsequent accretion layers, even in areas where the horns partially merge.

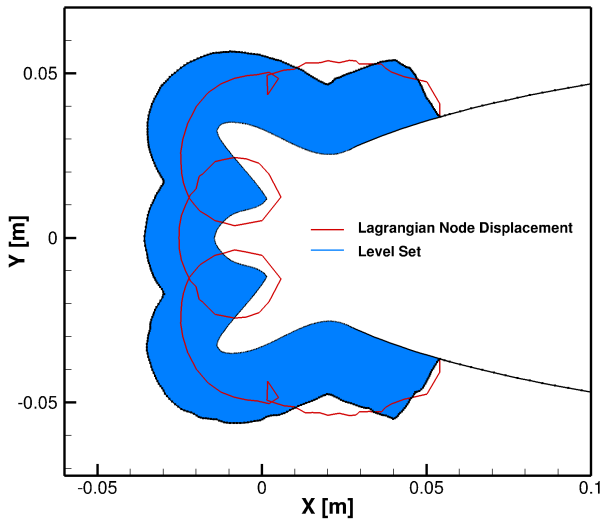
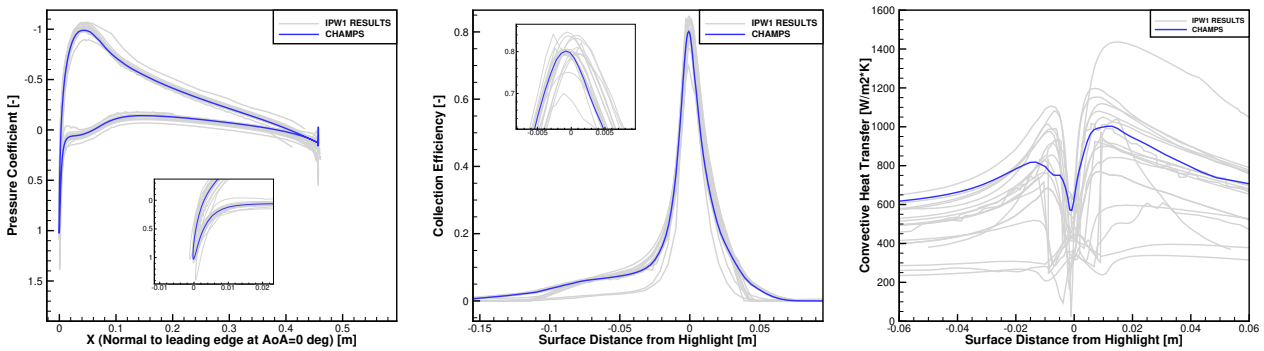


Figure 4.8 Comparison of the ice fronts obtained using Lagrangian node displacement and the proposed level-set approach for the manufactured three-horn configuration.

4.5.2 Case IPW1 241

The second case considered in this study corresponds to a rime ice case from IPW1. Rime ice forms when the temperature is significantly below the freezing point, such that water droplets freeze immediately upon impact. This leads to streamlined ice shapes. Figure 4.9 presents the icing-related variables and compares them with the IPW1 participant results. Overall, the predicted distributions are in good agreement with the participant envelope. Figure 4.10 shows the ice shape obtained for this case, again compared with the IPW1 results and with the experimental MCCS. The level-set solution agrees well with the experimental maximum ice height at the leading edge, with some mismatch on the upper and lower surfaces. These differences may be partly attributed to experimental uncertainties. Stochastic ice-accretion solvers reported in the literature, such as [164], may provide improved predictions in these regions. Figure 4.10 also shows the convergence of the ice shape with an increasing number of accretion layers for the level-set method, using 1, 10, and 20 layers. The solution is seen to converge by 10 layers, while the differences between 10 and 20 layers remain negligible.

Ice shape comparisons remain difficult to assess in a fully quantitative manner, since the agreement between numerical and experimental accretions is often judged not only through



(a) Pressure coefficient. (b) Collection efficiency. (c) Convective heat transfer.

Figure 4.9 Comparison of the predicted aerodynamic and icing-related surface quantities for IPW1 Case 241 with the range of results submitted by IPW1 participants.

global geometric measures but also through more qualitative considerations. In practice, validation therefore frequently relies on a combination of visual inspection and a limited number of representative geometric indicators. For the present rime ice case, no experimental ice mass data is available, which prevents the use of a direct mass-based comparison metric. Under these conditions, the maximum ice height at the leading edge constitutes a particularly relevant quantity for comparison, as it provides a simple and physically meaningful measure of the accretion. The values reported in Table 4.3 show that this metric is progressively improved as the number of accretion layers increases, with the relative error decreasing from 6.47% for the single-layer level-set prediction to 1.68% for the 10 layers simulation and 0.70% for the 20 layers simulation. These results indicate a clear convergence of the predicted rime ice shape toward the experimental reference with increasing temporal resolution of the multilayer accretion process.

Table 4.3 Comparison of the maximum ice height relative to the clean geometry.

Config.	Ice height [mm]	Relative error [%]
Exp.	10.87	–
01 layer	10.16	6.47
10 layers	10.69	1.68
20 layers	10.79	0.70

4.5.3 Case IPW1 242

The third case considered is a glaze ice case, for which the static temperature is not sufficiently low to ensure immediate freezing upon droplet impact. As a result, a liquid water film

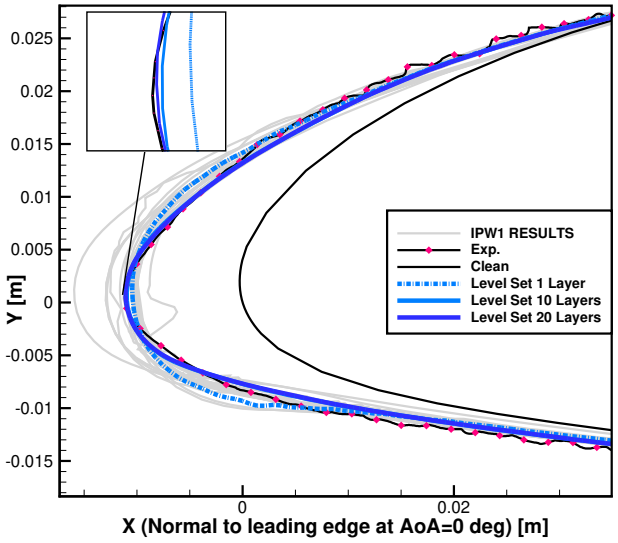
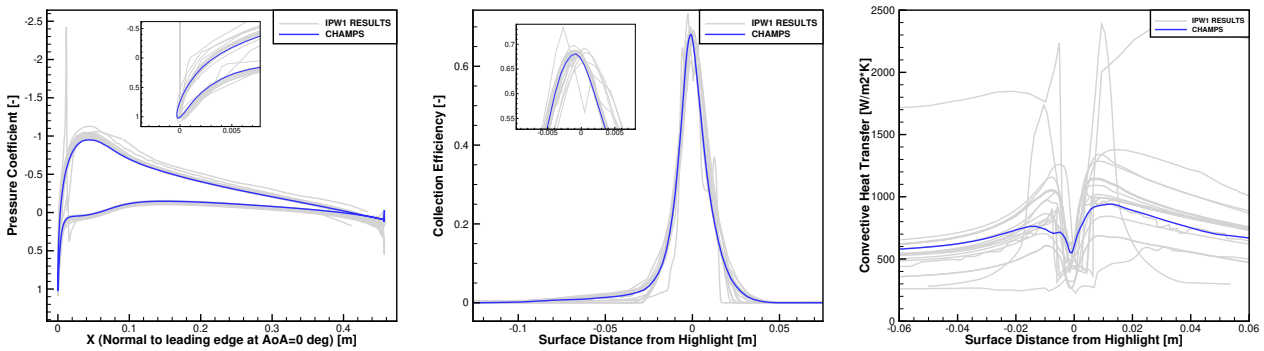


Figure 4.10 Predicted ice shapes for IPW1 Case 241. The clean and experimental profiles are shown with the IPW1 participants' results. CHAMPS level-set predictions using 1, 10, and 20 accretion layers are included to assess the effect of the multilayer approach.

forms on the surface, and part of the collected water runs back downstream before freezing. In contrast to rime ice, this process leads to less streamlined and more complex accreted shapes, often characterized by the development of ice horns [165]. The icing-related variables obtained for this case are presented in Figure 4.11. Overall, the predicted trends remain in reasonable agreement with the range of results reported by the other IPW1 participants. However, the computed heat transfer coefficient (HTC) lies on the lower end of the participant spread, which is consistent with the formation of an ice shape that does not fully match the experimental accretion, as further illustrated in Figure 4.12.

To assess whether the remaining discrepancy could be attributed to insufficient numerical convergence, a temporal convergence study was performed with respect to the number of accretion layers. The results showed that the predicted glaze ice shape reached convergence at approximately 20 layers, as illustrated in Figure 4.13a. This indicates that the remaining differences with the experimental ice shape cannot be primarily attributed to the number of accretion layers and instead point to limitations associated with the physical modeling.

This finding suggests that another physical parameter plays a major role in glaze ice shape formation. Because the predicted HTC strongly influences the local freezing fraction and water runback, the parameters controlling convective heat transfer require further investigation. One of the key input parameters directly affecting HTC is the equivalent sand-grain roughness of the surface. Surface roughness enhances near-wall momentum exchange and



(a) Pressure coefficient. (b) Collection efficiency. (c) Convective heat transfer.

Figure 4.11 Comparison of the predicted aerodynamic and icing-related surface quantities for IPW1 Case 242 with the range of results submitted by IPW1 participants.

promotes turbulence within the boundary layer, thereby increasing convective heat transfer at the wall. In the context of icing, a higher HTC promotes faster heat removal from the water film, thereby reducing liquid water runback and favoring earlier freezing closer to the impingement region. This mechanism is particularly important in glaze ice accretion, where the competition between water runback and local freezing largely governs horn development.

In the present simulations, no advanced roughness evolution model is yet available to update the roughness consistently from one accretion layer to the next. Consequently, the flow solver relies on a user-prescribed roughness value, which was initially set to $k_s/c = 0.001$ for the baseline solution, a value commonly adopted in the literature [166]. A second study was therefore performed to assess the influence of this parameter on horn formation. The results, presented in Figure 4.13c, show that increasing k_s increases the HTC, which in turn reduces water runback and promotes earlier freezing as expected. This behavior strengthens horn formation and shifts the accretion toward shapes that are more representative of the experimental glaze ice geometry. In particular, larger roughness values lead to more pronounced horn development and reduce the downstream displacement of the horn tips. For the present case, among the values examined, $k_s/c = 0.003$ provides the best overall agreement with the experimental profile, especially in terms of horn angle, as highlighted in Table 4.4. Some discrepancies nevertheless remain in horn thickness. This comparison suggests that roughness-driven HTC variations play a central role in glaze ice horn formation and should be modeled with greater physical fidelity. Future work will therefore investigate more elaborate roughness formulations.

In contrast to the previous case, for which the overall accretion remains relatively compact and the maximum ice height provides a meaningful scalar indicator, the characterization

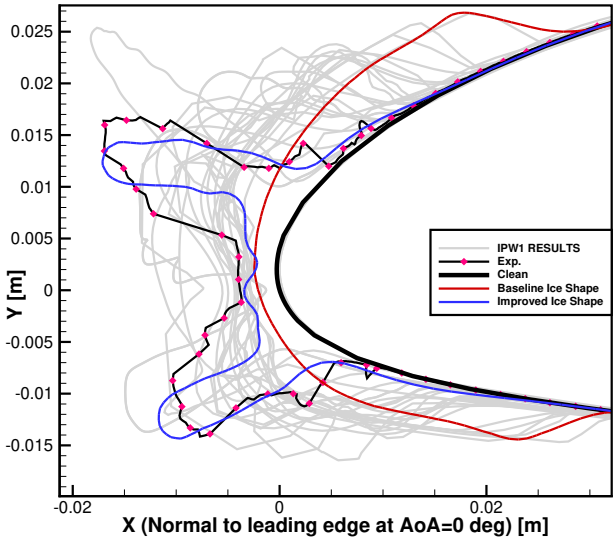
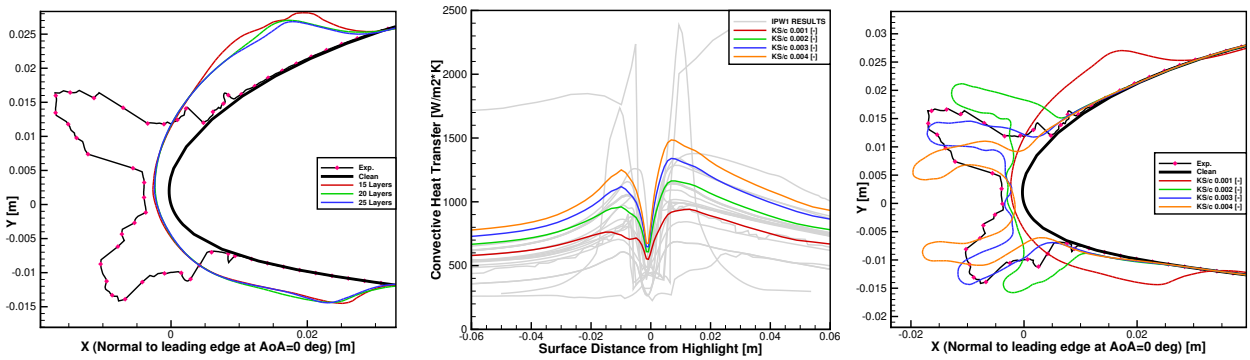


Figure 4.12 Comparison of the predicted ice shapes for Case 242 obtained with different prescribed roughness values. Increasing k_s/c modifies the heat-transfer distribution and improves the predicted horn orientation.



(a) Layer convergence. (b) HTC variation. (c) Ice shape variation.

Figure 4.13 Comparison of glaze ice predictions for Case 242: temporal convergence with respect to the number of accretion layers, variation of the heat transfer coefficient (HTC) with roughness magnitude k_s, and influence of k_s on the predicted ice shape.

of the present glaze ice case is more appropriately based on the orientation of the upper and lower horns. For this reason, the horn angles were selected as the primary quantitative metrics for this case. The smallest roughness value, $k_s/c = 0.001$, is not considered in the analysis, as no clear horn formation is observed for this configuration.

The horn angles were computed using reference points extracted from the experimental ice shape (Fig. 4.14). Specifically, for each horn, upper and lower, a representative point was identified at the midpoint of the corresponding horn region. The associated reference point on the clean airfoil was then defined as the closest point on the clean geometry to this experimental horn location. These experimentally derived reference points were subsequently used consistently for all roughness cases.

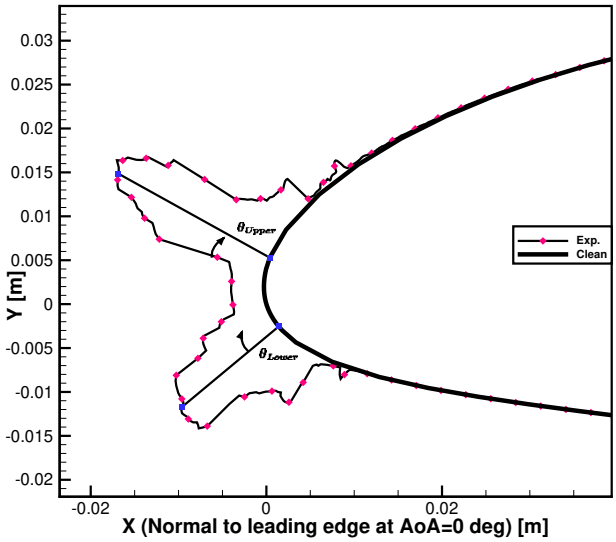


Figure 4.14 Definition of the horn angles on the experimental ice shape.

For a given roughness case k_s, the horn tip was identified as the point exhibiting the largest upstream extension, that is, the most negative x-coordinate. The horn angle was then computed as the angle between the vector connecting the clean reference point to the selected horn tip and the horizontal axis passing through.

The comparison reported in Table 4.4 shows that the predicted horn orientation is strongly affected by the prescribed roughness level. Among the values considered, the case $k_s/c = 0.003$ yields the best overall agreement with the experimental reference, with relative errors of 22.9% and 5.96% for the upper and lower horn angles, respectively. Lower roughness values underpredict the horn deviation and lead to angles that remain too close to the airfoil surface, whereas the largest roughness value does not further improve the agreement. Instead, it appears to over-promote local freezing and modifies the horn morphology such that the

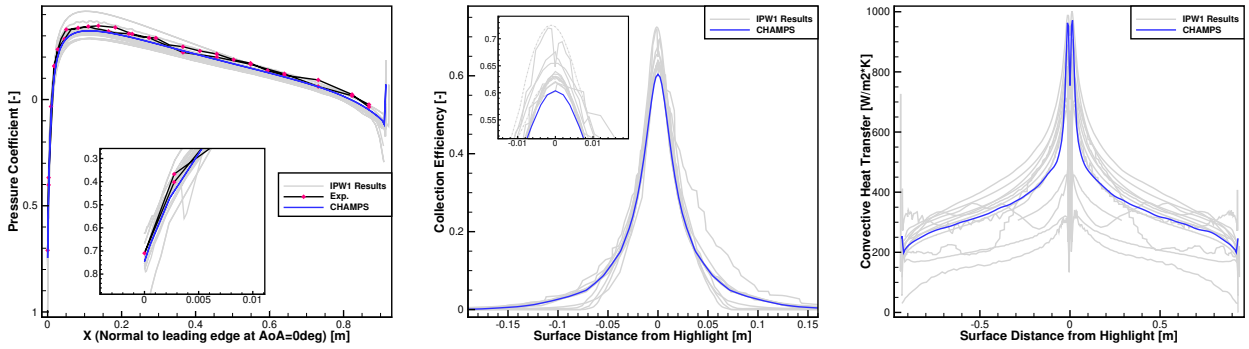
measured horn angles are again underpredicted.

Table 4.4 Comparison of the upper and lower horn angles for Case 242, together with the relative error with respect to the experimental reference.

Config.	$k_s/c\,[-]$	θ_{Upper} [°]	Upper rel. error [%]	θ_{Lower} [°]	Lower rel. error [%]
Exp.	–	29.00	–	39.77	–
2	0.002	46.69	60.96	66.89	68.21
3	0.003	22.34	22.99	37.39	5.96
4	0.004	0.63	97.83	17.34	56.39

4.5.4 Case IPW1 361

Case 361 involves a 30° swept NACA 0012 wing configuration tested in a wind tunnel. According to the workshop conditions, this case corresponds to a rime ice regime. Figure 4.15 presents the obtained predictions for the aerodynamic and icing-related surface quantities. Overall, a good agreement is observed with the envelope of the IPW1 participants' results, indicating that the main physical trends are well captured by the present framework.



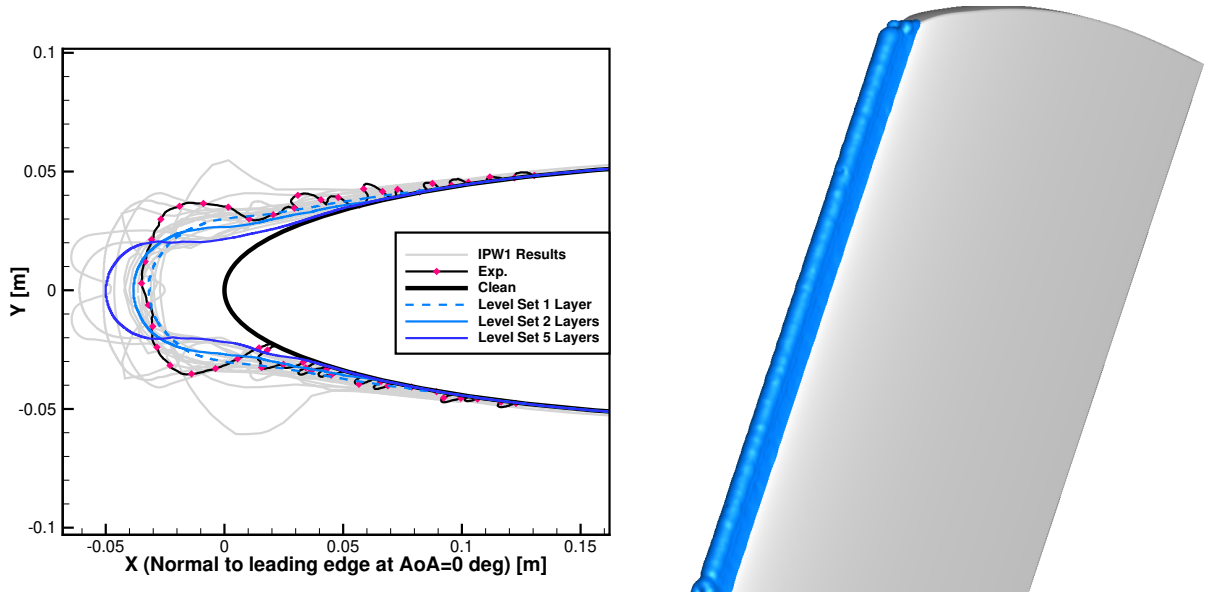
(a) Pressure coefficient. (b) Collection efficiency. (c) Convective heat transfer.

Figure 4.15 Comparison of the predicted aerodynamic and icing-related surface quantities for IPW1 Case 361 against the results submitted by IPW1 participants.

Figure 4.16 compares the ice shapes obtained with 1, 2, and 5 accretion layers against the experimental ice shape and the IPW1 participants' results. The predicted shapes were post-processed following the workshop guidelines by extracting maximum combined cross sections (MCCS) from the computed 3D ice shape. In contrast with the behavior observed for the previous cases, the single-layer prediction appears to remain the closest to the experimental ice shape. A similar trend was also reported by [89]. As the number of layers increases, the

predicted ice shape becomes progressively thicker and more streamlined, which is consistent with the behavior expected for a rime ice case.

This tendency suggests that the discrepancy may not primarily originate from the numerical treatment of the geometric update, but rather from the experimental accretion conditions themselves. More specifically, the flatter and less protruding experimental ice shape may indicate that the accretion did not correspond to an ideal purely rime regime, or that additional physical effects not included in the present model influenced the final geometry.



(a) Comparison of predicted ice shapes by the level-set

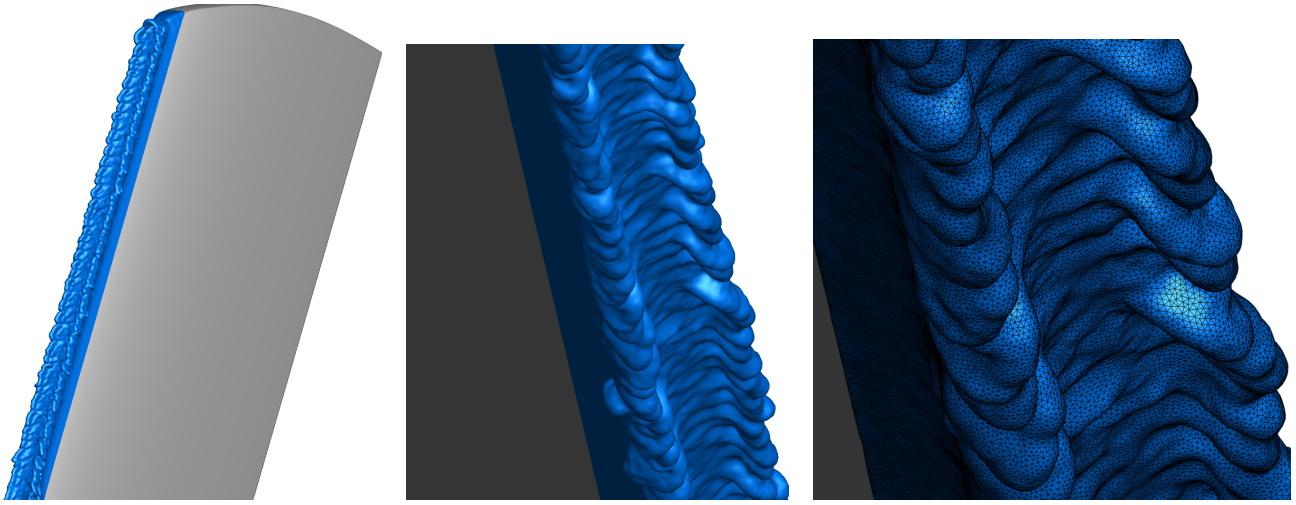
(b) Clipped view of the 5-layer ice shape

Figure 4.16 Predicted ice shapes for Case 361. The effect of the number of accretion layers on the final ice shape is shown for 1, 2, and 5 layers, together with the experimental ice shape and the IPW1 participants' results. A 3D view of the predicted accreted geometry is also presented.

4.5.5 Case IPW1 362

Following the previous rime ice case, Case 362 is characterized by similar aerodynamic and icing conditions, with the main difference being the higher static temperature. Although this change has only a limited effect on the aerodynamic flow field, it plays a central role in the resulting ice morphology. As discussed previously, increasing the temperature reduces the local freezing fraction and promotes liquid water runback along the surface. The water can then freeze farther downstream, leading to more complex glaze ice shapes. In the present

swept wing configuration, this behavior is associated with the formation of spanwise scallop structures with non-uniform size and spacing. Such features are consistent with the 3D nature of glaze ice accretion on swept wings, where the spanwise flow component, liquid-water runback, and local heat-transfer variations can promote non-uniform ice growth. These features are illustrated in Fig. 4.17.



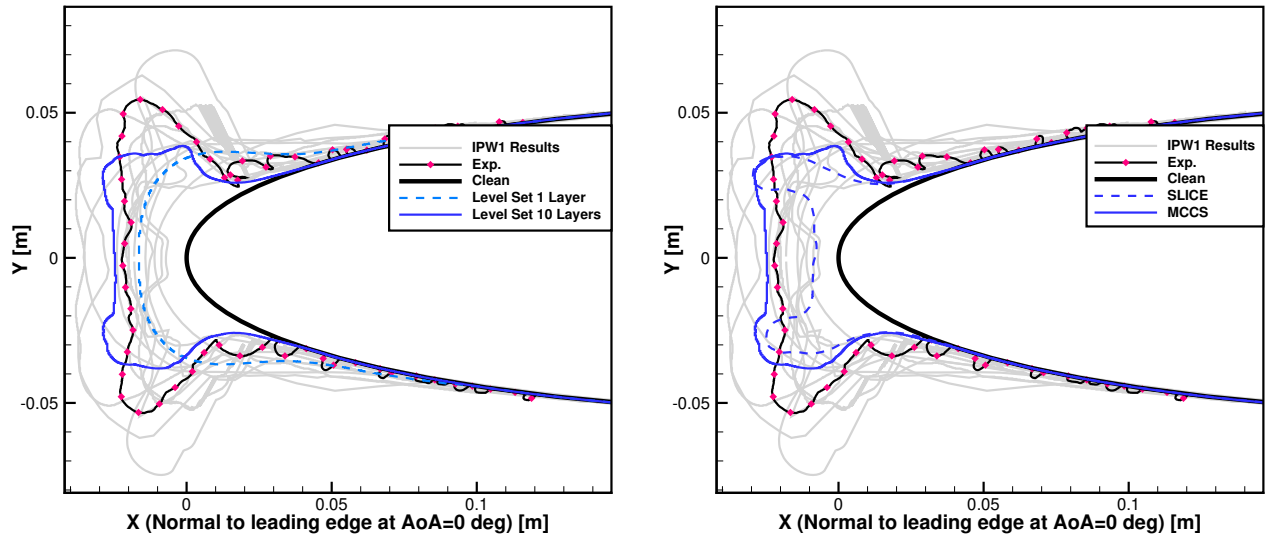
(a) Clipped view of the 10-layer ice shape (b) Spanwise scallop structures (c) Adapted surface mesh on the ice shape

Figure 4.17 Predicted 3D ice shape for IPW1 Case 362 after 10 accretion layers. The clipped views illustrate the spanwise scallop structures formed along the leading edge and the corresponding adapted surface mesh after level-set extraction.

In contrast with the rime ice case, a multilayer approach is essential for this glaze ice configuration in order to progressively capture the development of the complex scallop structures, as shown in Fig. 4.18a. The MCCS obtained with 10 accretion layers shows reasonably good overall agreement with the experimental ice shape. However, discrepancies remain near the impingement limits and in the predicted horn orientation. The simulation was performed using a constant roughness height of $k_s/c = 0.001$. As shown in the previous cases, the prescribed roughness is an important parameter controlling the horn orientation and downstream water runback. In the present 3D case, the surface features generated under this imposed roughness may have locally limited the runback process, which could partly explain the mismatch observed near the impingement limits.

Figure 4.18b presents a comparison of the predicted ice shape using a local spanwise slice extracted at the midspan location and the corresponding MCCS representation. Since the experimental data are only available in the form of an MCCS, a direct comparison with the local experimental morphology is not possible. However, the predicted results clearly

illustrate how the MCCS can smooth and damp local spanwise variations, masking important 3D features of the ice accretion. This emphasizes the limitations of assessing highly 3D ice shapes using only 2D cross-sectional representations, since they may not fully capture the local morphology of the accretion. Future work should therefore aim to develop improved 3D comparison tools and more representative quantitative metrics for evaluating the agreement between predicted and experimental ice shapes.



(a) Effect of the number of accretion layers

(b) Comparison between midspan slice and MCCS extraction

Figure 4.18 Predicted ice shapes for IPW1 Case 362. The experimental ice shape, clean airfoil profile, and IPW1 participant results are shown for reference. The level-set predictions obtained with 1 and 10 accretion layers are compared, together with the ice shape extracted from a single spanwise slice and the maximum combined cross-section (MCCS) extracted from the 3D ice shape.

4.6 Conclusion

This work presented a level-set based framework for 3D multilayer ice accretion simulations within the CHAMPS solver. The proposed methodology combines a conservative level-set formulation, a semi-implicit Rosenbrock–Wanner time integration strategy, local interface reconstruction for ice-volume evaluation, and zero level set extraction using MMG to generate conformal surface geometries. In addition, a smoothing procedure was introduced to regularize the extracted interface while preserving the overall geometric integrity of the accreted ice shape. Altogether, these elements provide a robust alternative to traditional Lagrangian

node-displacement methods, particularly for configurations involving complex geometric features and possible topological changes.

The proposed framework was assessed on both manufactured and experimental benchmark cases. The results showed that the method is capable of capturing complex ice-shape evolutions while maintaining good agreement with reference solutions and experimental data. In particular, the level-set formulation demonstrated clear advantages in terms of geometric robustness and conformity of the extracted interface. The analysis of the glaze ice case also highlighted the strong sensitivity of horn formation to the convective heat-transfer and, consequently, to the prescribed surface roughness.

Future work will focus on improving the physical modeling of glaze ice accretion. In particular, more advanced local roughness models will be investigated to represent the spatial and temporal evolution of surface roughness throughout successive accretion layers, rather than relying on a prescribed uniform equivalent sand-grain roughness. Such developments are expected to improve the prediction of local heat transfer, water runback, and ultimately glaze ice horn formation. In addition, future efforts should also address the comparison of highly 3D ice shapes with experimental measurements by developing improved comparison tools and quantitative metrics that better preserve local morphological features.

CHAPTER 5 An Immersed-Boundary Approach for Multilayer Ice Accretion Simulations

This chapter presents an immersed-boundary framework for aircraft ice accretion simulations. The methodology uses a signed-distance level-set representation of the geometry, with the immersed-boundary formulation applied to both the RANS flow solver and the Eulerian and Lagrangian droplet impingement solvers on a non-conformal background mesh. A hybrid strategy is then adopted for ice-shape evolution, where the level-set-based geometry update is performed on adapted body-fitted meshes. The approach is assessed using a cylinder verification case, an impingement benchmark, and rime and glaze ice accretion cases on a NACA 0012 airfoil, and a 3D multilayer ice-accretion benchmark. The cylinder case confirms that the immersed-boundary droplet formulations recover the analytical collection-efficiency solution. The impingement benchmark shows good agreement with experimental collection-efficiency data, while the rime and glaze cases demonstrate that the ice shapes agree well with experimental measurements. The 3D case is used to demonstrate the complete immersed-boundary multilayer methodology on a more complex configuration.

5.1 Introduction

The numerical ice-accretion workflow and its principal physical components, namely the aerodynamic flow, droplet impingement, surface thermodynamics, and geometry evolution, have been introduced in the literature review (Section 2.1). In a conventional body-fitted framework, successive accretion layers require the computational mesh to conform to the evolving ice surface. This requirement can become increasingly restrictive as complex ice geometries develop. To address these constraints, this chapter focuses on the implementation and assessment of an immersed-boundary icing framework within CHAMPS. The solid surface is embedded in a non-conformal background mesh through a signed-distance representation. The immersed-boundary flow solver was developed prior to this work and is presented here for completeness, while both Eulerian and Lagrangian droplet solvers are adapted to the immersed geometry. The thermodynamic model remains surface based, and geometry evolution is performed with the continuous level-set methodology presented in Chapter 4.

The resulting strategy is hybrid: the flow and droplet solutions are performed on the background IBM mesh, whereas the geometry-evolution problem is solved on its own adapted body-fitted mesh. This arrangement removes the need to repeatedly generate or deform a body-fitted RANS mesh during multilayer accretion, thereby alleviating the more restrictive

mesh-generation requirements associated with the flow and droplet solvers. Such constraints are considerably less limiting for the level-set problem, for which suitable adapted meshes can be generated more readily.

The chapter first describes the immersed-boundary representation, flow treatment, and droplet formulations in Sections 5.2–5.4, followed by their coupling with the thermodynamic and geometry-evolution models in Section 5.5. The complete framework is then assessed in Section 5.6 using a canonical cylinder impingement case, an SLD impingement benchmark, and rime and glaze ice-accretion cases on airfoils. Comparisons with body-fitted predictions and experimental data are used to isolate the effect of the immersed-boundary treatment. Finally, a 3D multilayer ice-accretion case is considered to demonstrate the complete immersed-boundary framework on a more complex configuration. The main findings are summarized in Section 5.7.

5.2 Immersed-Boundary Representation

In the present chapter, the solid geometry is represented implicitly in the volume mesh using a signed-distance level-set function ϕ:

$$\phi = \begin{cases} < 0, & \text{inside the solid region,} \\ = 0, & \text{at the immersed interface,} \\ > 0, & \text{inside the fluid domain.} \end{cases} \tag{5.1}$$

An example of the resulting signed-distance field around a circular cylinder is shown in Figure 5.1.

The initial level-set field is computed from the shortest distance between each volume-cell center and the immersed geometry. The sign of ϕ is determined using a ray-casting procedure [158]. The gradient of the level-set field is reconstructed using a weighted least-squares approach, from which the unit normal to the immersed interface is obtained as

$$\boldsymbol{n}_\phi = \frac{\nabla\phi}{\|\nabla\phi\|} \tag{5.2}$$

The geometric operations required by the immersed-boundary formulation involve repeated distance, closest-point, and intersection queries against the surface mesh. To accelerate these operations, an axis-aligned bounding-box (AABB) tree is constructed from the immersed-surface elements [158]. Rather than subdividing the spatial domain, the AABB tree recursively partitions the surface elements into smaller groups and encloses each group within a

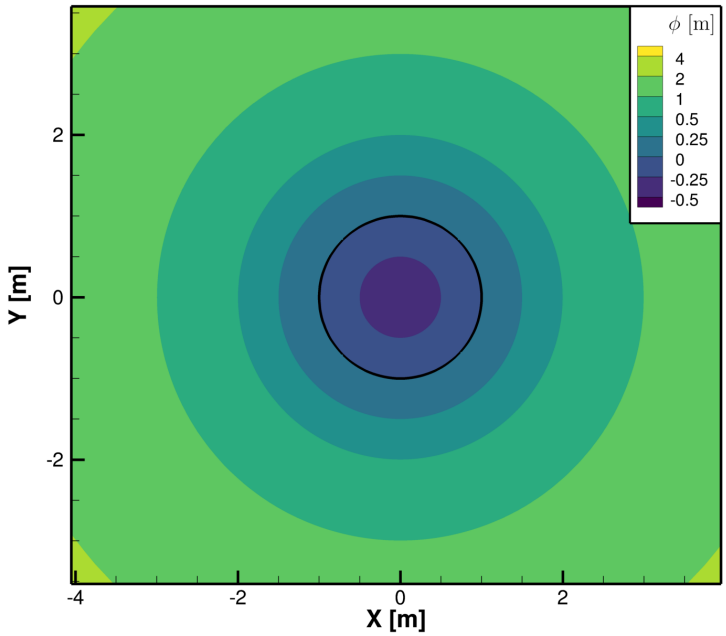
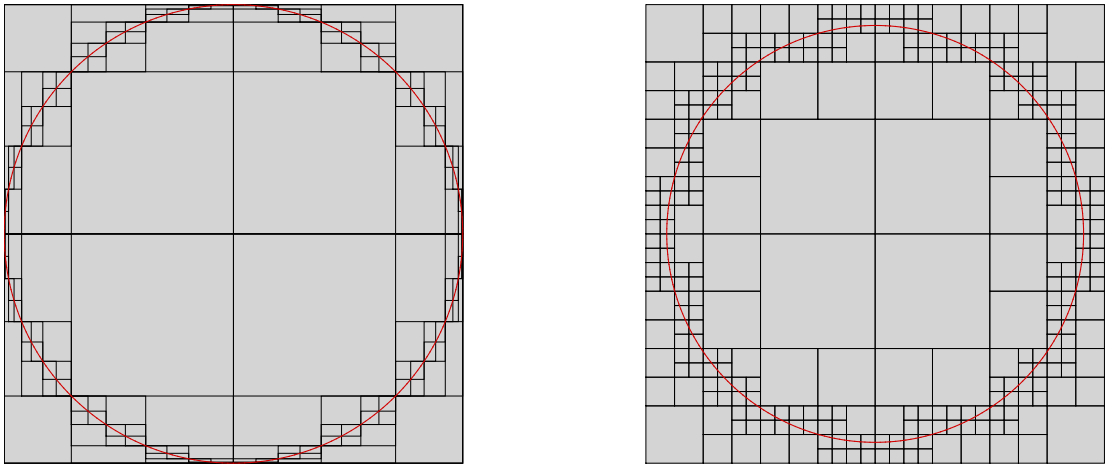


Figure 5.1 Signed level-set field ϕ around a circular cylinder. The zero level set defines the cylinder surface, while positive values correspond to the fluid region and negative values to the solid region.

tightly fitted bounding box. During a geometric query, complete branches of the tree can be discarded whenever their bounding boxes cannot contain a closer point or intersect the search segment, thereby avoiding explicit tests against the corresponding surface elements. This differs from a quadtree, which recursively subdivides the physical domain into four spatial regions independently of the distribution of the surface elements. As illustrated in Figure 5.2 for a circular immersed surface, the AABB hierarchy remains concentrated around the surface geometry, whereas the quadtree partitions the surrounding domain, including regions containing no surface elements.

As an illustrative example, 1,000 nearest-point queries were performed against the circular surface of Figure 5.2, discretized using 256 segments. A direct search requires 256 geometric tests per query. The AABB tree reduced this number to an average of ≈ 25 geometric tests per query, compared with ≈ 60 for the quadtree. Both hierarchical approaches therefore substantially reduce the geometric search effort, with the AABB tree requiring considerably fewer geometric tests for this example. The benefit of reducing these operations becomes increasingly important for 3D simulations, where the immersed surface will contain a substantially larger number of elements and geometric queries are performed repeatedly. Since the geometric operations required by the present formulation are primarily performed against



(a) AABB-tree decomposition.

(b) Quadtree decomposition.

Figure 5.2 Comparison of AABB-tree and quadtree search structures for a circular immersed surface. The AABB tree recursively partitions the surface elements and encloses each group within a bounding box, whereas the quadtree recursively partitions the surrounding spatial domain.

the immersed surface, the AABB tree was selected as the search structure.

Flow and droplet quantities at the immersed boundary are evaluated by projecting variables from neighboring volume cells to the interface. For each immersed-boundary surface element, the closest volume-cell center is first identified, and the variable is extrapolated as

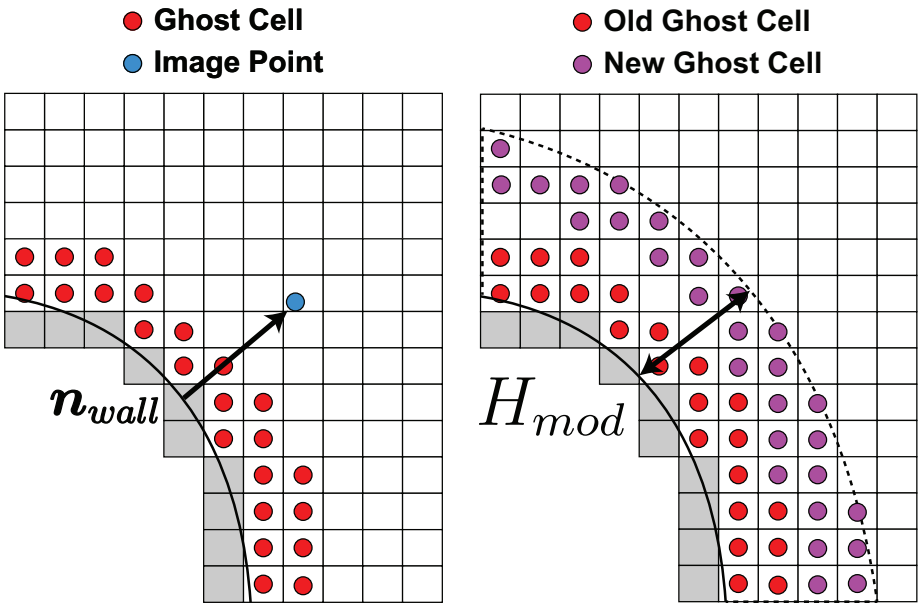
$$\boldsymbol{\varphi}_{IB} = \boldsymbol{\varphi}_V + \nabla\boldsymbol{\varphi}_V \cdot (\boldsymbol{x}_{IB} - \boldsymbol{x}_V) \tag{5.3}$$

where $\boldsymbol{\varphi}_{IB}$ is the value at the immersed boundary, $\boldsymbol{\varphi}_V$ is the value at the selected volume-cell center, $\nabla\boldsymbol{\varphi}_V$ is its reconstructed gradient, and $\boldsymbol{x}_{IB} - \boldsymbol{x}_V$ is the distance vector from the volume-cell center to the immersed-boundary point.

5.3 Immersed-Boundary Flow Treatment

The immersed-boundary treatment adopted for the flow equations follows the ghost-cell methodology of [167]. In this approach, the solid boundary condition is enforced by prescribing appropriate conservative variables $\boldsymbol{W}$ in cells located in the vicinity of the immersed interface. These cells, commonly referred to as ghost cells or target points, are selected near the solid–fluid interface, as illustrated in Figure 5.3a. The flow variables imposed in the ghost

cells are obtained from corresponding image points located in the fluid domain. Since these image points generally do not coincide with actual mesh nodes, their flow variables must be reconstructed from neighboring fluid cells. This reconstruction procedure is analogous to the interpolation strategy introduced in Section 5.2.



(a) Standard ghost-cell construction based on the wall-normal image point.

(b) Modified ghost-cell positioning used to enforce the RANS wall treatment at the prescribed distance H_{mod}.

Figure 5.3 Schematic representation of the ghost-cell treatment near an immersed boundary. The standard approach (a) defines ghost-cell values using image points along the local wall-normal direction, while the modified approach (b) introduces additional ghost cells to impose the RANS wall treatment at a prescribed distance from the immersed interface.

In conventional body-fitted RANS simulations, the mesh is typically highly stretched in the wall-normal direction in order to resolve the large velocity gradients that develop inside the boundary layer. In an immersed-boundary framework, however, imposing a comparable near-wall spacing throughout the background mesh would be excessively restrictive, especially for the high-Reynolds-number conditions typical of aircraft icing simulations, and would substantially increase the overall computational cost. For this reason, wall functions are employed to relax the near-wall resolution requirement. Their application within the immersed-boundary framework, however, requires particular attention to the placement of the ghost cells and their associated image points. The standard ghost-cell construction is not directly suitable for this purpose, since the location of the image point is usually chosen either arbitrarily or

according to a wall-distance criterion. When wall functions are used, such a choice does not guarantee that the target and image points lie in a consistent region of the boundary layer. This may compromise the consistency of the imposed wall-modeled boundary condition and lead to spurious oscillations in the pressure and skin-friction coefficients.

To address this issue, [167] proposed a modified ghost-cell placement strategy in which the location of the immersed-boundary target points is determined from a modeling height H_{mod}. This height is derived from flat-plate turbulent boundary-layer theory and is chosen so that the image points are located at an appropriate wall-modeling distance, while the target points define the modeled region below this threshold, as illustrated in Figure 5.3b. The modeling height is defined as

$$H_{mod} = \sqrt{2}\frac{y^+_{target}L}{Re\sqrt{0.058Re^{-0.2}}} \tag{5.4}$$

where y^+_{target} is the prescribed target wall coordinate, typically chosen in the logarithmic region of the boundary layer, with values between 50 and 300.

The boundary conditions imposed at the ghost cell are then constructed from the corresponding image points. First, the pressure is assumed to remain constant in the wall-normal direction. Denoting the ghost cell by the subscript GC and the image point by the subscript IM, this gives

$$p_{GC} = p_{IM} \tag{5.5}$$

The imposed isothermal wall temperature is estimated from the freestream total temperature using the isentropic total-temperature relation

$$\frac{T_0}{T_\infty} = 1 + \frac{1}{2}(\gamma - 1)M^2_\infty \tag{5.6}$$

where T_0 is the total temperature, T_∞ is the freestream static temperature, γ is the specific heat ratio, and M_∞ is the freestream Mach number. The wall temperature is then prescribed as $T_w = T_0$. Rather than imposing this temperature directly at the ghost-cell center, the ghost-cell density is reconstructed from the image-point state as

$$\rho_{GC} = 2\frac{p_{IM}}{RT_w} - \rho_{IM} \tag{5.7}$$

The ghost-cell temperature is then obtained from the perfect-gas equation of state

$$T_{GC} = \frac{p_{GC}}{R\rho_{GC}} \tag{5.8}$$

where R is the specific gas constant.

The tangential velocity at the ghost cell is obtained from the selected wall law. A useful feature of the method is that the ghost-cell construction is independent of the particular wall function used. In the present chapter, the Reichardt [168] wall function is adopted:

$$u^+ = \frac{\ln(1+\kappa y^+)}{\kappa} + 7.8\left[1 - \exp\left(-\frac{y^+}{11}\right) - \frac{y^+}{11}\exp\left(-\frac{y^+}{3}\right)\right] \tag{5.9}$$

where $\kappa \approx 0.41$ is the von Kármán constant.

For the $k-\omega$ turbulence model, the ghost-cell values of the turbulent variables are also reconstructed from wall-function-consistent near-wall relations. The specific dissipation rate is obtained by blending its viscous-sublayer and logarithmic-layer asymptotic limits

$$\omega_{\text{vis}} = \frac{6\nu}{\beta_\omega y^2} \qquad \omega_{\log} = \frac{u_\tau}{\sqrt{C_\mu}\kappa y} \tag{5.10}$$

where β_ω and C_μ are turbulence-model constants. The wall-function value imposed at the ghost cell is then defined as

$$\omega_{GC} = \left(\omega_{\text{vis}}^2 + \omega_{\log}^2\right)^{1/2} \tag{5.11}$$

The turbulent kinetic energy is subsequently recovered from the eddy-viscosity relation

$$\mu_t = \rho\frac{k}{\omega} \qquad k_{GC} = \frac{\mu_{t,GC}\omega_{GC}}{\rho_{GC}} \tag{5.12}$$

Although the cells located between the ghost-cell layer and the immersed geometry are not advanced by the flow solver, flow quantities are assigned to these cells at the end of the flow simulation using the same wall-function reconstruction. This additional reconstruction is required by the droplet impingement solver, which uses the near-wall flow field to evaluate droplet trajectories and surface impacts in the vicinity of the immersed boundary.

5.4 Immersed-Boundary Droplet Treatment

The Eulerian and Lagrangian droplet formulations used in CHAMPS, including their governing equations, trajectory integration, adaptive seeding, collection-efficiency evaluation, and SLD extensions, are described in the literature review (Sections 2.3.1–2.3.3) and in Chapter 3. The present section is restricted to the modifications required to use both droplet formulations with an immersed geometry.

5.4.1 Eulerian Formulation: Governing Equations and Penalization Strategy

The governing continuity and momentum equations of the Eulerian droplet formulation are

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \boldsymbol{u}_d) = 0 \tag{5.13}$$

$$\frac{\partial (\alpha \boldsymbol{u}_d)}{\partial t} + \nabla \cdot (\alpha \boldsymbol{u}_d \otimes \boldsymbol{u}_d) = \frac{C_D Re_D}{24 S_t} \alpha (\boldsymbol{u}_a - \boldsymbol{u}_d) + \alpha \left(1 - \frac{\rho_a}{\rho_d}\right) \frac{\boldsymbol{g}}{Fr^2}, \tag{5.14}$$

where the variables and non-dimensional parameters are defined in Section 2.3.1.

The immersed-boundary treatment follows the penalization strategy of [169]. The mask functions used to localize the penalization are shown in Figure 5.4. A solid mask χ, illustrated in Figure 5.4a, activates the penalization inside the immersed body:

$$\chi = \begin{cases} 1, & \phi \leq 0, \\ 0, & \phi > 0, \end{cases} \tag{5.15}$$

where negative values of ϕ correspond to the solid region and positive values to the fluid region. Since the penalization should be imposed only in the droplet shadow region, a second mask χ_d, shown in Figure 5.4b, is introduced:

$$\chi_d = \begin{cases} 1, & \boldsymbol{u}_d \cdot \boldsymbol{n}_\phi \leq 0, \\ 0, & \boldsymbol{u}_d \cdot \boldsymbol{n}_\phi > 0. \end{cases} \tag{5.16}$$

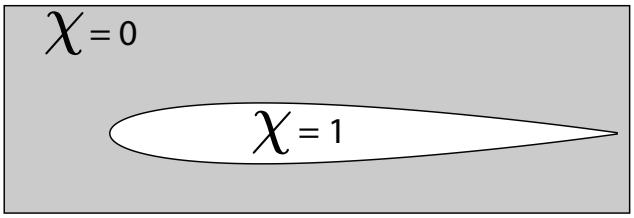
Thus, the penalization is active only where the immersed-solid and incoming-droplet regions overlap, through the combined mask $\chi\chi_d$, shown in Figure 5.4c.

The resulting penalized Eulerian droplet equations are

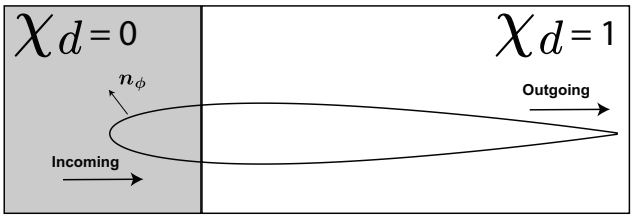
$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \boldsymbol{u}_d) = -\frac{\chi \chi_d}{\eta} \alpha \tag{5.17}$$

$$\frac{\partial (\alpha \boldsymbol{u}_d)}{\partial t} + \nabla \cdot (\alpha \boldsymbol{u}_d \otimes \boldsymbol{u}_d) = (1 - \chi) \left[\frac{C_D Re_D}{24 S_t} \alpha (\boldsymbol{u}_a - \boldsymbol{u}_d) + \alpha \left(1 - \frac{\rho_a}{\rho_d}\right) \frac{\boldsymbol{g}}{Fr^2} \right] - 2 \frac{\chi \chi_d}{\eta} \alpha \boldsymbol{u}_d \tag{5.18}$$

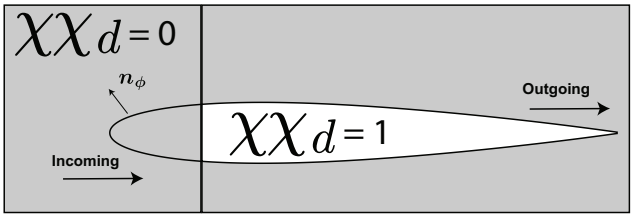
where η controls the penalization strength and is set to 10^{-8}. In the fluid region, the penalization terms vanish since $\chi = 0$. In the penalized region, where $\chi\chi_d = 1$, the source terms drive both the droplet volume fraction and momentum toward zero, corresponding to the



(a) Solid–fluid mask χ.



(b) Droplet-direction mask χ_d.



(c) Combined penalization indicator $\chi\chi_d$.

Figure 5.4 Mask functions used in the penalized Eulerian droplet formulation: (a) solid–fluid mask χ, (b) droplet-direction mask χ_d, and (c) combined penalization indicator $\chi\chi_d$, which restricts the penalization to the overlap between the immersed-solid and incoming-droplet regions.

immersed-boundary analogue of imposing a Dirichlet condition for the droplet phase.

5.4.2 Lagrangian Immersed-Surface Tracking

In a body-fitted formulation, droplet impacts are detected directly on boundary faces. In an immersed-boundary framework, however, the solid surface is not a boundary of the volume mesh, and a dedicated impact treatment is therefore required. The Lagrangian tracking algorithm presented in Chapter 3 remains unchanged; only the treatment used to detect and locate impacts on the immersed surface is modified. During preprocessing, cells intersected by the immersed geometry are tagged as potential impact cells and linked to the corresponding immersed-surface elements. This restricts the impact search during tracking to a local subset of surface elements.

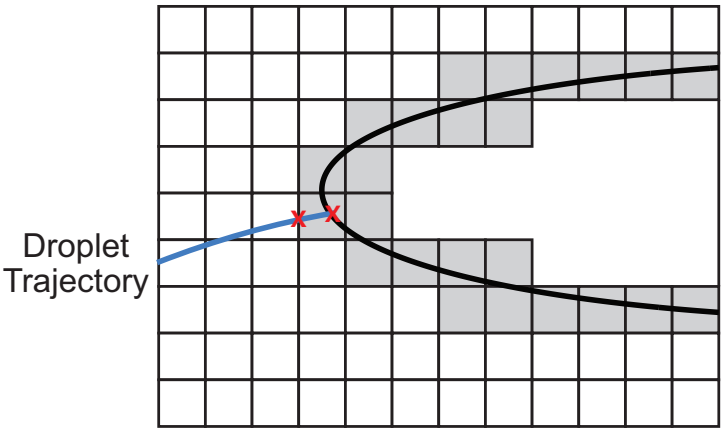


Figure 5.5 Droplet tracking in an immersed-boundary framework. The trajectory is advanced through the background mesh until it intersects the reconstructed immersed surface.

Droplets are seeded upstream and advanced through regular fluid cells using the volume-mesh connectivity. When the next crossed cell is tagged as an immersed-boundary cell, the trajectory ray is tested against the associated surface elements to determine the physical impact location, as illustrated in Figure 5.5. The local collection efficiency is then evaluated from the deformation of the droplet streamtube surrounding each impacting droplet using the methodology presented in Chapter 3. The SLD models require no modification for the immersed-boundary formulation, since droplet deformation is accounted for through the drag formulation and splashing is applied as a post-impact treatment. Only the splashing model of Wright et al. [64] is considered in the present chapter for both the Eulerian and Lagrangian simulations.

5.5 Immersed-Boundary Icing Coupling

For each accretion layer, the RANS and droplet solutions are first obtained on the non-conformal background mesh. The aerodynamic and droplet quantities required by the thermodynamic calculation are then evaluated at the physical immersed surface. The thermodynamic solver determines the local ice-growth distribution, which is subsequently transferred to the adapted level-set mesh. The ice–air interface is advanced using the continuous level-set methodology presented in Chapter 4, and the newly extracted ice surface replaces the previous immersed geometry. The flow and droplet solutions are then recomputed on the same background mesh using the updated immersed geometry for the following accretion layer.

The resulting multilayer procedure is hybrid with respect to the computational meshes. The RANS and droplet solvers retain the non-conformal background mesh as the ice geometry

evolves, while the level-set calculation uses a separate Delaunay mesh composed of simplicial tetrahedral elements for interface evolution. Unlike a body-fitted RANS mesh, the level-set mesh does not require the stringent near-wall first-cell spacing or boundary-layer refinement imposed by the aerodynamic calculation. It can therefore be generated comparatively easily and at low computational cost as the ice geometry evolves.

5.6 Results

The objective of this section is to assess the droplet impingement and icing capabilities of CHAMPS in an immersed-boundary setting. Both Eulerian and Lagrangian droplet formulations are considered, with and without supercooled-large-droplet (SLD) effects. A canonical cylinder case is first used for verification. The second case corresponds to the SLD impingement experiment of [141], which is used to evaluate the SLD effects on the predicted collection efficiency. Two NACA 0012 icing cases from wind-tunnel experiments [170] are then considered to assess the complete icing framework under rime and glaze conditions. Finally, the RG15 configuration considered during IPW2 is used as a demonstration of the complete immersed-boundary multilayer framework in 3D.

Table 5.1 Summary of the benchmark cases.

Parameter	SLD	001	004
Geometry	2D NACA 23012	2D NACA 0012	2D NACA 0012
Icing time $[s]$	N/A	420	384.0
LWC $[g/m^3]$	0.5	0.6	0.55
MVD $[\mu m]$	111.0	20.0	15.0
Ma [-]	0.232	0.325	0.317
AOA [°]	2.5	4.0	4.0
C_{ref} [inch]	36	36	21
T_{static} $[K]$	268.02	250.7	262.3
Re [-]	$4.57 \cdot 10^6$	$4.83 \cdot 10^6$	$4.44 \cdot 10^6$

5.6.1 Cylinder Impingement Verification

The cylinder case is used as a verification benchmark for the droplet impingement solvers. With drag and gravity neglected, droplets follow straight trajectories before impact, yielding an analytical solution for the collection efficiency [169]. For a cylinder of radius R centered at the origin and an incoming droplet velocity aligned with the x direction, the analytical solution can be written along the cylinder surface as

$$\beta_{\text{ana}} = \alpha_\infty U_\infty \max\left(-\cos\theta, 0\right) \tag{5.19}$$

where θ is the polar angle along the cylinder surface and α_∞ is the freestream droplet volume fraction. In the present verification case, α_∞ and U_∞ are both set to unity, so the analytical collection efficiency reaches its maximum value $\beta_{\text{ana}} = 1$ at $\theta = \pi$, decreases to zero at $\theta = \pi \pm \pi/2$, and remains zero over the rear side of the cylinder.

Figure 5.6 compares the Eulerian and Lagrangian predictions on both body-fitted and immersed-boundary meshes. All approaches reproduce the analytical solution, confirming the correctness of the droplet transport and impact detection procedures.

5.6.2 SLD Impingement

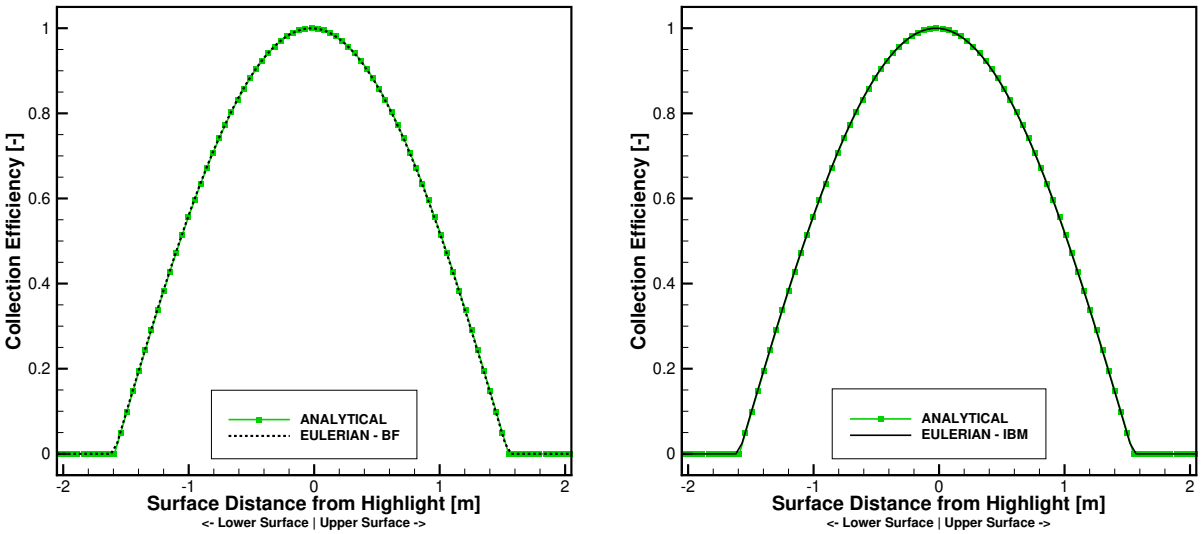
The NACA 23012 benchmark of [141] is considered to evaluate the SLD modelling. This experiment provides collection efficiency measurements for large droplets and highlights the limitations of conventional trajectory models when droplet splashing is neglected. In particular, simulations without SLD effects generally overpredict both the magnitude and the extent of water collection.

The collection efficiency distributions are shown in Figure 5.7. For both Eulerian and Lagrangian formulations, the baseline predictions without SLD effects overestimate the experimental data, especially away from the stagnation region. When the SLD model is activated, splashing removes part of the impacting water mass, reducing the deposited collection efficiency and improving agreement with the experimental measurements. The body-fitted and immersed-boundary formulations show similar responses to the SLD correction, confirming the consistency of the immersed-boundary implementation.

Some discrepancies remain near the stagnation region, where the peak collection efficiency is overpredicted. This behavior is consistent with the Wright et al. splashing model [64], which assumes limited mass loss for normal impacts, while splashing becomes more important farther from the stagnation point as the impact angle becomes more tangential to the surface. Since both formulations yield similar results, only the Lagrangian formulation is retained for the remainder of this section due to its computational efficiency for such cases.

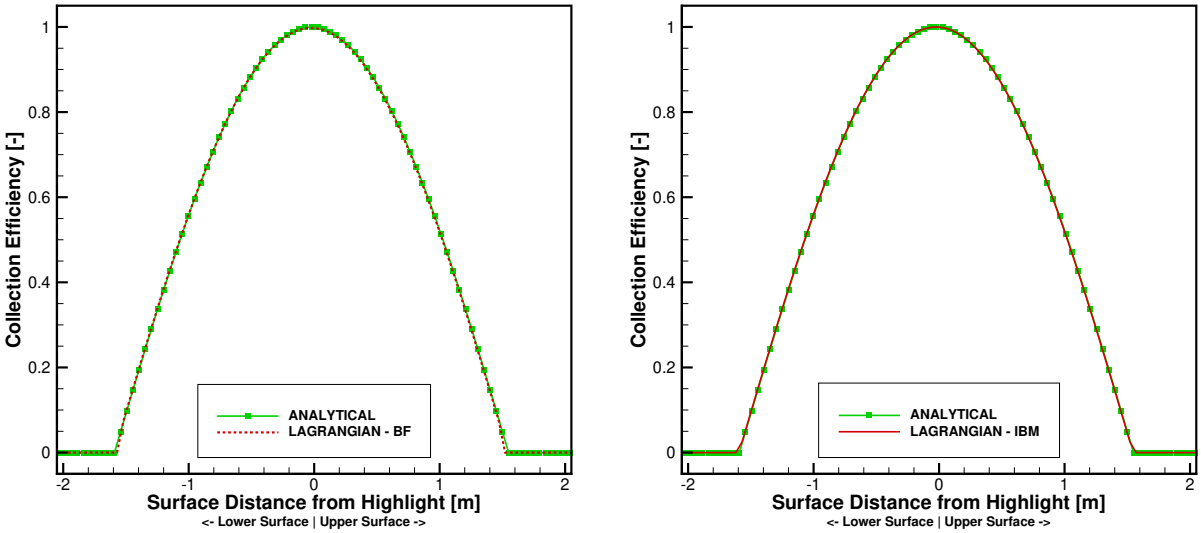
5.6.3 Case 01

Case 01 corresponds to a rime ice condition, where droplets freeze upon impact. Under these conditions, water runback is negligible and the ice shape is mainly governed by the collection



(a) Eulerian body-fitted.

(b) Eulerian immersed-boundary.



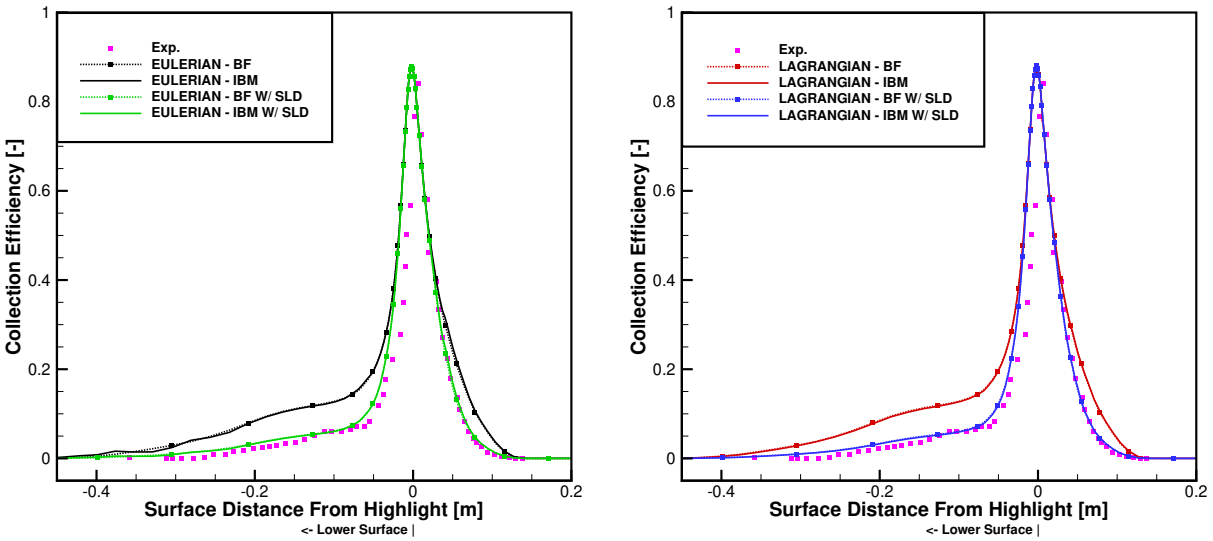
(c) Lagrangian body-fitted.

(d) Lagrangian immersed-boundary.

Figure 5.6 Collection efficiency distributions on the cylinder obtained with Eulerian and Lagrangian formulations on body-fitted and immersed-boundary meshes.

efficiency distribution. Since the impingement verification showed close agreement between body-fitted and immersed-boundary methods, similar ice shapes are expected.

The predicted ice shapes are shown in Figure 5.8. The body-fitted and immersed-boundary results are in good agreement for both single-layer and multilayer simulations. The five-layer prediction shows improved agreement with the experimental profile compared with the single-layer result. This supports the point discussed earlier in Section 5.1: updating the flow,



(a) Eulerian formulation.

(b) Lagrangian formulation.

Figure 5.7 Collection efficiency distributions for the Papadakis SLD impingement case obtained with the Eulerian and Lagrangian formulations, with and without SLD effects.

droplet, thermodynamic, and geometric fields over successive accretion steps provides a more accurate representation of the evolving ice shape than a single-step calculation. The overall agreement with the experimental data confirms that the immersed-boundary framework is able to reproduce impingement-driven rime ice accretion with good accuracy.

5.6.4 Case 04

Case 04 corresponds to a glaze ice condition, where only part of the collected water freezes locally, while the remaining liquid forms a water film that runs back along the surface and progressively freezes downstream. The ice shape therefore depends not only on droplet impingement, but also on the local thermodynamic balance and convective heat transfer. Compared with rime ice, glaze ice prediction is therefore more sensitive to the wall treatment.

The predicted ice shapes and the corresponding heat transfer coefficient distributions are presented in Figure 5.9. Although the same wall-function formulation is used, differences in the enforcement of the wall boundary condition lead to variations in the predicted heat transfer coefficient. The immersed-boundary solution predicts slightly larger heat transfer near the leading edge, where glaze ice growth is particularly sensitive to thermal effects.

These differences have a limited impact on the single-layer prediction, for which the body-fitted and immersed-boundary ice shapes remain close. In the multilayer simulation, however,

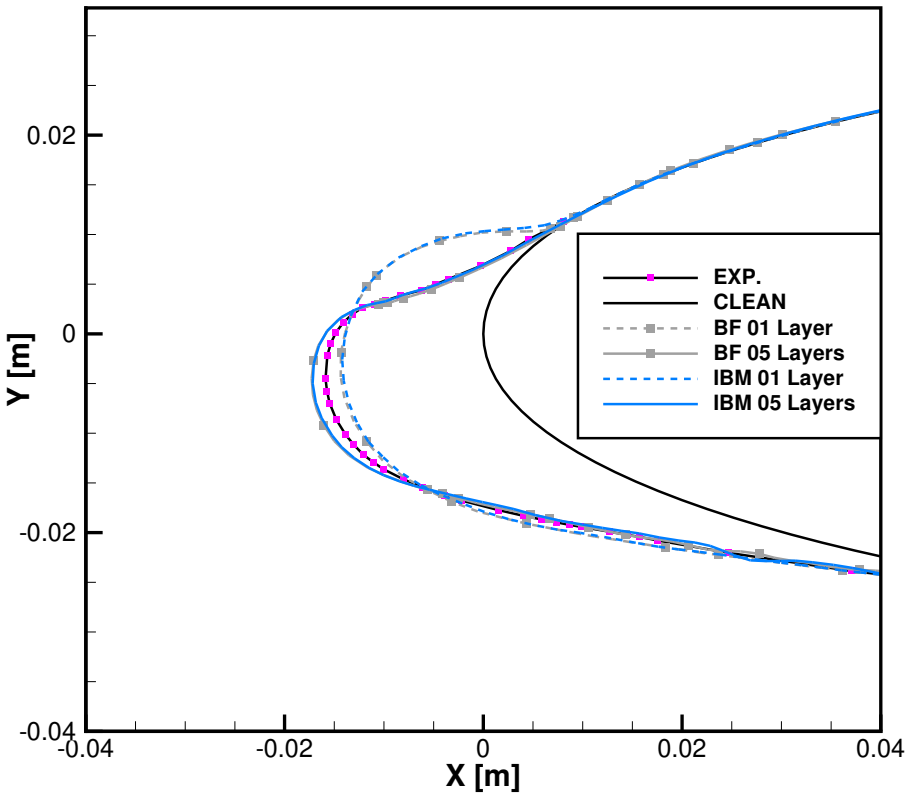


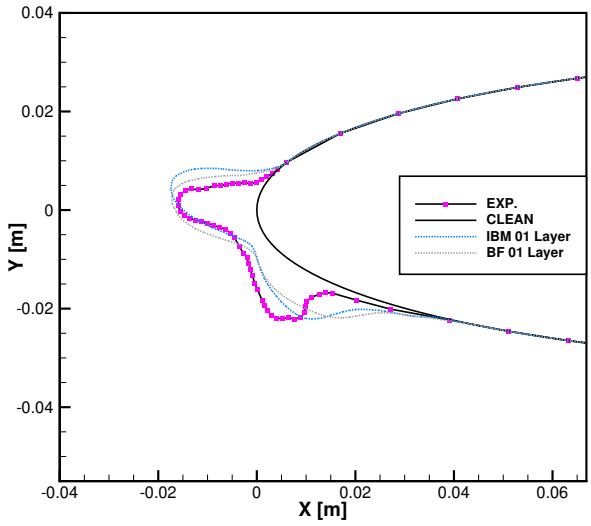
Figure 5.8 Predicted ice shapes for Case 01 using body-fitted and immersed-boundary formulations. The clean and experimental profiles are shown for reference, together with predictions obtained using 1 and 5 accretion layers.

small heat-transfer differences accumulate as the geometry is updated, leading to more visible deviations in the final ice shape. This effect is most noticeable near the lower horn, where the immersed-boundary solution predicts larger accretion.

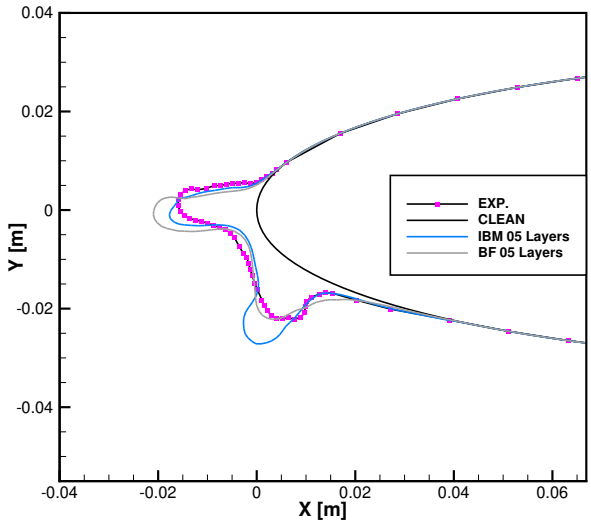
Despite these differences, both approaches remain in good agreement with the experimental measurements and correctly reproduce the main glaze ice morphology, including the upper and lower horns. The results indicate that the immersed-boundary framework can predict glaze ice accretion with accuracy comparable to the body-fitted formulation, while also highlighting differences in the local heat-transfer prediction that should be further investigated.

5.6.5 Case RG15

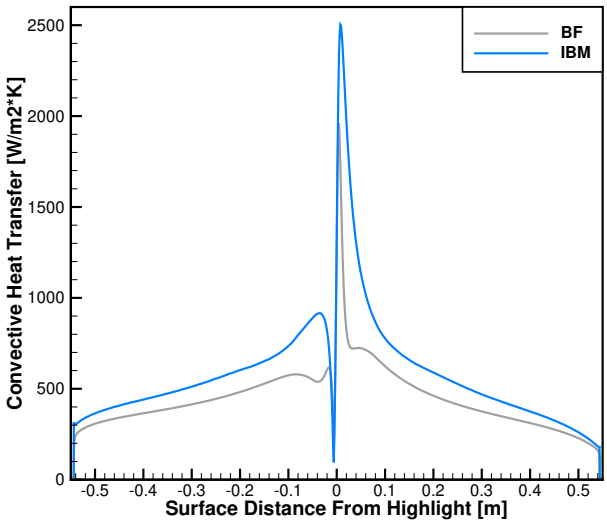
As a final demonstration, the immersed-boundary icing framework is applied to a 3D geometry based on the RG15 configuration considered during IPW2. The objective of this case is not to reproduce or validate the original workshop condition, but rather to provide a 3D



(a) Single-layer ice shape.



(b) Five-layer ice shape.



(c) Convective heat transfer coefficient.

Figure 5.9 Predicted ice shapes and convective heat transfer coefficient distributions for Case 04 using body-fitted and immersed-boundary formulations. The clean and experimental profiles are shown for reference in the ice-shape comparisons, together with predictions obtained using 1 and 5 accretion layers.

test of the complete numerical framework.

The freestream conditions are defined by a Mach number of $M_\infty = 0.07$, a Reynolds number of $Re = 200$, a static temperature of $T_\infty = 263.15$ K, and an angle of attack of $\alpha = 4°$. The reference chord is $C_{\text{ref}} = 0.3$ m. A liquid-water content of LWC $= 0.52$ g/m^3 is prescribed.

An icing time of 20 min is considered. Laminar-flow conditions are deliberately prescribed to alleviate the mesh-resolution requirements and associated element count.

Figure 5.10 first compares the collection-efficiency distributions obtained with the Eulerian and Lagrangian immersed-boundary formulations. The two approaches produce similar 3D impingement patterns along the leading edge, consistent with the agreement previously observed between the two droplet formulations using both body-fitted and immersed-boundary approaches.

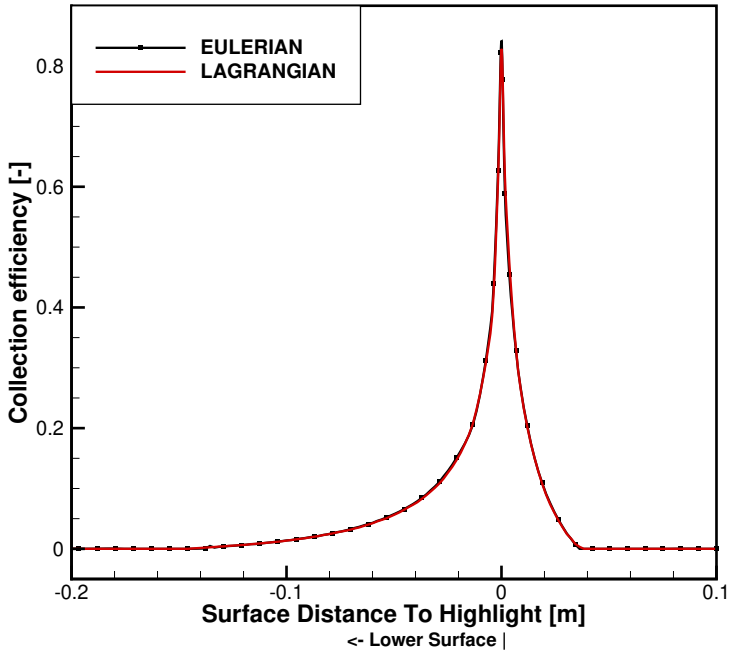
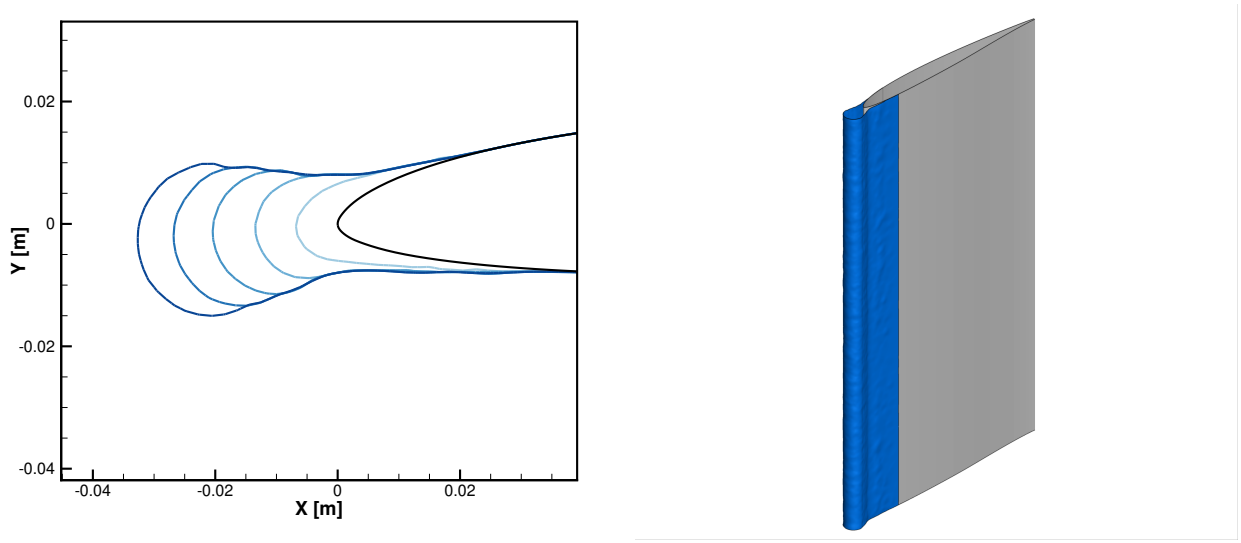


Figure 5.10 Comparison of the collection-efficiency distributions predicted by the Eulerian and Lagrangian immersed-boundary formulations for the RG15 configuration.

Figure 5.11a illustrates the evolution of the ice geometry through the successive accretion layers, while Figure 5.11b shows the clean wing together with the predicted ice geometry. The calculation demonstrates that the coupling between the immersed-boundary flow and droplet solvers and the level-set geometry-evolution methodology can be carried out for a 3D configuration. Following each geometry update, the extracted ice surface is introduced as the new immersed geometry while the aerodynamic and droplet calculations retain the non-conformal background mesh, thereby avoiding the generation of a new body-fitted RANS mesh.

At this stage, the case is intended as a demonstration of the 3D capability rather than a quantitative validation. The simplified laminar-flow assumption deliberately reduce the physical complexity of the original RG15 configuration and allow the numerical behavior of the cou-



(a) Evolution of the ice geometry through successive accretion layers.

(b) Clean wing and predicted ice shape.

Figure 5.11 Predicted 3D ice accretion on the RG15 wing: (a) evolution of the ice geometry through successive accretion layers and (b) clean wing and predicted ice geometry.

pled methodology to be examined independently of the additional complexities associated with turbulent wall modeling and glaze-ice thermodynamics. Nevertheless, the resulting 3D ice geometry demonstrates the feasibility of coupling the immersed-boundary formulation with the level-set geometry-evolution methodology for 3D ice accretion and provides a basis for future simulations under representative aerodynamic and icing conditions.

5.7 Conclusion

This chapter extended the capabilities of an existing ghost-cell-based RANS immersed-boundary flow solver to ice accretion simulations. Eulerian and Lagrangian droplet transport formulations were adapted to the immersed geometry and integrated with the thermodynamic and geometry-evolution methodologies previously presented, resulting in a coupled immersed-boundary icing framework. The objective is to reduce the dependency on body-fitted mesh generation and deformation while preserving the accuracy required for aircraft icing simulations.

The framework was first verified using a canonical cylinder case with an analytical collection-efficiency solution. Both Eulerian and Lagrangian droplet solvers reproduced the theoretical distribution on body-fitted and immersed-boundary meshes.

The methodology was then assessed under SLD conditions using the [141] benchmark. Including droplet splashing improved the agreement with experimental collection-efficiency data, with similar behavior observed for the body-fitted and immersed-boundary formulations.

Complete ice accretion simulations were then performed for rime and glaze conditions on a NACA 0012 airfoil from wind-tunnel experiments [170]. For the rime case, the body-fitted and immersed-boundary results were nearly identical and agreed well with the experimental profile. For the glaze case, both approaches reproduced the main ice-shape features, including horn formation, although differences in heat transfer led to deviations in the multilayer ice shapes.

Finally, the framework was applied to the 3D RG15 configuration under simplified laminar-flow condition. Consistent Eulerian and Lagrangian impingement distributions were obtained, and the multilayer simulation demonstrated the coupling of the immersed-boundary flow and droplet solvers with the level-set geometry-evolution methodology for a 3D configuration.

Future work will focus on improving the immersed-boundary aerodynamic and thermal wall treatments and assessing the methodology under more representative 3D aerodynamic and icing conditions.

CHAPTER 6 GENERAL DISCUSSIONS

The developments presented in this thesis introduce alternative numerical approaches at three distinct stages of the ice-accretion framework: Eulerian and Lagrangian formulations for droplet impingement, algebraic surface displacement and level-set methods for geometry evolution, and body-fitted and immersed-boundary approaches for the treatment of the evolving geometry. Each alternative introduces different compromises between computational efficiency, numerical robustness, implementation complexity, and modeling capabilities. This chapter discusses these trade-offs collectively, with particular emphasis on how the appropriate methodology depends on the complexity and objectives of the icing simulation.

6.0.1 Eulerian and Lagrangian Droplet Impingement

The first two Ice Prediction Workshops primarily focused on assessing the ability of numerical icing tools to reproduce experimental ice shapes. Although the final ice geometry provides an important validation metric, ice accretion results from the successive coupling of the different modules discussed throughout this thesis. Agreement or disagreement in the final ice shape cannot therefore necessarily be attributed to a particular component of the icing framework. The recent third Ice Prediction Workshop (IPW3) consequently placed greater emphasis on the verification and validation of the individual components of the icing framework. For droplet impingement, one of the questions considered was whether the Eulerian and Lagrangian formulations converge toward consistent predictions despite their different descriptions of the dispersed phase.

The IPW3 test cases included the NACA0012 and the ONERA M6 wing, both considered in a 3D setup. The corresponding reference conditions prescribed for the workshop are summarized in Table 6.1. In the present section, the NACA0012 configuration is used to compare the integrated collected-water mass predicted by the Eulerian and Lagrangian formulations across successive grid refinements. The ONERA M6 configuration provides a complementary benchmark for assessing the computational performance of both formulations and the time required to obtain the corresponding impingement solutions.

Figure 6.1 compares the integrated collected-water mass predicted by the Eulerian and Lagrangian formulations over the grid sequence summarized in Table 6.2. Results are shown for droplets corresponding to the MVD of 20 μm. The difference between the two formulations remains below 2% over the complete grid sequence and systematically decreases with spatial refinement. The figure also highlights a greater sensitivity of the Lagrangian prediction to

Table 6.1 Reference conditions for the IPW3 droplet-impingement benchmark cases.

	NACA0012	ONERA M6
M_∞	0.315	0.400
α [°]	4.0	2.0
Re	4.40×10^6	4.256×10^6
LWC [g/m^3]	0.55	0.51
MVD [μm]	20	20
c_{ref} [m]	0.5334	0.801673

Table 6.2 NACA0012 grid hierarchy.

Grid level	Elements
L4	5,861,000
L3	11,079,000
L2	26,988,000
L1	59,863,000

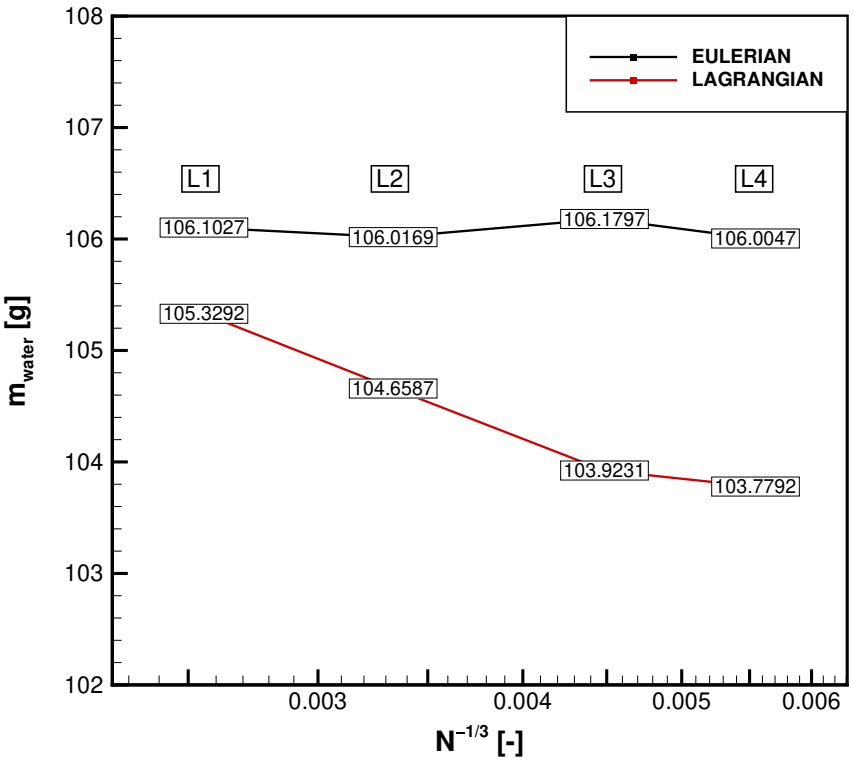


Figure 6.1 Spatial convergence of the integrated collected-water mass obtained with the Eulerian and Lagrangian droplet formulations for the NACA0012 IPW3 configuration using the MVD of 20 μm.

the underlying volume-mesh resolution. This behavior can be attributed in part to the repeated interpolation of the carrier-flow quantities along a large number of droplet trajectories following different paths through the computational domain. Consequently, changes in the spatial resolution of the underlying flow field can influence the integrated collected-water mass through the trajectory calculations. In contrast, the Eulerian droplet equations are solved directly on the volume mesh using the local flow solution, resulting in considerably less variation of the integrated collected-water mass over the present grid sequence. Nonetheless, the Eulerian and Lagrangian formulations converge toward consistent integrated impingement predictions with spatial refinement, reaching a difference of 0.8% on Level 1.

The computational performance of the two formulations is compared for the ONERA M6 configuration, with representative timings reported in Table 6.3. The Level 4 calculation was performed on a workstation equipped with an Intel Xeon w7-2595X processor with 26 physical cores. The Level 1 calculation was performed on 8 Trillium cluster compute nodes, each containing two AMD EPYC 9655 processors with 96 cores per socket, corresponding to 192 physical cores per node. The hardware information is provided for reference to distinguish between the shared-memory and distributed-memory calculations.

Table 6.3 Representative computational performance of the Eulerian and Lagrangian droplet formulations for the ONERA M6 configuration.

Grid	Elements	Eul. [s]	Lag. [s]	ETR* [%]	Hardware
L4	135,168	83.05	15.40	81.5	Xeon w7-2595X
L1	69,206,016	1501.81	1182.13	21.3	8 × 2 AMD EPYC 9655

Note: ETR: execution-time reduction.

The Lagrangian formulation is particularly attractive when the computational mesh can be efficiently accessed within shared memory, allowing trajectories to traverse the domain without requiring remote access to flow and mesh data. For the ONERA M6 calculations reported here, the Lagrangian formulation reduces the execution time relative to the Eulerian formulation by 81.5% on Level 4 and 21.3% on Level 1. For the larger distributed-memory calculation, the required flow and mesh data are replicated across the compute nodes to minimize communication during trajectory integration. This replication represents a necessary trade-off: although it substantially reduces remote data accesses, it increases the memory requirements on each compute node and can become restrictive as the mesh size increases. In addition, the irregular memory-access patterns associated with trajectory tracking make the Lagrangian formulation more susceptible to memory-bandwidth limitations and cache misses as the number of concurrent trajectories increases. These effects reduce the computational advantage of the Lagrangian formulation on the Level 1 case despite limiting inter-node

communication.

The Eulerian formulation naturally follows the spatial decomposition of the computational mesh, making its parallel execution relatively straightforward to configure. In contrast, the Lagrangian performance reported here was achieved after careful optimization of the adaptive seeding strategy, trajectory workload balancing, and execution configuration according to the underlying cluster architecture. Efficient Lagrangian simulations on large 3D problems therefore require greater implementation effort and knowledge of the computing environment, while the Eulerian formulation is generally easier to configure for distributed-memory calculations.

6.0.2 Algebraic and Level-Set Ice Accretion

Explicit algebraic surface displacement presents important limitations for 3D multilayer ice accretion as increasingly complex ice features develop. Figure 6.2 compares the algebraic and level-set approaches under the same icing conditions and using the same number of accretion layers ($N_{\text{layers}} = 5$), such that the differences observed between the final geometries originate primarily from the geometry-evolution procedure. With the algebraic approach, successive displacement of the surface nodes causes the mesh to become increasingly susceptible to local folding and element entanglement. Extensive surface smoothing is therefore required between successive accretion steps to maintain a valid computational surface. Although this stabilizes the geometry-evolution procedure, it also progressively damps the smaller-scale ice features generated during the accretion process. This effect is clearly visible in Fig. 6.2, where the algebraic solution exhibits a considerably smoother and more uniform ice structure. In contrast, the level-set formulation allows the interface to evolve without being directly constrained by the connectivity of the previous surface mesh and requires little to no surface smoothing, thereby preserving substantially more of the 3D structures that develop during successive accretion layers.

This difference has implications beyond the prediction of an individual ice shape. In practice, the robustness of the geometry-evolution procedure determines which numerical experiments can be performed reliably. With algebraic displacement, increasingly complex geometries or a larger number of accretion layers can require case-dependent smoothing and remeshing parameters, while severe surface distortion may still lead to failure before completion of the simulation. Consequently, the geometry-update procedure itself can become a limiting factor when attempting to investigate the sensitivity of the complete icing framework to physical and numerical parameters.

The level-set methodology substantially reduces this limitation and therefore provides a more

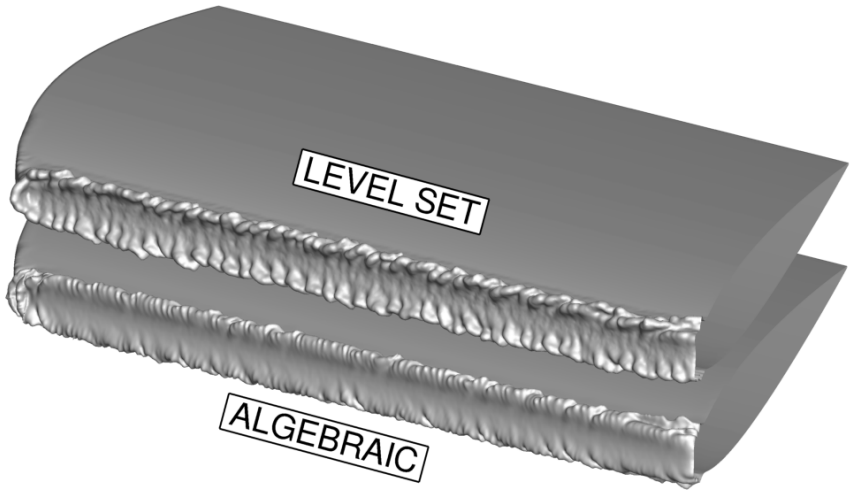


Figure 6.2 Comparison of algebraic and level-set ice shapes under identical icing conditions after five accretion layers.

suitable framework for systematic investigations of 3D multilayer ice accretion. Once the geometry evolution is sufficiently robust, different icing conditions, droplet-size distributions, spatial resolutions, and numbers of accretion layers can be investigated without the same concern that the evolving surface will fold onto itself or cause the simulation to fail midway through the accretion sequence. This capability is particularly important because the resulting 3D ice morphology can exhibit a strong sensitivity to these parameters. As illustrated in Fig. 6.3, different structures can emerge along the iced leading edge depending on the droplet-size distribution, the number of accretion layers, and the spatial resolution used to represent the evolving interface. The level-set methodology therefore provides not only a more robust geometry-evolution technique, but also an enabling tool for exploring the underlying numerical and physical mechanisms governing the development of complex 3D ice structures, such as scallop formation.

The level-set formulation nevertheless introduces a considerably more elaborate geometry-evolution procedure. Rather than directly displacing the surface mesh, the ice–air interface is represented implicitly as the zero contour of a level-set function defined over an auxiliary volume mesh. Each geometry update therefore requires the construction or adaptation of this volume mesh, initialization of the signed-distance field, propagation of the required surface quantities into the volume, advection of the level-set field, and extraction of the updated zero-level-set surface. These additional operations increase both the computational cost and the complexity of setting up a simulation compared with direct algebraic displacement. Much

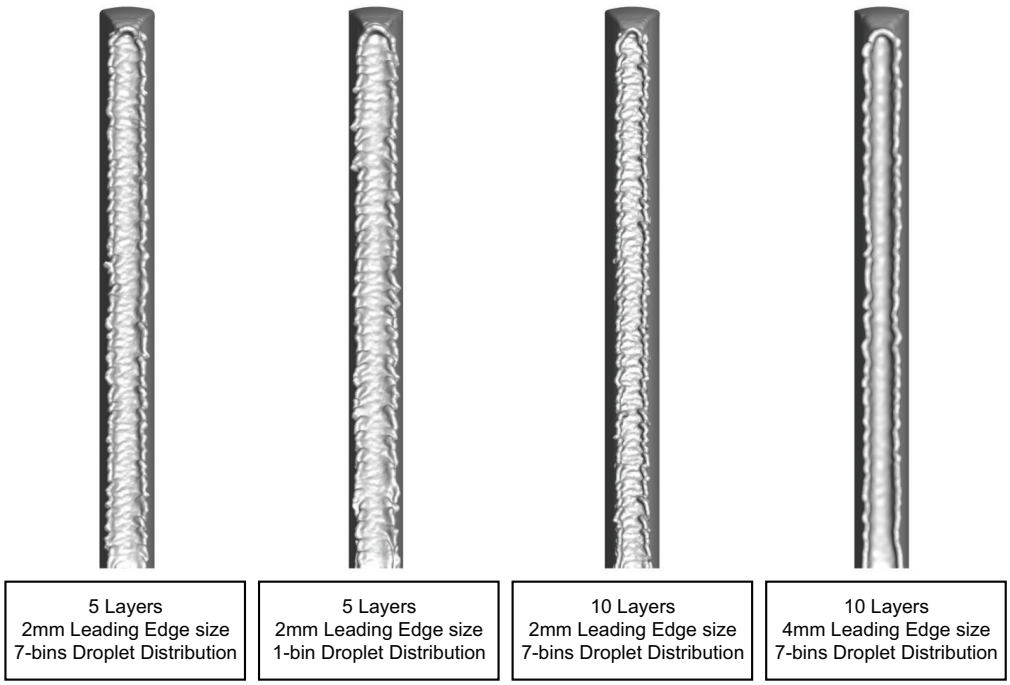


Figure 6.3 3D level-set ice shapes obtained for different numerical and physical configurations.

of this workflow can, however, be automated, such that the additional complexity does not necessarily translate into a comparable increase in user intervention.

6.0.3 Body-Fitted and Immersed-Boundary Approaches

As CFD applications are extended toward increasingly complex geometrical configurations, mesh generation can represent a substantial portion of the overall simulation workflow. While generating a computational mesh around a complex geometry is generally straightforward, obtaining a high-quality body-fitted mesh with adequate near-wall resolution and controlled element quality can require significant effort. This difficulty becomes particularly important for applications involving evolving geometries, such as ice accretion, where the computational mesh must accommodate increasingly complex ice surfaces over successive accretion layers.

Body-fitted meshes provide direct control of the near-wall resolution and are well suited to aerodynamic and icing calculations, for which wall shear stress and heat transfer depend strongly on the near-wall solution. Their main limitation for multilayer ice accretion is the need to maintain a high-quality geometry-conforming mesh as the ice surface evolves. As increasingly complex ice features develop, repeated deformation or regeneration of the volume mesh can therefore become a significant constraint on the robustness of the complete

simulation.

Immersed-boundary methods alleviate this constraint by embedding the evolving geometry within a non-conforming background mesh, thereby avoiding repeated generation of a body-fitted RANS mesh. This flexibility comes at the expense of the spatial resolution required near the immersed interface. Accurate representation of the geometry and reconstruction of near-wall quantities require sufficiently small cells around the surface, and the background mesh must remain adequately refined as the ice grows away from the clean geometry.

In Chapter 5, this requirement was addressed by prescribing a sufficiently large refined region around the anticipated ice-accretion region, as illustrated in Figure 6.4. While suitable for the cases considered, this approach requires prior knowledge or an estimate of the expected ice geometry and can introduce unnecessary cells in regions where fine resolution is ultimately not required.

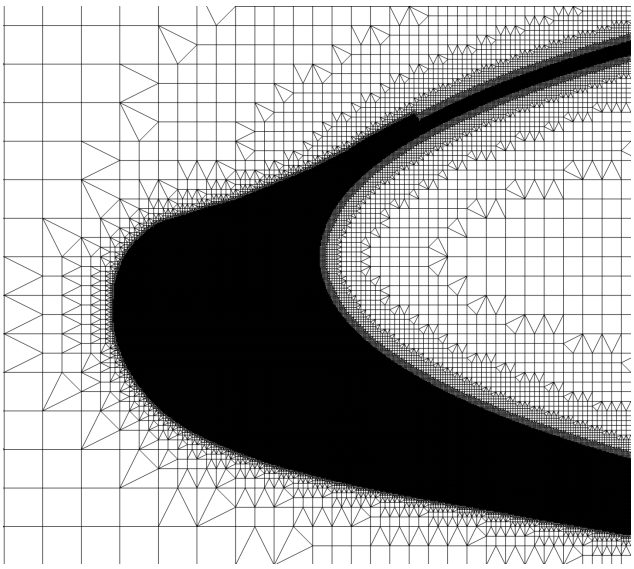


Figure 6.4 Example of the prescribed refined region surrounding the NACA0012 geometry and the anticipated ice-accretion region.

Adaptive mesh refinement provides a more general solution by allowing the background mesh to follow the evolving immersed interface while retaining coarser cells elsewhere. Combining immersed-boundary methods with efficient adaptive refinement would therefore preserve their principal advantage for multilayer icing—avoiding repeated body-fitted volume-mesh generation—without requiring a large predefined region of fine resolution. The main challenge is to perform this adaptation efficiently enough that its computational cost does not offset the advantages of the immersed-boundary approach.

CHAPTER 7 CONCLUSION

7.1 Summary of Works

The objective of this thesis was to improve the robustness of 3D multilayer ice accretion simulations, with particular emphasis on the numerical difficulties associated with droplet impingement, repeated geometry evolution, and the treatment of increasingly complex iced surfaces. The work was conducted within the CHAMPS icing framework and resulted in developments addressing three complementary components of the icing simulation framework: Lagrangian droplet tracking, level-set-based geometry evolution, and immersed-boundary ice accretion.

The first contribution concerned the development and assessment of a Lagrangian particle-tracking methodology for droplet impingement. A face-to-face trajectory algorithm was combined with ray-tracing-enhanced intersection detection, adaptive seeding, and surface interpolation strategies to improve the robustness and computational efficiency of the method. Supercooled Large Droplet effects were incorporated into the droplet formulations, including models accounting for droplet deformation, splashing, and secondary-droplet transport. The resulting predictions showed good agreement with reference and experimental data. Comparisons between the Eulerian and Lagrangian formulations further demonstrated that both approaches converge toward consistent collection-efficiency and collected-water-mass predictions.

The comparison nevertheless highlighted important differences between the two formulations. The Eulerian approach naturally follows the decomposition of the computational mesh and provides a predictable workload and efficient parallelization for large 3D simulations. The Lagrangian formulation can be considerably less expensive when a relatively small number of trajectories is sufficient, particularly in 2D calculations, but becomes more challenging on large distributed-memory configurations because of irregular trajectory lengths, load balancing, and access to mesh information distributed across computational domains. Its principal advantage lies in retaining the history and impact location of individual droplets, providing a natural framework for more detailed physical models involving splashing and re-impingement of the resulting droplets. The two formulations should therefore be regarded as complementary rather than competing approaches.

The central contribution of this thesis was the development of a continuous level-set methodology for 3D multilayer ice accretion. The evolving ice–air interface is represented implic-

itly as the zero contour of a signed-distance function defined on an auxiliary volume mesh. The proposed methodology combines surface-quantity propagation, semi-implicit level-set advection, local interface reconstruction for ice-volume evaluation, interface extraction, and surface smoothing. The method was assessed using representative icing benchmarks and demonstrated the ability to reproduce complex ice geometries while maintaining a conformal reconstructed surface.

The comparison with explicit algebraic surface displacement illustrates the principal motivation for the level-set approach. Direct displacement remains simple and computationally attractive for single-layer calculations and geometries undergoing limited deformation. However, its robustness progressively depends on the quality and connectivity of the evolving surface mesh as additional accretion layers are introduced. The implicit level-set representation removes this direct dependency during geometry evolution and consequently allows more complex geometries to develop without surface-element entanglement. Although this requires additional operations associated with volume-meshing, field propagation, advection, and interface extraction, these procedures can largely be automated and accelerated. The additional numerical complexity therefore provides an advantageous trade-off when robust multilayer evolution is required.

The third contribution extended the icing framework to an immersed-boundary representation. The methodology combines immersed-boundary aerodynamic and droplet treatments with the existing thermodynamic model and level-set geometry-evolution procedure. Verification on canonical configurations demonstrated consistency between body-fitted and immersed-boundary droplet predictions, while rime and glaze ice-accretion calculations showed encouraging agreement between the two approaches. These results demonstrate the feasibility of performing the complete icing sequence without requiring the generation of a new aerodynamic volume mesh at each accretion layer.

The immersed-boundary investigation also showed that eliminating body-fitted remeshing does not eliminate the mesh-resolution problem. Instead, the numerical challenge shifts toward maintaining sufficient spatial resolution around the evolving immersed interface. Prescribing a sufficiently large region of uniformly fine cells can rapidly become computationally expensive for 3D configurations. Adaptive mesh refinement therefore represents a natural complement to the immersed-boundary approach, allowing the computational resolution to follow the evolving ice geometry without requiring excessive refinement throughout the complete domain.

Taken together, the developments presented in this thesis address different but interconnected limitations of 3D icing simulations. Lagrangian particle tracking increases the flexibility avail-

able for modeling detailed droplet physics, the level-set formulation improves the robustness of multilayer geometry evolution, and the immersed-boundary methodology reduces the dependency of the flow solution on repeated geometry-conforming volume remeshing. Among these developments, the level-set formulation provides the central geometry-evolution framework around which the other methodologies can be combined. The resulting numerical framework provides a foundation for increasingly automated and robust 3D multilayer ice accretion simulations.

7.2 Limitations

Despite the developments achieved in this work, several limitations remain within the different components of the icing framework. The accuracy of ice-accretion predictions remains dependent on the physical models preceding geometry evolution. A robust geometric methodology cannot compensate for inaccuracies in aerodynamic quantities, droplet impingement, heat transfer, surface roughness, or thermodynamic modeling. This limitation is particularly important under glaze ice conditions, for which horn development exhibits substantial sensitivity to the convective heat-transfer prediction and the prescribed equivalent sand-grain roughness.

The Lagrangian formulation also retains computational limitations for large 3D configurations. The number of trajectories required to obtain a converged impingement field is not known a priori, and individual trajectories can have substantially different computational costs. Adaptive seeding reduces the number of unnecessary trajectories, but efficient execution on distributed-memory systems continues to depend on appropriate load balancing and data-locality management. Consequently, the computational advantage observed for smaller or shared-memory problems does not necessarily extend directly to large distributed-memory calculations.

Finally, the present immersed-boundary implementation remains limited by the absence of dynamic adaptive mesh refinement. Adequate resolution must currently be prescribed over a sufficiently large region to accommodate the expected evolution of the ice geometry. This requires prior knowledge or an estimate of the resulting ice shape and can lead to unnecessarily large computational meshes when refinement is maintained in regions that are not ultimately reached by the evolving interface. The development of an efficient adaptive strategy capable of refining and coarsening the background mesh as the ice surface evolves therefore represents an important step toward improving the computational efficiency of the immersed-boundary approach for large-scale 3D multilayer icing simulations.

7.3 Future Research

The developments presented in this thesis suggest several directions for future research. For Lagrangian droplet tracking, future developments should focus on improving computational efficiency for large 3D configurations, particularly through GPU acceleration. A GPU-oriented implementation would, however, require data structures and algorithms specifically designed to accommodate irregular memory-access patterns, mesh-intersection searches, and heterogeneous trajectory lengths. Combined with the adaptive cloud-refinement and load-balancing strategies developed in this work, such an implementation could substantially reduce the computational cost of large-scale 3D impingement calculations.

Further development of the level-set methodology should focus on improving the geometry-evolution workflow for large 3D icing simulations. MMG is a general-purpose meshing tool and is not specifically designed to generate surface meshes for CFD simulations. A dedicated surface-meshing strategy could provide greater control over surface-element size and quality, particularly in highly curved regions of the evolving ice geometry, while ensuring that the extracted surface remains suitable for subsequent volume-mesh generation and flow calculations. The current methodology also enforces conservation of the total ice volume at the global level. Future work should investigate local ice-mass conservation strategies to ensure that the spatial distribution of the accreted mass is preserved during level-set advection and interface reconstruction. More generally, the increased robustness of the level-set geometry-evolution framework provides an opportunity to investigate physical modeling assumptions more systematically. In particular, the influence of spatially and temporally varying surface roughness, variable ice-density formulations, and their interaction with the thermodynamic model can now be assessed with greater confidence over complex 3D multilayer configurations, without the robustness of the geometry-update procedure becoming a limiting factor.

Finally, further development of the immersed-boundary icing framework should focus on improving the near-wall heat-transfer treatment and incorporating adaptive mesh refinement. Improving the near-wall heat-transfer treatment is necessary to further bridge the gap between immersed-boundary and established body-fitted approaches for icing simulations. Adaptive mesh refinement would dynamically maintain the spatial resolution required near the immersed interface as the ice geometry evolves, while retaining coarser cells elsewhere. Together, these developments would improve the applicability of the immersed-boundary approach within CHAMPS for complex 3D multilayer icing simulations.

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APPENDIX A THERMODYNAMIC BALANCE EQUATIONS

This appendix provides the detailed expressions associated with the mass- and energy-balance equations introduced in Section 2.4. The balances are applied locally to a surface control volume containing the ice layer and, when present, a liquid-water film.

A.1 Mass Balance

The local conservation of water mass is expressed as

$$\dot{m}_{imp} + \sum \dot{m}_{in} = \sum \dot{m}_{out} + \dot{m}_{ev/su} + \dot{m}_{ice} \tag{A.1}$$

where $\dot{m}_{imp}$ is the impinging-water mass flow rate, $\dot{m}_{in}$ and $\dot{m}_{out}$ are the incoming and outgoing runback-water flow rates, respectively, $\dot{m}_{ev/su}$ represents mass loss through evaporation or sublimation, and $\dot{m}_{ice}$ is the ice-accretion rate. The evaporation or sublimation rate may be evaluated as

$$\dot{m}_{ev/su} = \frac{0.7}{C_{p,a}} HTC \left(\frac{p_{v,s} - p_{v,\infty}}{P_\infty} \right) \tag{A.2}$$

where $C_{p,a}$ is the specific heat capacity of air, $p_{v,s}$ and $p_{v,\infty}$ are the water-vapor pressures at the surface and in the freestream, respectively, and P_∞ is the freestream static pressure. The vapor pressure at the water or ice surface is evaluated as [171]

$$p_v = 3386 \left(0.0039 + 6.8096 \times 10^{-6} \bar{T}^{\,2} + 3.5579 \times 10^{-7} \bar{T}^{\,3} \right) \tag{A.3}$$

where p_v is expressed in Pa and

$$\bar{T} = 72 + 1.8 \left(T - 273.15 \right) \tag{A.4}$$

with T expressed in K.

A.2 Energy Balance

Conservation of energy over the same surface control volume is written as

$$\dot{Q}_{kin} + \dot{Q}_d + \dot{Q}_a + \sum \dot{Q}_{in} = \dot{Q}_{conv} + \dot{Q}_{rad} + \dot{Q}_{ev/su} + \sum \dot{Q}_{out} + \dot{Q}_{ice} \tag{A.5}$$

where the individual contributions are detailed below. The kinetic energy associated with the impinging droplets is

$$\dot{Q}_{kin} = \dot{m}_{imp} \frac{u_d^2}{2} \tag{A.6}$$

where u_d is the droplet impact velocity. The sensible heat associated with the impinging droplets is

$$\dot{Q}_d = \dot{m}_{imp} C_{p,w} \left(T_s - T_\infty \right), \tag{A.7}$$

where $C_{p,w}$ is the specific heat capacity of liquid water. The aerodynamic-heating contribution is expressed as

$$\dot{Q}_a = \frac{r\, HTC\, U_\infty^2}{2 C_{p,a}}, \tag{A.8}$$

where r is the recovery factor. The convective heat exchange is

$$\dot{Q}_{conv} = HTC \left(T_s - T_{rec} \right), \tag{A.9}$$

The energy transported by incoming and outgoing runback water is

$$\sum \dot{Q}_{in} = \sum \dot{m}_{in} C_{p,w} \left(T_s - T_{ref} \right), \tag{A.10}$$

and

$$\sum \dot{Q}_{out} = \sum \dot{m}_{out} C_{p,w} \left(T_s - T_{ref} \right), \tag{A.11}$$

respectively, where T_{ref} denotes the reference freezing temperature. Radiative heat transfer is given by

$$\dot{Q}_{rad} = 4 \epsilon \sigma_r T_\infty^3 \left(T_s - T_\infty \right), \tag{A.12}$$

where ϵ is the surface emissivity and σ is the Stefan–Boltzmann constant. The latent-energy

contributions associated with evaporation and sublimation are

$$\dot{Q}_{ev} = \dot{m}_{ev} L_v, \qquad \dot{Q}_{su} = \dot{m}_{su} L_s, \tag{A.13}$$

where L_v and L_s are the latent heats of vaporization and sublimation, respectively. The energy associated with ice formation may be expressed as

$$\dot{Q}_{ice} = \dot{m}_{ice} \left[C_{p,i} \left(T_s - T_{ref} \right) - L_f \right], \tag{A.14}$$

where $C_{p,i}$ is the specific heat capacity of ice and L_f is the latent heat of fusion.

APPENDIX B DROPLET-IMPINGEMENT VERIFICATION AND SLD MODELS

This appendix presents supplementary material for the droplet-impingement framework. It first documents the mesh-refinement study used to compare the Eulerian and Lagrangian collection-efficiency predictions and then provides the detailed parameters of the SLD splashing models.

B.1 Mesh Refinement Study

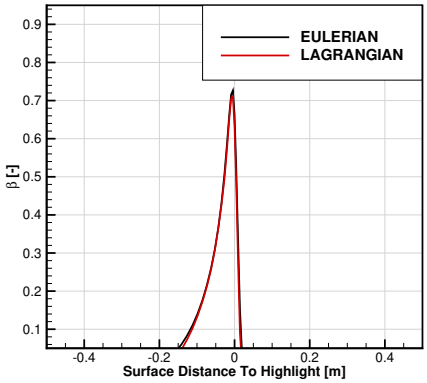
Figure B.1 compares the Eulerian and Lagrangian collection-efficiency predictions on four successively refined meshes. The corresponding flow conditions are summarized in Table B.1, and the maximum differences are reported in Table B.2.

Table B.1 Summary of the NACA 0012 droplet-impingement verification case.

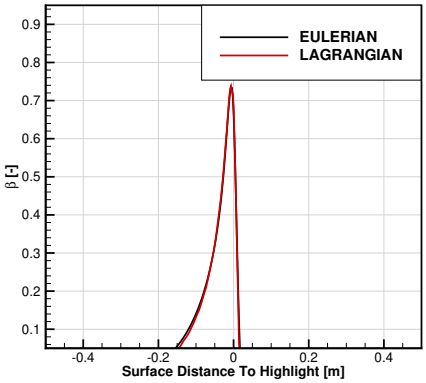
Parameter	NACA 0012
Geometry	2D NACA 0012
MVD $[\mu m]$	30.0
AOA [°]	5.0
Ma [-]	0.4
C_{ref} $[inch]$	1.0
Re [-]	$8.85 \cdot 10^6$

Table B.2 Difference Between Eulerian vs. Lagrangian results in terms of Collection Efficiency.

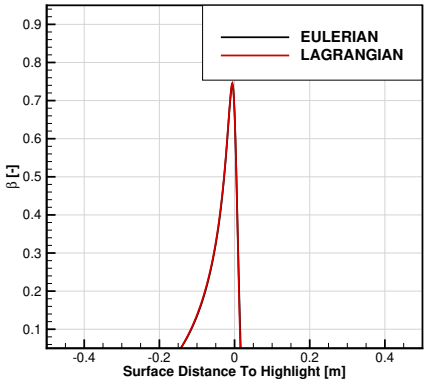
Mesh	Max Difference β [-]
129x129	0.069
257x257	0.059
513x513	0.037
1025x1025	0.035



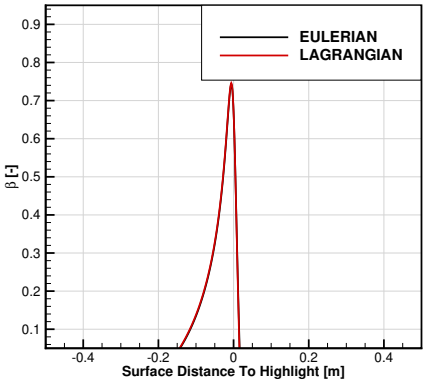
(a) Impingement Results on the 129x129 mesh.



(b) Impingement Results on the 257x257 mesh.



(c) Impingement Results on the 513x513 mesh.



(d) Impingement Results on the 1025x1025 mesh.

Figure B.1 Mesh refinement effect on the Eulerian vs. Lagrangian results in terms of the collection efficiency.

B.2 SLD Model Details

Table B.3 provides the parameters used for the SLD models described in Section 2.3.3.

Table B.3 SLD Splashing Models Details

	Wright et al. [64]	
θ_0	Droplet Angle of Impact	-
K_M	Splashing Parameter	$\frac{0.859\,K^{0.5}\,(\frac{\rho_w}{LWC})^{0.125}}{sin(\theta_0)^{1.25}}$
K	Mundo Parameter	$Oh\,Re_d^{1.25}$
Oh	Ohnesorge Number	$\frac{\mu_w}{\sqrt{\sigma_w\,\rho_w\,D}}$
	Trontin et al. [68]	
θ_c	Characteristic Droplet Angle of Impact	$25.0°$
θ_c	Critical Droplet Angle of Impact	$\theta_0\,tanh(Ca^{1.5})$
Ca	Capillary Number	$\frac{\mu_w\lvert\boldsymbol{u_d}\rvert}{\sigma_w}$
K_C	Cossali Number	$We\,Oh^{-0.4}$
We	Weber Number	$\frac{\rho_w\lvert\boldsymbol{u_d}\rvert^2 d}{\sigma_w}$
K_0	Critical Cossali Number	657.0
$g(\frac{\theta}{\theta_c})$	Angle of Impact Function	$1\,if\,\theta > \theta_c\,else\,\frac{\theta}{\theta_c}$
$f(\tilde{K_c})$	Cossali Number Function	$1 - 0.15\frac{\tilde{K_c}^2}{(100+\tilde{K_c}^2)}$
$\tilde{K_c}$	-	$max(\frac{K_c - K_0}{K_0}, 0)$

APPENDIX C NUMERICAL VERIFICATION

This appendix provides supplementary verification results for the numerical methodologies developed in this thesis.

C.1 Lagrangian Droplet Integration

The temporal accuracy of the Lagrangian droplet integration is assessed using a manufactured trajectory with an analytical solution. A droplet is transported in a prescribed uniform carrier velocity $\boldsymbol{U}$ and subjected to a constant body acceleration $\boldsymbol{g}$ according to

$$\frac{d\boldsymbol{u}_d}{dt} = 2\left(\boldsymbol{U} - \boldsymbol{u}_d\right) + \boldsymbol{g} \qquad \frac{d\boldsymbol{x}_d}{dt} = \boldsymbol{u}_d \tag{C.1}$$

The prescribed conditions are $\boldsymbol{U} = (1, 0.4, 0.2)$, $\boldsymbol{g} = (0, -0.3, 0.1)$, $\boldsymbol{u}_d(0) = (0.2, -0.1, 0.05)$, and $\boldsymbol{x}_d(0) = \boldsymbol{0}$. These conditions produce a time-dependent droplet velocity while retaining an analytical solution for the trajectory. Temporal convergence is evaluated using the Euclidean error in the droplet position at the final time:

$$e_x = \|\boldsymbol{x}_{\text{num}}(t_f) - \boldsymbol{x}_{\text{exact}}(t_f)\| \tag{C.2}$$

Figure C.1 shows the decrease in the final-position error with time-step refinement. The error follows the second-order reference slope, confirming second-order temporal convergence of the Lagrangian trajectory integration for this verification problem.

C.2 Level-Set Transport

The spatial and temporal accuracy of the level-set transport solver is assessed using the uniform translation of a circular level-set field over the unit-square domain $\Omega = [0, 1] \times [0, 1]$. The field has a radius $R = 0.2$ and is initially centered at $(0.35, 0.45)$. A constant velocity $\boldsymbol{V} = (0.2, 0.1, 0)$ is prescribed from $t = 0$ to $t = 0.5$, resulting in a displacement of $(0.1, 0.05)$ and a final center position of $(0.45, 0.50)$. Since the field undergoes pure translation without deformation, the analytical solution at the final time is obtained directly by translating the initial field.

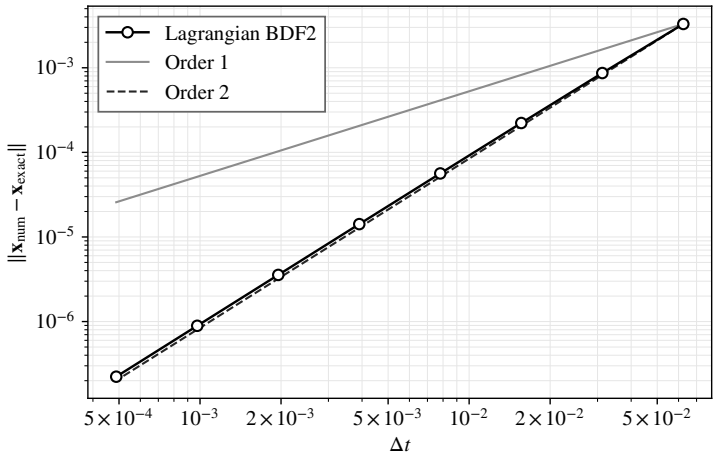


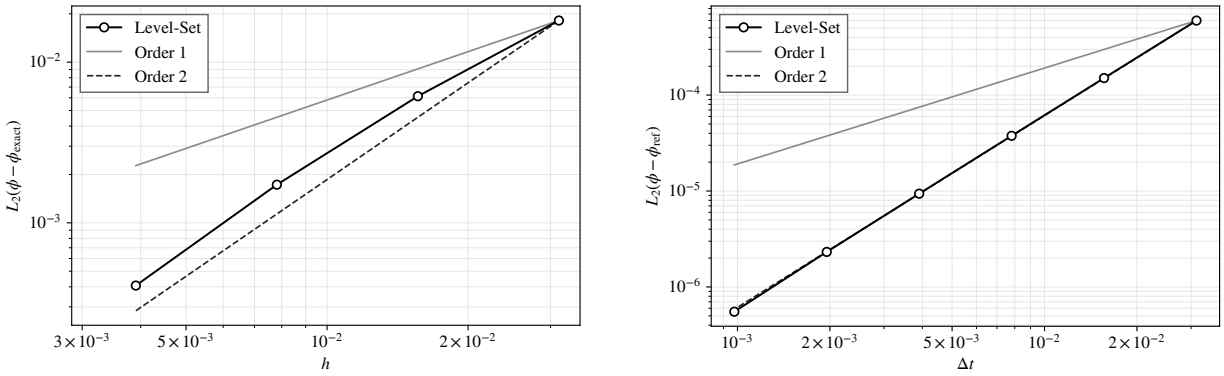
Figure C.1 Temporal convergence of the Lagrangian droplet trajectory integration using BDF2 with a BDF1 startup step.

C.2.1 Spatial Convergence

Spatial convergence is evaluated using four successively refined meshes with characteristic spacings $h = \Delta x = \Delta y = 0.03125$, 0.015625, 0.0078125, and 0.00390625, corresponding to $N = 32$, 64, 128, and 256 divisions along each coordinate direction. The time step is reduced consistently with mesh refinement such that the temporal error remains negligible relative to the spatial discretization error. Figure C.2(a) shows the decrease in the L_2 error with mesh refinement. The observed convergence approaches second order, consistent with the spatial reconstruction used by the level-set solver.

C.2.2 Temporal Convergence

Temporal convergence is evaluated at a fixed spatial resolution of $N = 256$. Starting from $\Delta t = 0.03125$, the time step is successively halved, corresponding to 16, 32, 64, 128, 256, and 512 time steps over the prescribed final time. The spatial discretization is kept fixed throughout the refinement study. The second-order Rosenbrock–Wanner scheme is used to advance the level-set transport equation. To isolate the temporal discretization error from the fixed spatial error, the numerical solutions are compared with a reference solution computed on the same spatial mesh using 2048 time steps. Figure C.2(b) shows the resulting L_2 error with time-step refinement. The measured convergence remains approximately second order, consistent with the formal temporal accuracy of the Rosenbrock–Wanner scheme.



(a) Spatial convergence.

(b) Temporal convergence.

Figure C.2 Spatial and temporal convergence of the level-set transport solver for the uniform translation of a circular level-set field: (a) spatial convergence with mesh refinement and (b) temporal convergence of the second-order Rosenbrock–Wanner scheme with time-step refinement.

C.3 Eulerian Immersed-Boundary Droplet Formulation

The spatial convergence of the Eulerian immersed-boundary droplet formulation is assessed using the analytical cylinder impingement problem introduced in Section 5.6.1. The Cartesian background mesh is successively refined from $h = 0.04$ to $h = 0.0025$, where $h = \Delta x$ denotes the uniform Cartesian cell size, while all other numerical parameters are maintained constant. The convergence is evaluated using the error in the integrated collection efficiency β_{tot} relative to the analytical solution.

Figure C.3 shows the decrease in the integrated collection-efficiency error with mesh refinement. The coarser meshes remain outside the asymptotic convergence regime, while the finer meshes exhibit approximately second-order convergence. A similar behavior was reported by Lavoie et al. [169] for immersed-boundary Eulerian droplet calculations. Although volume penalization generally limits the spatial accuracy near the immersed boundary, the penalization of the droplet equations is applied within the shadow region, where the collection efficiency is zero, rather than within the impingement region where β is evaluated. Consequently, the penalization does not necessarily limit the convergence rate of the predicted collection efficiency.

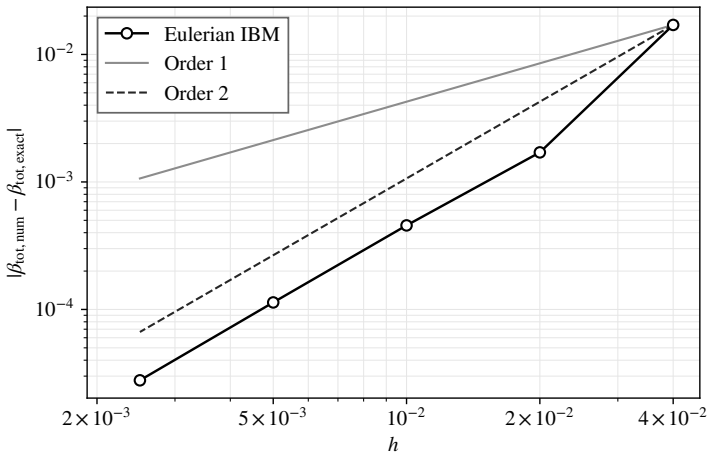


Figure C.3 Spatial convergence of the integrated collection efficiency for the Eulerian immersed-boundary cylinder impingement problem.